

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-31005-RP.pvdb

Date of Analysis : Aug 12, 2026 9:36am

PV Elite 28, January 2026



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FileName : 401-E-31005-RP

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Vessel Design Summary: Step: 29 9:36am Aug 12,2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1210.000 mm ID
Vessel Design Length, Tangent to Tangent 16848.70 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 110 °C
Channel Side Design Temperature 65 °C
Shell Side Design Pressure 16000.000 KPa.
Channel Side Design Pressure 1000.000 KPa.
Shop Shell Side Test Pressure 21254.553 KPa.
Shop Channel Side Test Pressure 1464.740 KPa.
Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-350 LF2	1	...	K03011	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-516 70	...	...	K02700	
Tubes	SA-334 1	...	...	K03008	+IT
Tubesheet	SA-350 LF1	1	...	K03009	+IT
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Chnl rear cover	1000.000	0.00	1108.000	3.0000	N/A	No
Chnl rear flange	1000.000	0.00	1135.000	3.0000	N/A	No
Chnl. rear	1011.890	0.00	3458.000	3.0000	N/A	No
Shell	16000.594	0.00	17945.000	1.5000	N/A	No
Chnl. front	1011.890	0.00	3458.000	3.0000	N/A	No
Chnl. front flange	1000.000	0.00	1135.000	3.0000	N/A	No
Chnl. front flange	1000.000	0.00	1108.000	3.0000	N/A	No

Liquid Level: 1210.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
4574.35	Bar 134.4 x 14.0	SR, 2600.0 Fr40



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Vessel Design Summary:

Step: 29 9:36am Aug 12,2026

12524.35

Bar 134.4 x 14.0

SR, 10400.0 Fr4

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	71.3	65.0	73.0	65.0	61.9	...	1.00	1.00
Body Flg	174.3	103.0	90.0	82.0	77.4	...	1.00	1.00
Cylinder	1674.3	1500.0	16.0	16.0	6.8	...	1.00	1.00
Cylinder	14924.3	13000.0	67.0	67.0	59.6	...	1.00	1.00
Cylinder	16674.3	1500.0	16.0	16.0	6.8	...	1.00	1.00
Body Flg	16783.7	103.0	90.0	82.0	77.4	...	1.00	1.00
Body Flg	16848.7	65.0	73.0	65.0	61.9	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3- Channel in	WRC-297	0.966	Shell	Passed
N2- Shell outl	WRC-297	0.966	Shell	Passed
N1- Shell inle	WRC-297	0.966	Shell	Passed
N4- Channel ou	WRC-297	0.924	Nozzle	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	300.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	970.000	mm
Wear Pad Width	350.000	mm
Wear Pad Thickness	30.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	2650.000	mm
Baseplate Length	1260.000	mm
Baseplate Thickness	75.000	mm
Baseplate Width	150.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	45.000	mm
Web Thickness	45.000	mm
Height of Center Web	306.000	mm
Number of Bolts in Baseplate	2	

Baseplate Sketch



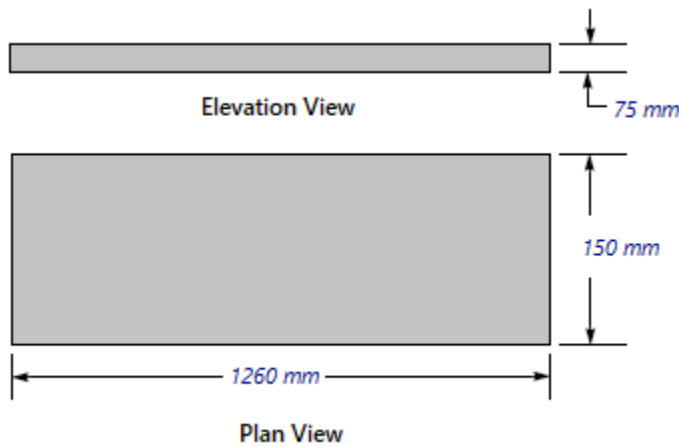
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Vessel Design Summary:

Step: 29 9:36am Aug 12,2026

**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	48859.73	Kgf
Maximum Transverse Saddle Shear Load	4186.85	Kgf
Maximum Longitudinal Saddle Shear Load	1021.70	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	43356.84	Kgf
Maximum Transverse Saddle Shear Load	804.28	Kgf
Maximum Longitudinal Saddle Shear Load	136.02	Kgf

Weights:

Fabricated - Bare w/o removable internals	64084.5	kg.
Shop Test - Fabricated + water (full)	80455.1	kg.
Shipping - Fab. + rem. intls.+ shipping app.	64084.5	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	64084.5	kg.
Empty - Fab. + intls. + details + wghts.	64084.5	kg.
Operating - Empty + operating liquid (no ca)	70962.3	kg.

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Nozzle Summary: Step: 27 9:36am Aug 12,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
N3- Channel inl	3263	...	...	OK	OK	Passed
N2- Shell outle	17945	...	...	OK	OK	Passed
N1- Shell inlet	17946	...	...	OK	OK	Passed
N4- Channel out	3031	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 17945 Nozzle : N2- Shell outle [Shellside]
 Minimum MAWP Nozzles : 3031 Nozzle : N4- Channel out [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
30	N3- Channel inle	924.350	90.000	217.325
40	N2- Shell outlet	2924.349	90.000	204.100
40	N1- Shell inlet	13924.349	270.000	204.100
50	N4- Channel outl	15924.349	90.000	217.325

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 26 9:36am Aug 12,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3- Channel inle	18.000 in	40	WNF	18.000	14.275	757.20	18.00	310.85	150
N2- Shell outlet	18.000 in Actual		None	18.000	26.000	...	...	457.07	...
N1- Shell inlet	18.000 in Actual		None	18.000	26.000	...	...	457.07	...
N4- Channel outl	18.000 in	40	WNF	18.000	14.275	857.20	16.00	310.85	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3- Channel	SA-106 B	14.000	10.000	12.000	18.000	...
N2- Shell o	SA-350 LF2	67.000	10.000	...	...	...
N1- Shell i	SA-350 LF2	67.000	10.000	...	...	...
N4- Channel	SA-106 B	14.000	10.000	12.000	16.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N3- Channel inle	924.350	90.0	250.00	0.00	Chnl. rear
N2- Shell outlet	2924.349	90.0	300.00	0.00	Shell
N1- Shell inlet	13924.349	270.0	300.00	0.00	Shell
N4- Channel outl	15924.349	90.0	250.00	0.00	Chnl. front

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Bill of Material:

Step: 25 9:36am Aug 12,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 73.0mm THK X 1380.0mm OD	SA-266 2
2	WELD NECK FLANGE: 90.0mm THK X 1380.0mm OD	SA-266 2
11	BAFFLES: 1200.0mm DIA X 8.0mm THK	SA-516 70
14	TIE RODS: 19mm DIA X 13000mm	SA-193 B7
2	TUBESHEET: 1344.0mm X 250.0mm THK	SA-350 LF1
1417	TUBES: 13500.0mm X 19.0mm DIA X 2.5mm THK	SA-334 1
2	SADDLE: 300.0mm X 120 DEG	SA-516 70
2	BAR RING STIFFENER: 1344.0mm ID X 1612.8mm OD X 14.0mm	SA-516 70
2	GASKET: 1271.0mm OD X 1239.0mm ID	...
64	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B7
128	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
2	CYLINDRICAL SEGMENT 16.0mm THK X 1509.9mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 781.4mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 67.0mm THK X 3051.6mm X 13000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 67.0mm THK X 749.7mm X 13000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 749.7mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 3051.6mm X 1500.0mm	SA-516 70
1	NAMEPLATE	...

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MDMT Summary:

Step: 28 9:36am Aug 12,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Shell [8]	B	14	12	Yes	0.945	67.000	1.00
SR, 2600.0 Fr[23]	B	-15	-15	Yes	1.000	14.000	1.00
SR, 10400.0 F[23]	B	-15	-15	Yes	1.000	14.000	1.00
Left Saddle [24]	B	-1	-1	Yes	1.000	30.000	1.00
Right saddle [24]	B	-1	-1	Yes	1.000	30.000	1.00
N2- Shell outl[1]	B	14	9	Yes	0.887	67.000	1.00
N1- Shell inle[1]	B	14	9	Yes	0.887	67.000	1.00
Tubesheet: SS[13]	B	14	6	Yes	0.841	67.000	1.00

Warmest MDMT: 14 12

Chnl rear cov[11]	B	6	0	No	0.860	16.250	1.00
Chnl rear fla[11]	B	4	-24	No	0.299	14.000	1.00
Chnl. front f[11]	B	4	-24	No	0.299	14.000	1.00
Chnl. front f[11]	B	6	0	No	0.860	16.250	1.00
Chnl. rear [8]	B	6	-22	No	0.297	16.000	1.00
Chnl. front [8]	B	6	-22	No	0.297	16.000	1.00
N3- Channel in[1]	B	6	-23	No	0.271	16.000	1.00
Nozzle Flg [5]	A	-18	-48	No			
N4- Channel ou[1]	B	6	-23	No	0.271	16.000	1.00
Nozzle Flg [5]	A	-18	-48	No			
Tubesheet: CS[14]	B	-104	-104	No	0.160	62.500	1.00
Bolting [21]		-48					
Bolting [21]		-48					

Warmest MDMT: 6 0

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	12.0	11.4	Fail
Channel/Tube	0.0	11.4	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



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Warnings and Errors:

Step: 0 9:36am Aug 12,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.

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Input Echo:

Step: 1 9:36am Aug 12,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	16000	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	110.0	°C
Channel Side Design Temperature	65.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-2	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070



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Input Echo:

Step: 1 9:36am Aug 12,2026

0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test

Hydrostatic

Pressure Test Position

Horizontal

Test Performed in Corroded Condition

No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000



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Input Echo: Step: 1 9:36am Aug 12,2026

Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Chnl rear cover
Distance "FROM" to "TO"	65 mm
Flange Outside Diameter	1380 mm
Element Thickness	65 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	73 mm
External Corrosion Allowance	0 mm



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Input Echo:

Step: 1 9:36am Aug 12,2026

Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	165	N./mm ²
Allowable Stress, Operating	156	N./mm ²
Allowable Stress, Hydrotest	237.5	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	233	N./mm ²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Chnl rear flange
Distance "FROM" to "TO"	103 mm
Flange Inside Diameter	1210 mm
Element Thickness	82 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	90 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Chnl. rear
Distance "FROM" to "TO"	1500 mm
Inside Diameter	1210 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	175 N./mm ²
Allowable Stress, Operating	164 N./mm ²
Allowable Stress, Hydrotest	247 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm



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Input Echo: Step: 1 9:36am Aug 12,2026

Yield Stress, Operating	246	N./mm ²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1210	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N3- Channel inle	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	18	in.
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	155.01	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	13000	mm
Inside Diameter	1210	mm
Element Thickness	67	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	67	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	16000	KPa.
Design Temperature Internal Pressure	110	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	2650	mm
Width of Saddle	300	mm
Height of Saddle at Bottom	970	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	350	mm
Thickness of Wear Plate	30	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	



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Input Echo:

Step: 1 9:36am Aug 12,2026

Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	10600	mm
Width of Saddle	300	mm
Height of Saddle at Bottom	970	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	350	mm
Thickness of Wear Plate	30	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.3000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1210	mm
Liquid Density	0.49920E-04	kg./cm ³

Element From Node	40	
Detail Type	Ring	
Detail ID	SR, 2600.0 Fr40	
Dist. from "FROM" Node / Offset dist	2650	mm
Inside Diameter of Ring	1344	mm
Thickness of Ring	14	mm
Outside Diameter of Ring	1612.8	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	40	
Detail Type	Ring	
Detail ID	SR, 10400.0 Fr4	
Dist. from "FROM" Node / Offset dist	10600	mm
Inside Diameter of Ring	1344	mm
Thickness of Ring	14	mm
Outside Diameter of Ring	1612.8	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2- Shell outlet	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	486.84	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-350 LF2 +IT	



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Input Echo:

Step: 1 9:36am Aug 12,2026

Element From Node	40
Detail Type	Nozzle
Detail ID	N1- Shell inlet
Dist. from "FROM" Node / Offset dist	12000 mm
Nozzle Diameter	18 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	486.84 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF2 +IT

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Chnl. front
Distance "FROM" to "TO"	1500 mm
Inside Diameter	1210 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1210 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	50
Detail Type	Nozzle
Detail ID	N4- Channel outl
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	18 in.
Nozzle Schedule	40
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	166.6 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Flange
Description	Chnl. front flange
Distance "FROM" to "TO"	103 mm
Flange Inside Diameter	1210 mm



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Input Echo:

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Element Thickness	82	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	90	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	165	N./mm ²
Allowable Stress, Operating	156	N./mm ²
Allowable Stress, Hydrotest	237.5	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	233	N./mm ²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange
Description	Chnl. front flange
Distance "FROM" to "TO"	65 mm
Flange Outside Diameter	1380 mm
Element Thickness	65 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	73 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



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XY Coordinate Calculations: Step: 2 9:36am Aug 12,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Chnl rear cover		71.35	...	65	...
Chnl rear flang		174.35	...	103	...
Chnl. rear		1674.35	...	1500	...
Shell		14924.3	...	13000	...
Chnl. front		16674.3	...	1500	...
Chnl. front fla		16783.7	...	103	...
Chnl. front fla		16848.7	...	65	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: FLANGE

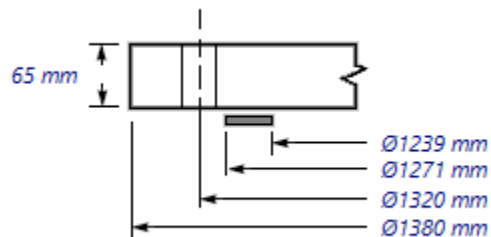
Flng: 9 9:36am Aug 12,2026

Flange Input Data Values

Description: FLANGE :

Chnl rear cover

Flange Type		Blind	
Design Pressure	P	1000.00	KPa.
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1380.000	mm
Flange Thickness	t	65.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	156.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	4146.903	mm
Diameter of Bolt Circle	C	1320.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1282.000	mm
Flange Face Inside Diameter	Fid	1210.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1271.000	mm
Gasket Inside Diameter	Gi	1239.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	16.000 mm
Basic Gasket Width,	$bo = N/2$	8.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1256.745 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: FLANGE

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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1257^2 \cdot 1000$$

$$= 126497.188 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1257 \cdot 3 \cdot 1000$$

$$= 17217.760 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(126497 + 17218 + 0, 0)$$

$$= 143714.953 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1257 + 0 \cdot 0 \cdot 0$$

$$= 197849.266 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((143715 + 0 + 4 \cdot 0 / 1257) / 172, 197849 / 172)$$

$$= 112.809 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	112.809	113.755	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	129.383	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 114 \cdot 172 / (68.9 \cdot \pi (1271 + 1239))$$

$$= 3.599 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172 (113 + 114) / 2$$

$$= 198678.77 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 143714.95 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1320 - 1257) / 2$$

$$= 31.6274 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 143715 \cdot 31.6 \cdot 1 / (62^2 \cdot 1257) + 0.3 \cdot 1 \cdot 1000 \cdot 1257^2 / 62^2$$

$$= 140.80 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 198679 \cdot 31.6 \cdot 1 / (62^2 \cdot 1257)$$



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Flg Calc [Int P]: FLANGE

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= 24.24 N./mm²

Bolt Stress, Operating [BS_o]:

= W_{m1} / A_b

= 143715/114

= 123.90 N./mm²

Bolt Stress, Seating [BS_a]:

= (W_{m2} / A_b)

= (197849/114)

= 170.57 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	140.80	156.00	24.24	165.00
Bolting	123.90	172.00	170.57	172.00

Reqd. Blind Flange Thickness at Center 61.902 mm

Reqd. Blind Flange Thickness at Gasket 26.762 mm

Estimated M.A.W.P. (Operating) 1108 KPa.

Estimated Finished Weight of Flange at given Thk. 753.5 kg.

Estimated Unfinished Weight of Forging at given Thk 753.5 kg.

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : 0 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : B

From Figure 3.7M- Non PWHT, the MDMT is : 6 °C

Temperature reduction per Graph 3.12 : 6 °C

The computed stress ratio is : 0.860

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	128.10	165.00	22.05	165.00
Bolting	123.90	172.00	170.57	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 57.277 mm

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Flg Calc [Int P]: FLANGE

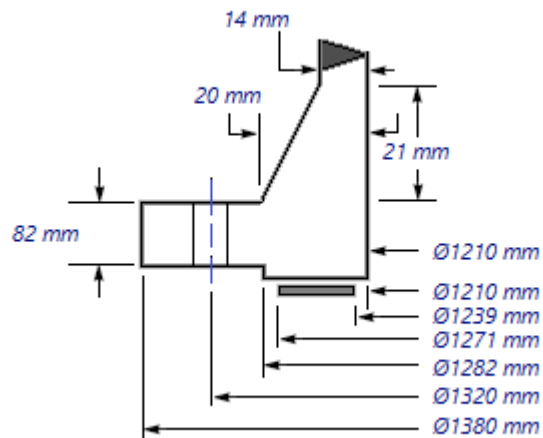
Flng: 10 9:36am Aug 12,2026

Flange Input Data Values

Description: FLANGE :

Chnl rear flange

Flange Type	Integral Weld Neck		
Design Pressure	P	1000.00	KPa.
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1210.000	mm
Flange Outside Diameter	A	1380.000	mm
Flange Thickness	t	82.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	21.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	156.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1320.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1282.000	mm
Flange Face Inside Diameter	Fid	1210.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1271.000	mm
Gasket Inside Diameter	Gi	1239.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: FLANGE

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (1000(1210+3) / (156-0.5 \cdot 1000)) + (3)$$

$$= 6.9102 \text{ mm}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 156 \cdot \log((2 \cdot 11 + 608) / 608)$$

$$= 2796.979 \text{ KPa.}$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 165 \cdot \log((2.0 \cdot 14 + 605) / 605)$$

$$= 3774.396 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	79.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1216.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	17.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	35.000 mm

Gasket Contact Width, $N = (G_o - G_i) / 2$	16.000 mm
Basic Gasket Width, $b_o = N / 2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1256.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1257^2 \cdot 1000$$

$$= 126497.188 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1257 \cdot 3 \cdot 1000$$

$$= 17217.760 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1216^2 \cdot 1000$$

$$= 118427.781 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 126497 - 118428$$

$$= 8069.409 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(126497 + 17218 + 0, 0)$$

$$= 143714.953 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1257 + 0 \cdot 0$$

$$= 197849.266 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((143715 + 0 + 4 \cdot 0 / 1257) / 172, 197849 / 172)$$

$$= 112.809 \text{ cm}^2$$



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Flg Calc [Int P]: FLANGE

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 79/(3 + 0.5)$$

$$= 186.229 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1320 \cdot \sin(\pi / 32)$$

$$= 129.383 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{129/(2 \cdot 25.4 + 79)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	112.809	113.755	
Radial Distance between Hub and Bolts:	34.925	35.000	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	129.383	186.229

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 114 \cdot 172 / (68.9 \cdot \pi(1271 + 1239))$$

$$= 3.599 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172(113 + 114) / 2$$

$$= 198678.77 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 143715 - 126497$$

$$= 17217.76 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1320 - 1257) / 2$$

$$= 31.6274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1320 - 1216) / 2 + 31.6)$$

$$= 41.8137 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1320 - 1216 - 17) / 2$$

$$= 43.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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Flg Calc [Int P]: FLANGE

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End Pressure, Md	118428.	43.5000	1.0000	5152.
Face Pressure, Mt	8069.	41.8137	1.0000	337.
Gasket Load, Mg	17218.	31.6274	1.0000	545.
Gasket Seating, Matm	198679.	31.6274	1.0000	6284.

Total Moment for Operation, Mo 6034. Kg-m.

Total Moment for Gasket seating, Mg 6284. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 115.655 mm

Hub Ratio, h/h0 = HL / H0 0.182

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.545

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.902

Factor V per 4.16.5 0.416

Factor f per 4.16.5 1.660

Stress factor from Table 4.16.4 [K]:

= A / B

= 1380/1216

= 1.135

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 (K² • log10(K))) / (K² - 1))/(K - 1)

= (0.66845 + (5.7169 (1.13² • log10(1.13))) / (1.13² - 1))/(1.13 - 1)

= 15.374

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))

= (1.13²(1 + 8.55246 • log10(1.13)) - 1)/((1.0472 + 1.9448 • 1.13²) • (1.13 - 1))

= 1.864

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))

= (1.13²(1 + 8.55246 • log10(1.13)) - 1)/(1.36136(1.13² - 1) • (1.13 - 1))

= 16.895

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)

= (1.13² + 1)/(1.13² - 1)

= 7.946

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V

= 16.9 • 11² • 116/0.42

= 567759.938 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0

= 0.9/116

= 0.008 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d

= (79 • 0.0078 + 1)/1.86 + 79³/567760

= 1.735

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)

= (1.66•6034/1216)/(1.74•17²)

= 161.05 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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Flg Calc [Int P]: FLANGE

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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.66 \cdot 6284 / 1216) / (1.74 \cdot 17^2)$$

$$= 167.72 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.82 \cdot 6034 / 1216) / (1.74 \cdot 79^2)$$

$$= 8.18 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.82 \cdot 6284 / 1216) / (1.74 \cdot 79^2)$$

$$= 8.52 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (15.4 \cdot 6034 / (79^2 \cdot 1216)) - 7.95 \cdot 8.18$$

$$= 54.84 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (15.4 \cdot 6284 / (79^2 \cdot 1216)) - 7.95 \cdot 8.52$$

$$= 57.11 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (161 + \max(8.18, 54.8)) / 2$$

$$= 107.94 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (168 + \max(8.52, 57.1)) / 2$$

$$= 112.42 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 143715 / 114$$

$$= 123.90 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (197849 / 114)$$

$$= 170.57 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	161.05	234.00	167.72	247.50
Radial Flange	8.18	156.00	8.52	165.00
Tangential Flange	54.84	156.00	57.11	165.00
Maximum Average	107.94	156.00	112.42	165.00
Bolting	123.90	172.00	170.57	172.00

Minimum Required Flange Thickness [Rigidity] 77.419 mm

Estimated M.A.W.P. (Operating) 1135 KPa.

Estimated Finished Weight of Flange at given Thk. 230.4 kg.

Estimated Unfinished Weight of Forging at given Thk 276.1 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot \text{Ma} / \text{Bsc} \cdot \text{Cnvfac} \cdot V / (\lambda \cdot \text{Eamb} \cdot \text{go}^2 \cdot \text{ho} \cdot \text{Ki})$$



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Flg Calc [Int P]: FLANGE

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$$= 52.14 \cdot 6387 / 1.02 \cdot 9807 \cdot 0.42 / (1.57 \cdot 202713 \cdot 11^2 \cdot 116 \cdot 0.3)$$

$$= 0.999 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6133 / 1.02 \cdot 9807 \cdot 0.42 / (1.57 \cdot 200199 \cdot 11^2 \cdot 116 \cdot 0.3)$$

$$= 0.971 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6284 / 1 \cdot 9807 \cdot 0.42 / (1.74 \cdot 202713 \cdot 11^2 \cdot 116 \cdot 0.3)$$

$$= 0.906 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6034 / 1 \cdot 9807 \cdot 0.42 / (1.74 \cdot 200199 \cdot 11^2 \cdot 116 \cdot 0.3)$$

$$= 0.880 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -24 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 4 °C
 Temperature reduction per Graph 3.12 : 28 °C
 The computed stress ratio is : 0.299

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	131.18	247.50	132.49	247.50
Radial Flange	9.22	165.00	9.31	165.00
Tangential Flange	40.48	165.00	40.89	165.00
Maximum Average	85.83	165.00	86.69	165.00
Bolting	123.90	172.00	170.57	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 54.864 mm

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Flg Calc [Int P]: New Flange

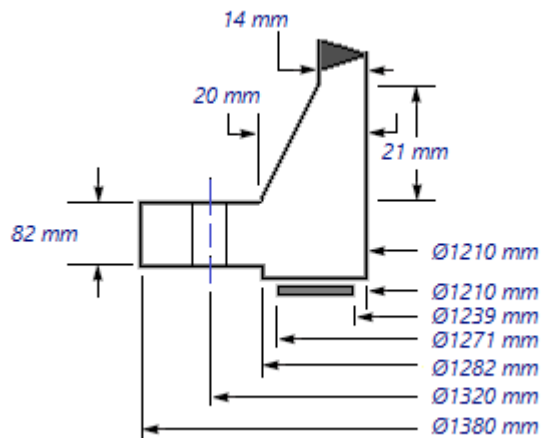
Flng: 11 9:36am Aug 12,2026

Flange Input Data Values

Description: New Flange :

Chnl. front flange

Flange Type	Integral Weld Neck		
Design Pressure	P	1000.00	KPa.
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1210.000	mm
Flange Outside Diameter	A	1380.000	mm
Flange Thickness	t	82.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	21.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	156.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1320.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1282.000	mm
Flange Face Inside Diameter	Fid	1210.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1271.000	mm
Gasket Inside Diameter	Gi	1239.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (1000(1210+3) / (156-0.5 \cdot 1000)) + (3)$$

$$= 6.9102 \text{ mm}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 156 \cdot \log((2 \cdot 11 + 608) / 608)$$

$$= 2796.979 \text{ KPa.}$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 165 \cdot \log((2.0 \cdot 14 + 605) / 605)$$

$$= 3774.396 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	79.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1216.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	17.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	35.000 mm

Gasket Contact Width, $N = (G_o - G_i) / 2$	16.000 mm
Basic Gasket Width, $b_o = N / 2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1256.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1257^2 \cdot 1000$$

$$= 126497.188 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1257 \cdot 3 \cdot 1000$$

$$= 17217.760 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1216^2 \cdot 1000$$

$$= 118427.781 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 126497 - 118428$$

$$= 8069.409 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(126497 + 17218 + 0, 0)$$

$$= 143714.953 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1257 + 0 \cdot 0$$

$$= 197849.266 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((143715 + 0 + 4 \cdot 0 / 1257) / 172, 197849 / 172)$$

$$= 112.809 \text{ cm}^2$$



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Flg Calc [Int P]: New Flange

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 79/(3 + 0.5)$$

$$= 186.229 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1320 \cdot \sin(\pi / 32)$$

$$= 129.383 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{129/(2 \cdot 25.4 + 79)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	112.809	113.755	
Radial Distance between Hub and Bolts:	34.925	35.000	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	129.383	186.229

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 114 \cdot 172 / (68.9 \cdot \pi(1271 + 1239))$$

$$= 3.599 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172(113 + 114) / 2$$

$$= 198678.77 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 143715 - 126497$$

$$= 17217.76 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1320 - 1257) / 2$$

$$= 31.6274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1320 - 1216) / 2 + 31.6)$$

$$= 41.8137 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1320 - 1216 - 17) / 2$$

$$= 43.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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Flg Calc [Int P]: New Flange Flng: 11 9:36am Aug 12, 2026

End Pressure, Md	118428.	43.5000	1.0000	5152.
Face Pressure, Mt	8069.	41.8137	1.0000	337.
Gasket Load, Mg	17218.	31.6274	1.0000	545.
Gasket Seating, Matm	198679.	31.6274	1.0000	6284.

Total Moment for Operation, Mo 6034. Kg-m.
Total Moment for Gasket seating, Mg 6284. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 115.655 mm
Hub Ratio, h/h0 = HL / H0 0.182
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.545

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.902
Factor V per 4.16.5 0.416
Factor f per 4.16.5 1.660

Stress factor from Table 4.16.4 [K]:

= A / B
= 1380/1216
= 1.135

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.13² • log10(1.13))) / (1.13² - 1))/(1.13 - 1)
= 15.374

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / ((1.0472 + 1.9448 • 1.13²) • (1.13 - 1))
= 1.864

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / (1.36136(1.13² - 1) • (1.13 - 1))
= 16.895

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.13² + 1) / (1.13² - 1)
= 7.946

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 16.9 • 11² • 116 / 0.42
= 567759.938 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.9/116
= 0.008 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (79 • 0.0078 + 1) / 1.86 + 79³/567760
= 1.735

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.66•6034/1216)/(1.74•17²)
= 161.05 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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Flg Calc [Int P]: New Flange

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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.66 \cdot 6284 / 1216) / (1.74 \cdot 17^2)$$

$$= 167.72 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.82 \cdot 6034 / 1216) / (1.74 \cdot 79^2)$$

$$= 8.18 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.82 \cdot 6284 / 1216) / (1.74 \cdot 79^2)$$

$$= 8.52 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (15.4 \cdot 6034 / (79^2 \cdot 1216)) - 7.95 \cdot 8.18$$

$$= 54.84 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (15.4 \cdot 6284 / (79^2 \cdot 1216)) - 7.95 \cdot 8.52$$

$$= 57.11 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (161 + \max(8.18, 54.8)) / 2$$

$$= 107.94 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (168 + \max(8.52, 57.1)) / 2$$

$$= 112.42 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= \text{Wm1} / \text{Ab}$$

$$= 143715 / 114$$

$$= 123.90 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (197849 / 114)$$

$$= 170.57 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	161.05	234.00	167.72	247.50
Radial Flange	8.18	156.00	8.52	165.00
Tangential Flange	54.84	156.00	57.11	165.00
Maximum Average	107.94	156.00	112.42	165.00
Bolting	123.90	172.00	170.57	172.00

Minimum Required Flange Thickness [Rigidity] 77.419 mm

Estimated M.A.W.P. (Operating) 1135 KPa.

Estimated Finished Weight of Flange at given Thk. 230.4 kg.

Estimated Unfinished Weight of Forging at given Thk 276.1 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot \text{Ma} / \text{Bsc} \cdot \text{Cnvfac} \cdot V / (\lambda \cdot \text{Eamb} \cdot \text{go}^2 \cdot \text{ho} \cdot \text{Ki})$$



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Flg Calc [Int P]: New Flange

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$$= 52.14 \cdot 6387 / 1.02 \cdot 9807 \cdot 0.42 / (1.57 \cdot 202713 \cdot 11^2 \cdot 116 \cdot 0.3)$$

$$= 0.999 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6133 / 1.02 \cdot 9807 \cdot 0.42 / (1.57 \cdot 200199 \cdot 11^2 \cdot 116 \cdot 0.3)$$

$$= 0.971 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6284 / 1 \cdot 9807 \cdot 0.42 / (1.74 \cdot 202713 \cdot 11^2 \cdot 116 \cdot 0.3)$$

$$= 0.906 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6034 / 1 \cdot 9807 \cdot 0.42 / (1.74 \cdot 200199 \cdot 11^2 \cdot 116 \cdot 0.3)$$

$$= 0.880 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -24 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : B

From Figure 3.7M- Non PWHT, the MDMT is : 4 °C

Temperature reduction per Graph 3.12 : 28 °C

The computed stress ratio is : 0.299

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	131.18	247.50	132.49	247.50
Radial Flange	9.22	165.00	9.31	165.00
Tangential Flange	40.48	165.00	40.89	165.00
Maximum Average	85.83	165.00	86.69	165.00
Bolting	123.90	172.00	170.57	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 54.864 mm

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Flg Calc [Int P]: New Flange

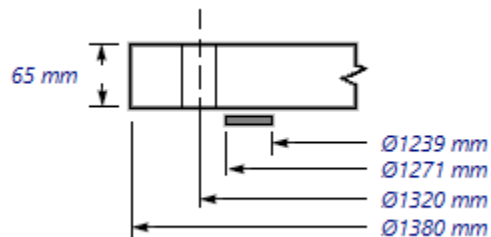
Flng: 12 9:36am Aug 12,2026

Flange Input Data Values

Description: New Flange :

Chnl. front flange

Flange Type		Blind	
Design Pressure	P	1000.00	KPa.
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1380.000	mm
Flange Thickness	t	65.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	156.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	4146.903	mm
Diameter of Bolt Circle	C	1320.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1282.000	mm
Flange Face Inside Diameter	Fid	1210.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1271.000	mm
Gasket Inside Diameter	Gi	1239.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 2, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	16.000 mm
Basic Gasket Width,	$bo = N/2$	8.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1256.745 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: New Flange

Flng: 12

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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1257^2 \cdot 1000$$

$$= 126497.188 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1257 \cdot 3 \cdot 1000$$

$$= 17217.760 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(126497 + 17218 + 0, 0)$$

$$= 143714.953 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1257 + 0 \cdot 0 \cdot 0$$

$$= 197849.266 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((143715 + 0 + 4 \cdot 0 / 1257) / 172, 197849 / 172)$$

$$= 112.809 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	112.809	113.755	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	129.383	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 114 \cdot 172 / (68.9 \cdot \pi (1271 + 1239))$$

$$= 3.599 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172 (113 + 114) / 2$$

$$= 198678.77 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 143714.95 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1320 - 1257) / 2$$

$$= 31.6274 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 143715 \cdot 31.6 \cdot 1 / (62^2 \cdot 1257) + 0.3 \cdot 1 \cdot 1000 \cdot 1257^2 / 62^2$$

$$= 140.80 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 198679 \cdot 31.6 \cdot 1 / (62^2 \cdot 1257)$$



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Flg Calc [Int P]: New Flange

Flng: 12

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= 24.24 N./mm²

Bolt Stress, Operating [BSO]:

= Wm1 / Ab

= 143715/114

= 123.90 N./mm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)

= (197849/114)

= 170.57 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	140.80	156.00	24.24	165.00
Bolting	123.90	172.00	170.57	172.00

Reqd. Blind Flange Thickness at Center 61.902 mm

Reqd. Blind Flange Thickness at Gasket 26.762 mm

Estimated M.A.W.P. (Operating) 1108 KPa.

Estimated Finished Weight of Flange at given Thk. 753.5 kg.

Estimated Unfinished Weight of Forging at given Thk 753.5 kg.

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : 0 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : B

From Figure 3.7M- Non PWHT, the MDMT is : 6 °C

Temperature reduction per Graph 3.12 : 6 °C

The computed stress ratio is : 0.860

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	128.10	165.00	22.05	165.00
Bolting	123.90	172.00	170.57	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 57.277 mm

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Internal Pressure Calculations: Step: 7 9:36am Aug 12,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Chnl rear cover		1000	73	3	1380	156	1
Chnl rear flange		1000	90	3	1210	156	2
Chnl. rear		1011.9	16	3	1210	164	3
Shell		16001	67	1.5	1210	175	4
Chnl. front		1011.9	16	3	1210	164	5
Chnl. front fln		1000	90	3	1210	156	6
Chnl. front fln		1000	73	3	1380	156	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Chnl rear cover		1000	1108	1288	65	61.9025	1
Chnl rear flange		1000	1135	1388	82	77.4192	2
Chnl. rear		1000	3458	4568	16	6.7632	3
Shell		16000	17945	18379	67	59.5723	4
Chnl. front		1000	3458	4568	16	6.7632	5
Chnl. front fln		1000	1135	1388	82	77.4192	6
Chnl. front fln		1000	1108	1288	65	61.9025	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 17004 KPa.
 Shell Side MAPnc = 18379 KPa.
 Channel Side MAWP = 1108 KPa.
 Channel Side MAPnc = 1288 KPa.

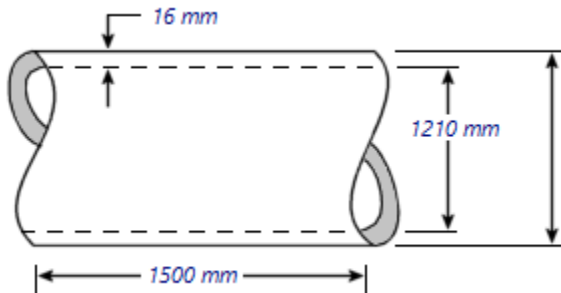
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Chnl. rear

Cylindrical Shell Design Information:			
Design Pressure P	1011.890 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	164.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	175.0 N./mm ²
Inside Diameter Di	1210.000 mm	Actual Stress	47.3 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	58527.9 cm ²	SG of Contents	1.000
Element Volume	1725161.0 cm ³	Weight of Contents	1724.1 Kg
Empty Weight	716.4 Kg	Operating Weight	2440.6 Kg
		Comp. All. Ope. Stress	109.4 N./mm ²
Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned}
 &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\
 &= 0.5 \cdot 1216 \cdot (\exp(1012/(164 \cdot 1)) - 1) + 3 + 0 \\
 &= 3.76 + 3 + 0 = 6.76 \text{ mm Equation 4.3.1}
 \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned}
 &= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq} \\
 &= 164 \cdot 1 \cdot \ln((2 \cdot 13 + 1216)/1216) - 11.9 \\
 &= 3469 - 11.9 = 3458 \text{ KPa.}
 \end{aligned}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$\begin{aligned}
 &= S \cdot E \cdot \ln((2 \cdot t + D)/D) \\
 &= 175 \cdot 1 \cdot \ln((2 \cdot 16 + 1210)/1210) \\
 &= 4568 \text{ KPa.}
 \end{aligned}$$

Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned}
 &= P \cdot D / (E(D_o - D)) \\
 &= 1012 \cdot 1216 / (1(1242 - 1216)) \\
 &= 47.3281 \text{ N./mm}^2
 \end{aligned}$$



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Element: Chnl. rear

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Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 1012 / (1 \cdot \ln((2 \cdot 13 + 1216) / 1216))$$

$$= 47.8324 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 1.305 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -22 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : B

From Figure 3.7M- Non PWHT, the MDMT is : 6 °C

Temperature reduction per Graph 3.12 : 28 °C

The computed stress ratio is : 0.297

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (3.86 \cdot 1) / (16 - 3 - 0)$$

$$= 0.297$$

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1107.893 KPa.
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	16.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.297	Temperature Difference	28 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

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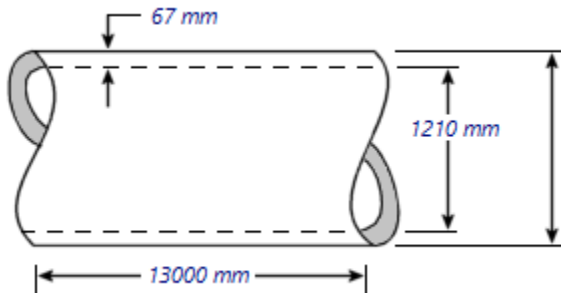
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Element: Shell

Step: 9 9:36am Aug 12,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	16000.594 KPa.	Int. Design Temperature	110 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	13000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.5000 mm	Allowable Stress (ope) S	175.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	175.0 N./mm ²
Inside Diameter Di	1210.000 mm	Actual Stress	148.2 N./mm ²
Spec. Min. Thk t,e, or sa	67.0000 mm	Spec. Nominal Thk.	67.000 mm
Surface Area	548899.1 cm ²	SG of Contents	0.050
Element Volume	9727577.0 cm ³	Weight of Contents	486.1 Kg
Empty Weight	27082.5 Kg	Operating Weight	27568.6 Kg
		Comp. All. Ope. Stress	122.7 N./mm ²
Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1213 \cdot (\exp(16001/(175 \cdot 1)) - 1) + 1.5 + 0$
 $= 58.1 + 1.5 + 0 = 59.6 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 175 \cdot 1 \cdot \ln((2 \cdot 65.5 + 1213) / 1213) - 0.59$
 $= 17946 - 0.59 = 17945 \text{ KPa.}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 175 \cdot 1 \cdot \ln((2 \cdot 67 + 1210) / 1210)$
 $= 18379 \text{ KPa.}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 16001 \cdot 1213 / (1(1344 - 1213))$
 $= 148.1668 \text{ N./mm}^2$



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Element: Shell

Step: 9 9:36am Aug 12,2026

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 16001 / (1 \cdot \ln((2 \cdot 65.5 + 1213) / 1213))$$

$$= 156.0308 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 5.247 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 12 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : B

From Figure 3.8M- with PWHT, the MDMT is : 14 °C

Temperature reduction per Graph 3.13 : 2 °C

The computed stress ratio is : 0.945

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}})$$

$$= (61.9 \cdot 1) / (67 - 1.5 - 0)$$

$$= 0.945$$

Warning: Impact Testing Is Mandatory!**Note:**

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	17003.643 KPa.
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	67.000 mm	Basic MDMT	14 °C
Stress/Thk. Ratio	0.945	Temperature Difference	2 °C
		Computed Min. Temp.	12 °C
		Joint Efficiency E*	100 %

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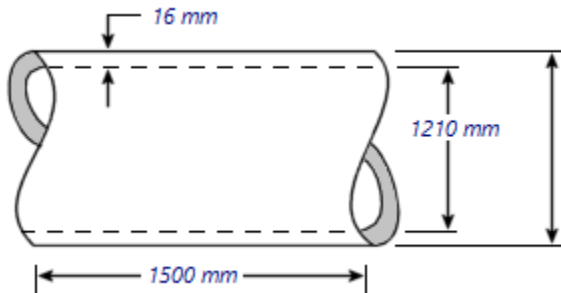
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Element: Chnl. front

Step: 10 9:36am Aug 12,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Chnl. front

Cylindrical Shell Design Information:			
Design Pressure P	1011.890 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	164.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	175.0 N./mm ²
Inside Diameter Di	1210.000 mm	Actual Stress	47.3 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	58527.9 cm ²	SG of Contents	1.000
Element Volume	1725161.0 cm ³	Weight of Contents	1724.1 Kg
Empty Weight	716.4 Kg	Operating Weight	2440.6 Kg
		Comp. All. Ope. Stress	122.7 N./mm ²
Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1216 \cdot (\exp(1012/(164 \cdot 1)) - 1) + 3 + 0$$

$$= 3.76 + 3 + 0 = 6.76 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$$

$$= 164 \cdot 1 \cdot \ln((2 \cdot 13 + 1216)/1216) - 11.9$$

$$= 3469 - 11.9 = 3458 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D)/D)$$

$$= 175 \cdot 1 \cdot \ln((2 \cdot 16 + 1210)/1210)$$

$$= 4568 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 1012 \cdot 1216 / (1(1242 - 1216))$$

$$= 47.3281 \text{ N./mm}^2$$



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Element: Chnl. front

Step: 10 9:36am Aug 12,2026

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 1012 / (1 \cdot \ln((2 \cdot 13 + 1216) / 1216))$$

$$= 47.8324 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 1.305 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -22 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : B

From Figure 3.7M- Non PWHT, the MDMT is : 6 °C

Temperature reduction per Graph 3.12 : 28 °C

The computed stress ratio is : 0.297

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (3.86 \cdot 1) / (16 - 3 - 0)$$

$$= 0.297$$

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1107.893 KPa.
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	16.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.297	Temperature Difference	28 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

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Shop Pressure Test Results:

Step: 11 9:36am Aug 12,2026

Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:

= Test Factor • ShellSide MAWP(St/S)

= 1.25 • 17004 (1)

= 21254.553 KPa.

Hydrotest Pressure Calculation [Channel Side]:

= Test Factor • Channel Side MAWP(St/S)

= 1.25 • 1108 (1.06)

= 1464.740 KPa.

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Chnl. rear	56.6	247.0	0.229	1476.60
Shell	202.5	247.0	0.820	21266.41
Chnl. front	56.6	247.0	0.229	1476.60

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3- Channel inle	Nozzle	161.00	151.00	1.066
N3- Channel inle	Pad	175.00	164.00	1.067
N2- Shell outlet	Nozzle	165.00	149.80	1.101
N1- Shell inlet	Nozzle	165.00	149.80	1.101
N4- Channel outl	Nozzle	161.00	151.00	1.066
N4- Channel outl	Pad	175.00	164.00	1.067

Minimum 1.066

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
SR, 2600.0 Fr40	175.00	158.20	1.106
SR, 10400.0 Fr4	175.00	158.20	1.106

Minimum 1.106

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Chnl rear cover	165.00	156.00	1.058
Chnl rear flange	165.00	156.00	1.058
Chnl. rear	175.00	164.00	1.067
Shell	175.00	175.00	1.000
Chnl. front	175.00	164.00	1.067
Chnl. front flange	165.00	156.00	1.058
Chnl. front flange	165.00	156.00	1.058

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):



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Shop Pressure Test Results:

Step: 11 9:36am Aug 12,2026

Description	Ambient	Operating	Ratio
Tube Material	138.00	125.20	1.102
Tubesheet Material	138.00	125.20	1.102

Minimum 1.102**Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):**

Description	Ambient	Operating	Ratio
N3- Channel inle	22.90	228.00	0.100
N2- Shell outlet	176.14	237.50	0.742
N1- Shell inlet	176.14	237.50	0.742
N4- Channel outl	22.90	228.00	0.100

Elements Suitable for Test Pressure.

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Ring MDMT Calcs:

Step: 12 9:36am Aug 12,2026

MDMT Calculations for Stiffening Rings:**MDMT Calculations for: SR, 2600.0 Fr40, Shell to Ring Junction, Curve: B**

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-15 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-15 °C
Temperature reduction per Graph 3.13	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$
$$= (61.9 \cdot 1) / (14 - 1.5 - 0)$$
$$= 1.000$$
MDMT Calculations for: SR, 10400.0 Fr4, Shell to Ring Junction, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-15 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-15 °C
Temperature reduction per Graph 3.13	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$
$$= (61.9 \cdot 1) / (14 - 1.5 - 0)$$
$$= 1.000$$
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ASME TS Calc: Pressure Test

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		21254.55 KPa.
Shell Thickness	ts	67.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1210.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	175.00 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Test Pressure		1464.74 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1210.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	175.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1417
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13000.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		138.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	575.174 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	7296.3 cm ²



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ASME TS Calc: Pressure Test

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Perimeter of Tube Layout Cp 3400.0 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-350 LF1
Tubesheet Material UNS Number		K03009
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	250.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1344.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00

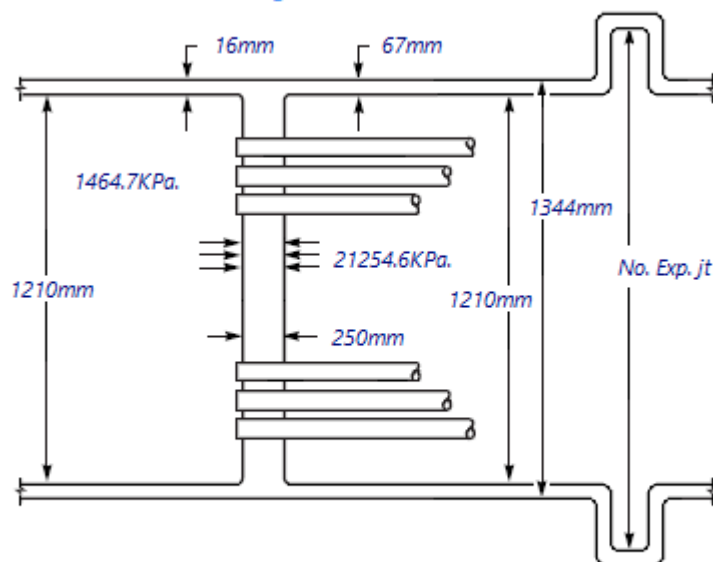
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option	Perform Plastic Calculation	

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Tubesheet Integral w/Shell and Channel



Configuration a:

Messages regarding the Tubesheet & Exchanger Input Data:



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Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

$$\begin{aligned} &\text{Computed Minimum Required Thickness [t]:} \\ &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i \quad \text{Equation 4.3.1} \\ &= 0.5 \cdot 14 \cdot (\exp(1465/(138 \cdot 1)) - 1) + 0 \\ &= 0.075 + 0 \\ &= 0.075 \text{ mm} \end{aligned}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$\begin{aligned} &= L / \sqrt{R_o \cdot t} \\ &= 13000 / \sqrt{9.5 \cdot 2.5} \\ &= 2667.544 \end{aligned}$$

Compute the Factor [Ch]:

$$\begin{aligned} &\text{Since } Mx > 2 \cdot (Do/t) \cdot 0.94, \text{ Ch} = 0.55 / (Do/t) \\ &= 0.55 \cdot (19/2.5) \\ &= 0.0724 \end{aligned}$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / Do)$$



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$$= 1.6 \cdot 0.072 \cdot 202701232 \text{ (2.5/19)}$$

$$= 3088.425 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 3088.4/205.0$$

$$= 15.066$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 205/379$$

$$= 0.541$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 3088/202701232$$

$$= 0.015235$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.54) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.134$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 205 (1 + 0.0020) / \ln(1 + 0.0020)^{0.13403422}$$

$$= 472.527 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1 - 0.54)$$

$$= 0.276$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (379 \cdot \exp(0.28)) / (0.28^{0.28})$$

$$= 712.658 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.541.5 - 0.5 \cdot 0.542.5 - 0.543.5$$

$$= 0.373$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-2.474	0.216E-07	0.209E-07	0.785E-07	-0.738E-07	0.192E+08	0.536E-02
2	284	0.447	0.297E-06	-0.412E-06	0.228E-06	0.549E-06	0.150E+07	0.174E+00
3	237	-1.014	0.915E-07	0.334E-07	0.141E-06	0.858E-08	0.358E+07	0.510E-01
4	213	-1.744	0.463E-07	0.389E-07	0.107E-06	-0.783E-07	0.842E+07	0.101E-01
5	201	-2.109	0.320E-07	0.296E-07	0.920E-07	-0.804E-07	0.128E+08	0.518E-03
6	195	-2.291	0.264E-07	0.250E-07	0.851E-07	-0.777E-07	0.157E+08	0.277E-02
7	198	-2.200	0.291E-07	0.272E-07	0.885E-07	-0.792E-07	0.142E+08	0.123E-02
8	199	-2.154	0.305E-07	0.284E-07	0.903E-07	-0.798E-07	0.135E+08	0.382E-03
9	200	-2.132	0.313E-07	0.290E-07	0.912E-07	-0.801E-07	0.131E+08	0.612E-04
10	200	-2.143	0.309E-07	0.287E-07	0.907E-07	-0.800E-07	0.133E+08	0.162E-03
11	200	-2.137	0.311E-07	0.288E-07	0.909E-07	-0.801E-07	0.132E+08	0.508E-04
12	200	-2.135	0.312E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.507E-05
13	200	-2.136	0.311E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.229E-04
14	200	-2.135	0.311E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.892E-05
15	200	-2.135	0.312E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.193E-05



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16	200	-2.135	0.312E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.157E-05
----	-----	--------	-----------	-----------	-----------	------------	-----------	-----------

Factor of Safety [FS]:

$$\begin{aligned} & \text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y \\ & = 2.407 - 0.741 \cdot F_{ic} / F_y \\ & = 2.407 - 0.741 \cdot 201/205 \\ & = 1.682 \end{aligned}$$

Allowable Circumferential Compressive Stress [Fha]:

$$\begin{aligned} & = F_{ic}/FS \\ & = 201/1.68 \\ & = 119.291 \text{ N./mm}^2 \end{aligned}$$

Allowable External Pressure at the given Thickness [Pa]:

$$\begin{aligned} & = 2 \cdot F_{ha} (t / D_o) \\ & = 2 \cdot 119284 (2.5/19) \\ & = 31390.572 \text{ KPa.} \end{aligned}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.8811 mm
Allowable Internal Pressure at Corroded thickness	42139.45 KPa.
Required Internal Design Pressure	1464.74 KPa.
Allowable External Pressure at Corroded thickness	31390.57 KPa.
Required External Design Pressure	21254.55 KPa.
Required Thickness due to Shell Side pressure	1.8811 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$\begin{aligned} & = l_{tx} / h \text{ (modified for corrosion if present)} \\ & = 0/250 \\ & = 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d^*]:

$$\begin{aligned} & = \max(d_t - 2 \cdot t_t (E_t T / E) \cdot (S_t T / S) \cdot (\rho); d_t - 2 \cdot t_t) \\ & = \max(19 - 2 \cdot 2.5 (202701232/202701232) \cdot (138/138) \cdot (0); 19 - 2 \cdot 2.5) \\ & = 19.0000 \text{ mm} \end{aligned}$$

Equivalent outer tube limit circle diameter [D_o]:

$$\begin{aligned} & = 2 \cdot r_o + d_t \\ & = 2 \cdot 575 + 19 \\ & = 1169.348 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} & = (p - d_t) / p \\ & = (25 - 19) / 25 \\ & = 0.240 \end{aligned}$$

Equivalent outer tube limit radius [a_o]:

$$\begin{aligned} & = D_o / 2 \\ & = 1169/2 \\ & = 584.6740 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$



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$$= 25 \sqrt{(1 - 4 \cdot \min(0 \cdot 100; 4 \cdot 1169 \cdot 25) / (\pi \cdot 1169^2))}$$

$$= 25.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25 - 19) / 25$$

$$= 0.2400$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 605 / 585$$

$$= 1.0348$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 605 / 585$$

$$= 1.0348$$

Compute parameter [xt]:

$$= 1 - Nt \left(\frac{dt - 2 \cdot tt}{2 \cdot a_o} \right)^2$$

$$= 1 - 1417 \left(\frac{19 - 2 \cdot 2.5}{2 \cdot 585} \right)^2$$

$$= 0.7969$$

Determine parameter [xs]:

$$= 1 - Nt \left(\frac{dt}{2 \cdot a_o} \right)^2$$

$$= 1 - 1417 \left(\frac{19}{2 \cdot 585} \right)^2$$

$$= 0.6259$$

Determine the value [h'g]:

$$= \max(h_g - CATC; 0) \text{ (For pressure only cases)}$$

$$= \max(0 - 0; 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 67 (1210 + 67) 202701232 / 13000$$

$$= 427390.7500 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2.5 (19 - 2.5) 202701232 / 13000$$

$$= 206.0551 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 427391 / (1417 \cdot 206)$$

$$= 1.46377$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_t)$$

$$= 1 / (1 + 427391 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= \left(\frac{12(1 - \nu_s^2)}{D_s + t_s} \right)^{1/4} \sqrt[4]{\frac{t_s}{(D_s + t_s)t_s}}$$

$$= \left(\frac{12(1 - 0.3^2)}{1210 + 67} \right)^{1/4} \sqrt[4]{\frac{67}{(1210 + 67)67}}$$

$$= 0.0062 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0062 \cdot 202701232 \cdot 67^3 / (6(1 - 0.3^2))$$

$$= 69392244736.0000 \text{ KPa} \cdot \text{mm}^2$$



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Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1210 + 67) \cdot 69.39E+09 / 250^3 \cdot (1 + 250 \cdot 0.0062 + (250 \cdot 0.0062)^2 / 2)$$

$$= 127966448.0000 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1210^2 / (4 \cdot 202701232 \cdot 67) \cdot (1 - 1210 \cdot 0.3 / (2(1210 + 67)))$$

$$= 0.23119E-01 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1210 + 16) 16 }$$

$$= 0.0130 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2))$$

$$= 0.013 \cdot 202701232 \cdot 16^3 / (6(1 - 0.3^2))$$

$$= 1973668096.0000 \text{ KPa.} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1210 + 16 / \cos(0)) \cdot 1973668096 / 250^3 \cdot (1 + 250 \cdot 0.013 \cdot (1 + 1) / 2 + (250 \cdot 0.013)^2 / 2 \cdot 1)$$

$$= 8835716.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1210^2 / (4 \cdot 202701232 \cdot 16) \cdot (1 - 1210 \cdot 0.3 / (2(1210 + 16)))$$

$$= 0.96145E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 10.000000$; $\mu^* = 0.240000$; $E^* = 41356984 \text{ KPa.}$

$$E^*/E = 0.204029$$
 ; $v^* = 0.407183$

Note: As h/p (10.000) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - v^*) N_t \cdot E_t \cdot t(dt - tt) a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.41^2) 1417 \cdot 202701232 \cdot 2.5 (19 - 2.5) 585^2) / (41356984 \cdot 13000 \cdot 250^3))^{1/4}$$

$$= 1.7626$$

Values from Table 4.18.3

$$Z_d = 0.256793$$
 ; $Z_v = 0.162234$; $Z_m = 0.664956$

$$Z_a = 0.696172E+00$$
 ; $Z_w = 0.162234$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1344 / 1169$$

$$= 1.1494$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 41356984 \cdot (127966448 + 8835716 +$$



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$$202701232 \cdot \ln(1.15) \\ = 2.3654$$

Compute parameter [φ]:

$$= (1 + v \cdot F) \\ = (1 + 0.41) \cdot 2.37 \\ = 3.3286$$

Compute parameter [Q1]:

$$= (\psi - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ = (1.03 - 1 - 3.33 \cdot 0.16) / (1 + 3.33 \cdot 0.66) \\ = -0.157233$$

Compute Parameter [Qz1]:

$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4 \\ = (0.26 + -0.16 \cdot 0.16) / 2 \cdot 1.76^4 \\ = 1.1163$$

Compute parameter [Qz2]:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4 \\ = (0.16 + -0.16 \cdot 0.66) / 2 \cdot 1.76^4 \\ = 0.2784$$

Compute parameter [U]:

$$= (Z_w + (\psi - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ = (0.16 + (1.03 - 1) \cdot 0.66) \cdot 1.76^4 / (1 + 3.33 \cdot 0.66) \\ = 0.5568$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ωs]:

$$= \psi \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ = 1.03 \cdot 69.39E+09 \cdot 0.0062 \cdot 0.02311936 (1 + 250 \cdot 0.0062) \\ = 26347.7500 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= a o^2 (\psi^2 - 1) \cdot (\psi - 1) / 4 - \omega_s \\ = 585^2 (1.03^2 - 1) \cdot (1.03 - 1) / 4 - 26348 \\ = -26137.5879 \text{ mm}^2$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ \zeta c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)] \\ = 1.03 \cdot 1973668096 [0.096145086 (1 + 0.013 \cdot 1 \cdot 250) - \\ 0.000 (1/250 + 1 \cdot 0.013/2)] \\ = 10818.7617 \text{ mm}^2$$

Parameter [ω*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\psi - 1) / 2] - \omega_c \\ = 585^2 [(1.03^2 + 1) \cdot (1.03 - 1) / 4 - (1.03 - 1) / 2] - 10819 \\ = -10608.5986 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:



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ASME TS Calc: Pressure Test

Case: 1 9:36a Aug 12, 2026

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (Ds/Do)^2] v_s - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 21255 \{ 0.63 + 2(1 - 0.63) 0.3 + [2/1.46 \cdot (1210/1169)^2] 0.3 - [(1.03^2 - 1)/(1 \cdot 1.46)] - [(1 - 1)/(2 \cdot 1 \cdot 1.46)] [(0^2 - (1210)^2)/1169^2] \}$$

$$= 26375.4219 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.8 + 2(1 - 0.8) 0.3 + 1/(1 \cdot 1.46)] 1465$$

$$= 2346.4021 \text{ KPa.}$$

Pressure [Pγ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 1417 \cdot 206 \cdot 0 / (3.14 \cdot 585^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.56 \cdot 0 / (2 \cdot \pi \cdot 585^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= -(0.56/585^2) \cdot (-26138 \cdot 21255 - -10609 \cdot 1465)$$

$$= 879.5407 \text{ KPa.}$$

Calculate the pressure [Pω]:

$$= (U/a_o^2) \cdot (\omega s \cdot P^* s - \omega c \cdot P^* c)$$

$$= (0.56/585^2) \cdot (26348 \cdot 0 - 10819 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot (P's - P't + P\gamma + P\omega + Pw + Prim)$$

$$= 0.1000E+01 \cdot 1.46 / (1 + 1 \cdot 1.46 (1.12 + (1.03 - 1) 0.28)) \cdot (26375 - 2346 + 0 + 0 + -0 + 880)$$

$$= 13768.4277 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega s \cdot P^* s - \omega c \cdot P^* c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-26138 \cdot 21255 - -10609 \cdot 1465) - (26348 \cdot 0 - 10819 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 3.33 \cdot 0.66)$$

$$= -17137.0488 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.16 + 2 \cdot -17137 / (13768 \cdot 585^2)$$

$$= -0.22864$$

Fm Value from Table 4.18.3 = 0.114321

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.11 / 0.24) \cdot (2 \cdot 585 / (250 - 0))^2 \cdot 13768$$

$$= 215.2403 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 205 = 292 \text{ N./mm}^2, 4.18.233$$



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ASME TS Calc: Pressure Test

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Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot h) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.11 / 0.24) \cdot (2 \cdot 585 / (210 - 0))^2 \cdot 13457$$

$$= 292.0936 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 210 + 0 + 0$$

$$= 209.9021 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$$

$$= (2 \cdot 0.24 \cdot 250 / 585) \cdot 156$$

$$= 31.9762 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (7296 \cdot 7642) / (0.24 \cdot 44.7 \cdot 3400)$$

$$= 152.7532 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 44.7 + 0 + 0$$

$$= 44.7343 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(210; 44.7)$$

$$= 209.9021 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.89, F_{tmax} = 1.06$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((21255 \cdot 0.63 - 1465 \cdot 0.8) - (13768) \cdot 0.89) / (0.8 - 0.63)$$

$$= -0.8961 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((21255 \cdot 0.63 - 1465 \cdot 0.8) - (13768) \cdot 1.06) / (0.8 - 0.63)$$

$$= -14.4073 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 14.4 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta T \cdot S_{y,t} = 194.7465 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.26 + -0.23 \cdot 0.16)1.76^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2)$$



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ASME TS Calc: Pressure Test

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Determine the factor [rt]:

$$= (\sqrt{(dt^2 + (dt - 2 \cdot tt)^2)})/4$$

$$= (\sqrt{(19^2 + (19 - 2 \cdot 2.5)^2)})/4$$

$$= 5.9002 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Et/Syt)}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202701232/205)}$$

$$= 142.3572$$

Determine the factor [Ft]:

$$= k \cdot L/r$$

$$= 0 \cdot 0/5.9$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (Sy/t/Fs) \cdot (1 - Ft/(2 \cdot Ct))$$

$$= (205/2) \cdot (1 - 0/(2 \cdot 142))$$

$$= 194.746 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 14.4 \cdot 1.3$$

$$= 190.38 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (19 - 2.5) 2.5$$

$$= 1.296 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.5 (19 - 2.5) 195$$

$$= 2573.444 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 195$$

$$= 2157.2014 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 195/195$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 195$$

$$= 1395.8363 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd} - 0.75 \cdot do)$$

$$= 2(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 2.5 \cdot (19 - 2.5)1 \cdot 1} - 0.75 \cdot 19)$$

$$= 2.945 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.5000 mm

Minimum Required Groove Weld Leg [agr]: 1.5000 mm



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ASME TS Calc: Pressure Test

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Tube-Tubesheet Jt allowable, 2573 is >= tube strength 2573 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 260 = 370 \text{ N./mm}^2$$

Axial Membrane Stress in Shell $[\sigma_s, m]$:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 585^2 / ((1210 + 67) 67) \cdot [13768 + (1.03^2 - 1)(21255 - 1465)] + 605^2 \cdot 1465 / ((1210 + 67) 67)$$

$$= 66.8738 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 67 mm, the minimum Shell length adjacent to the tubesheet should be 512.510 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	215.2 <=	292.1 N./mm ²	Pass
Tubesheet Shear Stress	49.2 <=	155.8 N./mm ²	Pass
Maximum Tube Stress	14.4 <=	194.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-14.4 <=	-194.7 N./mm ²	Pass
Maximum Force on any one Tube	190.4 <=	2573.4 Kgf	Pass
Axial Membrane Stress in Shell	66.9 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	209.902	250.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.500	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.500	2.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	22.241 ...	20	292	5 156	Ps+Pt-Th D1	Ok
D2uc	219.094 ...	231	292	54 156	Ps+Pt-Th D2	Ok
D3uc	209.902 ...	215	292	49 156	Ps+Pt-Th D3	Ok
Max:	219.0939 ... mm	0.792	0.346	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+/-)Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx



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[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:		Fail
D1uc	1	370	...	...	:	...	...	...	...	:		Ok
D2uc	66	370	...	...	:	...	...	...	...	:		Ok
D3uc	67	370	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.180		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	2	194	...	-194	21	2573	...	370	...	370	Ok
D2uc	16	194	-16	-194	212	2573	...	370	...	370	Ok
D3uc	14	194	-14	-194	190	2573	...	370	...	370	Ok
Max RATIO	0.082		0.082		0.082		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	219.094	250.000	Ok
Tube thickness :	1.881	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (67.000) adj. to tubesheet: 512.510 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -1.428 mm
 Due to Pressure + Thermal Shell Compresses by : -1.428 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0186	...	...	...	...	...
2	-1.4284	...	...	...	...	...
3	-1.4098	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...



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ASME TS Calc: Pressure Test

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7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculations: Step: 13 9:36am Aug 12,2026

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	62	No Calc	No Calc
20	30	No Calc	...	79	No Calc	No Calc
30	Tsht	1602.97	1242	13	No Calc	No Calc
Tsht	40	No Calc	...	13	No Calc	No Calc
40	Ring	2650.03	1344	65.5	No Calc	No Calc
Ring	Ring	7950	1344	65.5	No Calc	No Calc
Ring	Tsht	2399.96	1344	65.5	No Calc	No Calc
Tsht	50	No Calc	...	65.5	No Calc	No Calc
50	60	1603.04	1242	13	No Calc	No Calc
60	70	No Calc	...	79	No Calc	No Calc
70	80	No Calc	...	62	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	65	26.7622	...	No Calc
20	30	82	77.3684	...	No Calc
30	Tsht	16	No Calc	...	1516
Tsht	40	16	No Calc	...	No Calc
40	Ring	67	No Calc	...	12337
Ring	Ring	67	No Calc	...	9668
Ring	Tsht	67	No Calc	...	12603
Tsht	50	67	No Calc	...	No Calc
50	60	16	No Calc	...	1516
60	70	82	77.3684	...	No Calc
70	80	65	26.7622	...	No Calc

Minimum

1516

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1602.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Ring	2650.03	No Calc	No Calc	No Calc
Ring	Ring	7950	No Calc	No Calc	No Calc
Ring	Tsht	2399.96	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	1603.04	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.



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External Pressure Calculations: Step: 13 9:36am Aug 12, 2026

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Element and Detail Weights:

Step: 14

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	846.254	...	846.254	...	84.6254
20	30	251.885	128691	242.049	128931	25.1885
30	40	716.443	1725161	583.533	1742312	71.6443
40	50	27082.5	9727577	26507.3	9801808	2708.25
50	60	716.443	1725161	583.533	1742312	71.6443
60	70	251.885	128691	242.049	128931	25.1885
70	80	846.254	...	846.254	...	84.6254
Total		30711	13435282.00	29850	13544295.00	3071



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
30	Liqd	1724.11	750	...	...	Liquid: 30
30	Nozl	170.507	750	-819.325	...	N3- Channel inle
40	Sadl	692.937	2650	739	...	Left Saddle
40	Sadl	692.937	10600	739	...	Right saddle
40	Liqd	486.082	6500	...	...	Liquid: 40
40	Ring	74.5061	2650	...	...	SR, 2600.0 Fr40
40	Ring	74.5061	10600	...	...	SR, 10400.0 Fr4
40	Nozl	535.525	1000	-833.6	...	N2- Shell outlet
40	Nozl	535.525	12000	833.6	...	N1- Shell inlet
50	Liqd	1724.11	750	...	...	Liquid: 50
50	Nozl	183.262	750	-819.325	...	N4- Channel outl
30	FTsh	2167.48	1625	...	...	Fixed tubesheet
30	Tube	21134.9	8250	...	...	Tubes
30	RTsh	2167.48	14875	...	...	Tubesheet
40	Baf1	170.125	1097	...	...	Baffles
40	Baf1	170.125	2197	...	...	Baffles
40	Baf1	170.125	3297	...	...	Baffles
40	Baf1	170.125	4397	...	...	Baffles
40	Baf1	170.125	5497	...	...	Baffles
40	Baf1	170.125	6597	...	...	Baffles
40	Baf1	170.125	7697	...	...	Baffles
40	Baf1	170.125	8797	...	...	Baffles
40	Baf1	170.125	9897	...	...	Baffles
40	Baf1	170.125	10997	...	...	Baffles
40	Baf1	170.868	12097	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	1385.9
Liquid	3934.3
Stiffeners	149.0
Nozzles	1424.8
Exchanger Components	27342.0
Liquid in Tubes	2943.5
Sum of the Detail Weights	37179.4 kg.



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Element and Detail Weights:

Step: 14 9:36am Aug 12,2026

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	33782.9	33782.9	33782.9	33782.9	33782.9	33782.9
Saddles	1385.9	1385.9	1385.9	1385.9	1385.9	1385.9
Stif. Rings	149.0	149.0	149.0	149.0	149.0	149.0
Nozzles	1424.8	1424.8	1424.8	1424.8	1424.8	1424.8
Exchanger	27342.0	27342.0	27342.0	27342.0	27342.0	27342.0
Ope. Liquid	...	...	...	...	...	3934.3
Tube Ope Lqd	...	...	...	...	...	2943.5
Test Liquid	...	13427.1	...	...	...	...
Tube Tst Lqd	...	2943.5	...	...	...	...
Totals	64084.5	80455.1	64084.5	64084.5	64084.5	70962.3

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	64084.5 kg.
Shop Test Wt.	- Fabricated weight + water (full)	80455.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	64084.5 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	64084.5 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	64084.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	70962.3 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	70015.5 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	2945285.8 cm ³
Weight of Ope Liq in tubes :	2943.5 kg.
Weight of Water in tubes :	2943.5 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	18122	...
20	30	8184.64	4219.47
30	40	58527.9	57019.9
40	50	548899	494173
50	60	58527.9	57019.9
60	70	8184.64	4219.47
70	80	18122	...

Total	718568.0	616651.4 cm ²
Total	773.5	663.8 ft ²

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Nozzle Flange MAWP:

Step: 15 9:36am Aug 12,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3- Channel in	1868.0	1960.0	65	150	GR 1.1	...	...	...	...
N4- Channel ou	1868.0	1960.0	65	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1868.034 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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Wind Load Calculation:

Step: 16 9:36am Aug 12,2026

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020
Basic Wind Speed for IS-875	180 Km/hr
Wind Zone Number	5
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Risk Factor	[k1] 1.0
Terrain Category	2
Equipment Class	Class A
Topography Factor	[k3] 1.0
Importance Factor	[k4] 1.15
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Use Gust Response Factor (Dynamic Analysis)	No

Calculated Values:

Vessel Length to Diameter ratio: 13.126

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50 • 1 • 1 • 1 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1984, 1 • 0.91 • 1 • 0.6 • 57.5²)

= 1798.0 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.11 • 0.93 • 1798

= 180.2 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	1270.00	1.0000	1.0000	1.0000	57.50	0.9313
20	1270.00	1.0000	1.0000	1.0000	57.50	0.9313
30	1270.00	1.0000	1.0000	1.0000	57.50	0.9313
40	1270.00	1.0000	1.0000	1.0000	57.50	0.9313
50	1270.00	1.0000	1.0000	1.0000	57.50	0.9313
60	1270.00	1.0000	1.0000	1.0000	57.50	0.9313
70	1270.00	1.0000	1.0000	1.0000	57.50	0.9313

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm²	Wind Pressure Kgs/m²	Element Wind Load Kgf
10	20	1270	1656	1076.4	183.418	20.2174
20	30	1270	1452	1495.56	183.418	28.0903
30	40	1270	1490.4	22356	183.418	419.9
40	50	1270	1612.8	209664	183.418	3938
50	60	1270	1490.4	22356	183.418	419.9
60	70	1270	1452	1495.56	183.418	28.0903
70	80	1270	1656	1076.4	183.418	20.2174



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Wind Load Calculation:

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The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 17 9:36am Aug 12,2026

Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User	I	2.0000
Zone Number		2
Soil Type	Rocky or Hard Soil	
Response reduction factor (Force factor)	R	2.0000
Time Period of Vibration (program or user)		0.0303 Seconds
Value of Sa/g from user Spectrum Data		0.0939
Percent Damping		2.0 %
Considering MCE for load determination		No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:

$$= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$$

$$= (0.094 \cdot 1) \cdot 1$$

$$= 0.094$$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs

Computed Sa/g = 0.094

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.

The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:

$$= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$$

$$= 7626 \cdot 0.094 \cdot 1.3$$

$$= 930.4 \text{ Kgf}$$

Vertical load per 8.4:

$$= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$$

$$= 2/3 \cdot 0.094 \cdot 68630 \cdot 1.3$$



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Earthquake Load Calculation:

Step: 17

9:36am Aug 12,2026

= 5582.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	690	7625.52	930.411
20	30	605	7625.52	930.411
30	40	605	7625.52	930.411
40	Sad1	605	7625.52	930.411
Sad1	50	605	7625.52	930.411
40	50	605	7625.52	930.411
50	60	605	7625.52	930.411
60	70	605	7625.52	930.411
70	80	690	7625.52	930.411

The Earthquake Weight does not include the weight of the saddles.

The Earthquake Loads calculated and printed in the Earthquake Load calculation report have been factored by the input scalar/load reduction factor of: 1.300.

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Center of Gravity Calculation: Step: 18 9:36am Aug 12,2026

Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	8549.349 mm
Center of Gravity of the Liquid	8424.350 mm
Center of Gravity of the Stiffening Rings	8549.349 mm
Center of Gravity of the Nozzles	8491.492 mm
Center of Gravity of the Tubesheet(s)	8424.350 mm
Center of Gravity of the Tubes	8424.350 mm
Center of Gravity of the Baffles	8523.532 mm
Center of Gravity of Bare Shell New and Cold	8424.576 mm
Center of Gravity of Bare Shell Corroded	8424.581 mm
Vessel CG in the Operating Condition	8431.216 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	8431.853 mm
Vessel CG in the Test Condition	8430.337 mm

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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

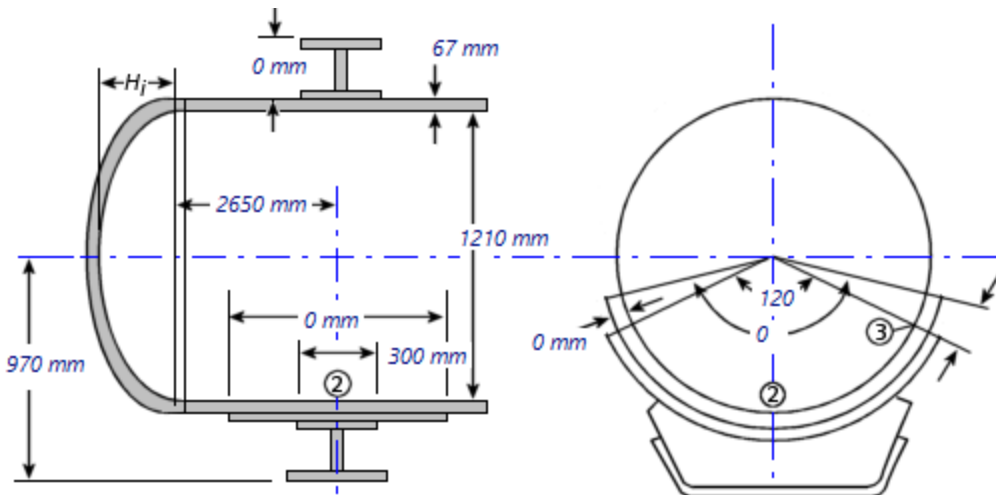
$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{ 639 \cdot 65.5 }, 2 \cdot 2650)$$

$$= 619.2131 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	639.25	mm
Shell Thickness used in this Case	t	67.000	mm
Stiffened Vessel Length per 4.15.6	L	13000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2650.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		175.00	N./mm ²
Head Allowable Stress used in Calculation		156.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	970.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	48166.79	Kgf
Pressure used in Analysis	P	16000.296 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	74.62	175.00
Long. Stress at Bottom of Midspan	81.54	175.00
Long. Stress at Top of Saddles	81.03	175.00
Long. Stress at Bottom of Saddles	75.14	175.00



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12, 2026

Tangential Shear in Shell	2.13	139.64
Circ. Compressive Stress in Shell	1.51	175.00
Stiffener Circ. Stress at Shell	16.45	218.75
Stiffener Circ. Stress at Tip	97.72	197.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (4874/2 + 0) \cdot 970/1216$$

$$= 5832.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(412, 0) \cdot 970/7950$$

$$= 50.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(8374, 0) \cdot 970/7950$$

$$= 1021.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (8374/2 + 0) \cdot 970/1216$$

$$= 10020.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 38146 + \max(50.3, 5833, 1022, 10020)$$

$$= 48166.8 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1799 (0 + \pi/4(1.34 \cdot 1.2)^2)$$

$$= 4042.046 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 48859.73 Kgf

Transverse Shear Load 4186.85 Kgf

Longitudinal Shear Load 8373.70 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1210/2 + 67 + 30) \sin(120/2), 2 \cdot 639)$$

$$= 1215.900 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -48167 \cdot 2650 [1 - (1 - 2650/13000 + (639^2 - 0^2) / (2 \cdot 2650 \cdot 13000)) / (1 + (4 \cdot 0) / (3 \cdot 13000))]$$

$$= -25262.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12, 2026

$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\
 &= 48167 \cdot 13000 / 4 \cdot ((1 + 2(639^2 - 0^2) / (13000^2)) / (1 + (4 \cdot 0) / (3 \cdot 13000)) - 4 \cdot 2650 / 13000) \\
 &= 29657.7 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 16000 \cdot 639 / (2 \cdot 65.5) - 29658 / (\pi \cdot 639^2 \cdot 65.5) \\
 &= 74.62 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 16000 \cdot 639 / (2 \cdot 65.5) + 29658 / (\pi \cdot 639^2 \cdot 65.5) \\
 &= 81.54 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (\pi \cdot Rm^2 \cdot t) \\
 &= 16000 \cdot 639 / (2 \cdot 65.5) - 25263 / (\pi \cdot 639^2 \cdot 65.5) \\
 &= 81.03 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (\pi \cdot Rm^2 \cdot t) \\
 &= 16000 \cdot 639 / (2 \cdot 65.5) + 25263 / (\pi \cdot 639^2 \cdot 65.5) \\
 &= 75.14 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 48167 \cdot (13000 - 2 \cdot 2650) / (13000 + (4 \cdot 0 / 3)) \\
 &= 28529.6 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$\begin{aligned}
 &= T / (\pi \cdot Rm \cdot t) \\
 &= 28530 / (\pi \cdot 639 \cdot 65.5) \\
 &= 2.13 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{639 \cdot 65.5} \\
 &= 159.607 \text{ mm}
 \end{aligned}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k/A \\
 &= -0.76 \cdot 48167 \cdot 0.1/237 \\
 &= -1.51 \text{ N./mm}^2
 \end{aligned}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$\begin{aligned}
 &= -K8 \cdot Q/A + K6 \cdot Q \cdot Rm \cdot c1/l \\
 &= -0.34 \cdot 48167 / 237 + 0.053 \cdot 48167 \cdot 639 \cdot 40.7 / 2794 \\
 &= 16.45 \text{ N./mm}^2
 \end{aligned}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$\begin{aligned}
 &= -K8 \cdot Q/A - K6 \cdot Q \cdot Rm \cdot c2/l \\
 &= -0.34 \cdot 48167 / 237 - 0.053 \cdot 48167 \cdot 639 \cdot 159 / 2794 \\
 &= -97.72 \text{ N./mm}^2
 \end{aligned}$$

Distance between Saddle Supports [Ls]:

$$= 7950.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$\begin{aligned}
 &= \alpha \cdot Ls (\text{Design Temperature} - \text{Ambient Temperature}) \\
 &= 0.12167 \text{E-04} \cdot 7950 (110 - 21.1) \\
 &= 8.598 \text{ mm}
 \end{aligned}$$



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12,2026

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	45.0000	mm
Web Thickness	Webtk	45.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(606 \cdot 65.5)} + 350)65.5^3)/12$$

$$= 1547.735 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1548 + 433(103 - 32.8)^2$$

$$= 22947.371 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	660.9	65.5	32.8	432.9	1417774.2	0.229E+05
Wearplate	350.0	30.0	80.5	105.0	845249.9	613.
Web	45.0	193.0	192.0	86.9	1667520.0	0.957E+04
BasePlate	150.0	75.0	326.0	112.5	3667499.8	0.564E+05
Totals	...	...	...	737.3	7598044.0	0.896E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7598044/737$$

$$= 103.058 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2)/(\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 48167$$

$$= 9802.9863 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 3.1587 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2)/(\text{theta}/2 \text{ in radians})$$

$$= 970 - 606 \cdot \sin(120/2)/1.05$$

$$= 468.429 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4592.0889 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.1814 \text{ N./mm}^2$$



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12,2026

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{2 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(48167 + 693) \cdot 150}{2 \cdot 1260 \cdot 175}}$$

$$= 22.112 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) \cdot (605 + 67 + 30)$$

$$= 1215.900 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1216 / (5 - 1)$$

$$= 303.975 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 304 \cdot 150 / 2$$

$$= 227.981 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 48860 / (1260 \cdot 150)$$

$$= 25.852 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 228 \cdot 25.9$$

$$= 5893.703 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 115 + 68.4$$

$$= 183.144 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 5894 / 183$$

$$= 3.156 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	45.0	255.0	172.5	114.8	1979437.2	0.982E+04
Web	152.0	45.0	22.5	68.4	153887.3	0.616E+04
Totals	...	...	...	183.1	2133324.8	0.160E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300 - 45$$

$$= 255.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12,2026

= 2133324/183
= 116.483 mm

Distance to Centroid [C1]:
= max(ytot, Saddle Width - ytot)
= max(116, 300 - 116)
= 183.517 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{15975/183}$
= 93.396 mm

Length of Outer Rib [L]:
= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 970 - cos(120/2)(605 + 67 + 30) - 75
= 544.000 mm

Intermediate Term [Cc]:
= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/262)}$
= 122.739

Slenderness ratio [KL/r]:
= KL/r
= 1 • 544/93.4
= 5.825

Bending Moment [Rm]:
= FI / (2 • Bplen) • e • L / 2
= 8374 / (2 • 1260) • 304 • 544/2
= 274.746 Kg-m.

Compressive Allowable, KL/r < Cc (5.82 < 123) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (5.82)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (5.82) / (8 \cdot 123) - (5.82^3) / (8 \cdot 123^3))$
= 155.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= Sc/Sca + (Rm • C1 / I) / Sba
= 3.16/155 + (275 • 184/159753152) / 175
= 0.038

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	45.0	255.0	172.5	114.8	1979437.2	0.139E+05
Web	304.0	45.0	22.5	136.8	307774.6	0.664E+04
Totals	...	...	...	251.5	2287212.0	0.205E+05

Distance to Centroid from Datum [ytot]:
= AY / A
= 2287212/252
= 90.929 mm

Distance to Centroid [C1]:
= max(ytot, abs(Saddle Width - ytot))
= max(90.9, abs(300 - 90.9))
= 209.071 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12,2026

$$= 970 - \cos(702/2)(75 + 0 + 0) - 0$$

$$= 193.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{20489/252}$$

$$= 90.253 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 193/90.3$$

$$= 2.138$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8374 / (2 \cdot 1260)$$

$$= 3.323 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.32 \cdot 304 \cdot 193$$

$$= 194.949 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 195/980015$$

$$= 1.951 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($2.14 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (2.14)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (2.14) / (8 \cdot 123) - (2.14)^3 / (8 \cdot 123^3))$$

$$= 156.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 6.7/157 + (195 \cdot 209/20489) / 175$$

$$= 0.054$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.29 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	7.8800 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	6.4579 cm ²
Saddle Load Q0 (Weight)	Q0	38839.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1021.7 Kg
Maximum Transverse Force	Ft	4186.8 Kg
Maximum Longitudinal Force	F1	8373.7 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 11027.6 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 11028 / (6.46 \cdot 2)$$

$$= 83.7 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12, 2026

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4187 / (6.46 \cdot 2)$$

$$= 31.8 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 8374 / (108 \cdot 2)$$

$$= 3.8063 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 970 \cdot 4187 + 0$$

$$= 4061.32 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 4061 / 38839$$

$$= 104.57 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 3.81, 0]$$

$$= 3.8063 \text{ cm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-1 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-1 °C
Temperature reduction per Graph 3.13	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (61.9 \cdot 1) / (30 - 1.5 - 0)$$

$$= 1.000$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{ 639 \cdot 65.5 }, 2 \cdot 2400)$$

$$= 619.2131 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius Rm 639.25 mm



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12,2026

Shell Thickness used in this Case	t	67.000	mm
Stiffened Vessel Length per 4.15.6	L	13000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2400.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		175.00	N./mm ²
Head Allowable Stress used in Calculation		156.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	970.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		46086.00	Kgf
Pressure used in Analysis	P	16000.296	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	73.43	175.00
Long. Stress at Bottom of Midspan	82.74	175.00
Long. Stress at Top of Saddles	80.38	175.00
Long. Stress at Bottom of Saddles	75.79	175.00
Tangential Shear in Shell	2.17	139.64
Circ. Compressive Stress in Shell	1.45	175.00
Stiffener Circ. Stress at Shell	15.74	218.75
Stiffener Circ. Stress at Tip	93.50	197.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (4874/2 + 0) \cdot 970/1216$
 $= 5832.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(412, 0) \cdot 970/7950$
 $= 50.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(8374, 0) \cdot 970/7950$
 $= 1021.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (8374/2 + 0) \cdot 970/1216$
 $= 10020.3 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 36066 + \max(50.3, 5833, 1022, 10020)$
 $= 46086.0 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1799 (0 + \pi/4 (1.34 \cdot 1.2)^2)$
 $= 4042.046 \text{ N}$



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12, 2026

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	46778.94	Kgf
Transverse Shear Load	4186.85	Kgf
Longitudinal Shear Load	8373.70	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1210/2 + 67 + 30)\sin(120/2), 2 \cdot 639)$$

$$= 1215.900 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -46086 \cdot 2400 [1 - (1 - 2400/13000 + (639^2 - 0^2)/(2 \cdot 2400 \cdot 13000))/(1 + (4 \cdot 0)/(3 \cdot 13000))]$$

$$= -19695.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \{ (1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L \}$$

$$= 46086 \cdot 13000/4 \{ (1 + 2(639^2 - 0^2)/(13000^2))/(1 + (4 \cdot 0)/(3 \cdot 13000)) - 4 \cdot 2400/13000 \}$$

$$= 39898.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 16000 \cdot 639/(2 \cdot 65.5) - 39898/(\pi \cdot 639^2 \cdot 65.5)$$

$$= 73.43 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 16000 \cdot 639/(2 \cdot 65.5) + 39898/(\pi \cdot 639^2 \cdot 65.5)$$

$$= 82.74 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t)$$

$$= 16000 \cdot 639/(2 \cdot 65.5) - 19696/(\pi \cdot 639^2 \cdot 65.5)$$

$$= 80.38 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m/(2t) + M_1/(\pi \cdot R_m^2 t)$$

$$= 16000 \cdot 639/(2 \cdot 65.5) + 19696/(\pi \cdot 639^2 \cdot 65.5)$$

$$= 75.79 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 46086(13000 - 2 \cdot 2400)/(13000 + (4 \cdot 0/3))$$

$$= 29069.6 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T/(\pi \cdot R_m \cdot t)$$

$$= 29070/(\pi \cdot 639 \cdot 65.5)$$

$$= 2.17 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{639 \cdot 65.5}$$



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= 159.607 mm

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

= $-K_5 \cdot Q \cdot k/A$

= $-0.76 \cdot 46086 \cdot 0.1/237$

= -1.45 N./mm^2

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

= $-K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$

= $-0.34 \cdot 46086/237 + 0.053 \cdot 46086 \cdot 639 \cdot 40.7/2794$

= 15.74 N./mm^2

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

= $-K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$

= $-0.34 \cdot 46086/237 - 0.053 \cdot 46086 \cdot 639 \cdot 159/2794$

= -93.50 N./mm^2

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	45.0000	mm
Web Thickness	Webtk	45.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

= $((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$

= $((1.56 \cdot \sqrt{(606 \cdot 65.5)} + 350)65.5^3)/12$

= 1547.735 cm^4

Resolved Inertia for the Shell [I,shell]:

= $\text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$

= $1548 + 433(103 - 32.8)^2$

= 22947.371 cm^4

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	660.9	65.5	32.8	432.9	1417774.2	0.229E+05
Wearplate	350.0	30.0	80.5	105.0	845249.9	613.
Web	45.0	193.0	192.0	86.9	1667520.0	0.957E+04
BasePlate	150.0	75.0	326.0	112.5	3667499.8	0.564E+05
Totals	...	...	...	737.3	7598044.0	0.896E+05

Distance to Centroid [C1]:

= AY / A

= $7598044/737$

= 103.058 mm

Angle [β]:

= $180 - \text{Saddle Angle}/2$

= $180 - 120/2$

= 120.0

Saddle Splitting Coefficient [K1]:



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$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 46086$$

$$= 9379.5010 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.0223 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 970 - 606 \cdot \sin(120/2) / 1.05$$

$$= 468.429 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4393.7124 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.9576 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(46086 + 693)150 / (2 \cdot 1260 \cdot 175)}$$

$$= 21.636 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (605 + 67 + 30)$$

$$= 1215.900 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1216 / (5 - 1)$$

$$= 303.975 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 304 \cdot 150 / 2$$

$$= 227.981 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 46779 / (1260 \cdot 150)$$

$$= 24.751 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 228 \cdot 24.8$$

$$= 5642.708 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 115 + 68.4$$

$$= 183.144 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$



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= 5643/183
= 3.022 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	45.0	255.0	172.5	114.8	1979437.2	0.982E+04
Web	152.0	45.0	22.5	68.4	153887.3	0.616E+04
Totals	...	...	...	183.1	2133324.8	0.160E+05

Rib dimension [D]:

= Saddle Width - Web Thickness
= 300 - 45
= 255.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 2133324/183
= 116.483 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
= max(116, 300 - 116)
= 183.517 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{15975/183}$
= 93.396 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 970 - cos(120/2)(605 + 67 + 30) - 75
= 544.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/262}$
= 122.739

Slenderness ratio [KL/r]:

= KL/r
= 1 • 544/93.4
= 5.825

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 8374 / (2 • 1260) • 304 • 544/2
= 274.746 Kg-m.

Compressive Allowable, KL/r < Cc (5.82 < 123) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
= (1 - (5.82)² / (2 • 123²)) 262 /
(5/3 + 3 • (5.82) / (8 • 123) - (5.82³) / (8 • 123³))
= 155.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 3.02/155 + (275 • 184/159753152) / 175
= 0.037

Check of Inside Ribs:



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12,2026

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	45.0	255.0	172.5	114.8	1979437.2	0.139E+05
Web	304.0	45.0	22.5	136.8	307774.6	0.664E+04
Totals	...	...	...	251.5	2287212.0	0.205E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 2287212/252

= 90.929 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))

= max(90.9, abs(300 - 90.9))

= 209.071 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 970 - cos(702/2) (75 + 0 + 0) - 0

= 193.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{20489/252}$

= 90.253 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 193/90.3

= 2.138

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 8374/(2 • 1260)

= 3.323 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 3.32 • 304 • 193

= 194.949 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 195/980015

= 1.951 N./mm²

Compressive Allowable, KL/r < Cc (2.14 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (2.14)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (2.14) / (8 \cdot 123) - (2.14^3) / (8 \cdot 123^3))$

= 156.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 6.41/157 + (195 • 209/20489)/175

= 0.052

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12,2026

Bolt Allowable Stress	Stba	137.29 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	7.8800 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	6.4579 cm ²
Saddle Load Q0 (Weight)	Q0	36758.6 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1021.7 Kgf
Maximum Transverse Force	Ft	4186.8 Kgf
Maximum Longitudinal Force	Fl	8373.7 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8374 / (6.46 \cdot 2)$$

$$= 63.6 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4187 / (6.46 \cdot 2)$$

$$= 31.8 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 8374 / (108 \cdot 2)$$

$$= 3.8063 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 970 \cdot 4187 + 0$$

$$= 4061.32 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 4061 / 36759$$

$$= 110.48 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 3.81, 0]$$

$$= 3.8063 \text{ cm}^2$$

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-1 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-1 °C
Temperature reduction per Graph 3.13	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$



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Saddle Calcs: Operating Case: Step: 19 9:36am Aug 12, 2026

$$\begin{aligned} &= (61.9 \cdot 1) / (30 - 1.5 - 0) \\ &= 1.000 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

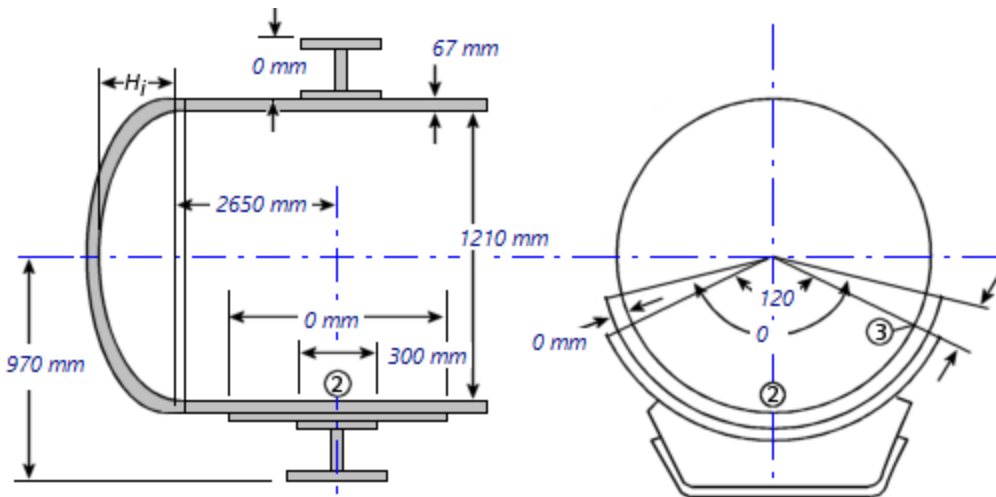
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{638 \cdot 67}, 2 \cdot 2650)$$

$$= 622.6580 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	638.50	mm
Shell Thickness used in this Case	t	67.000	mm
Stiffened Vessel Length per 4.15.6	L	13000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2650.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	970.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	42663.89	Kgf
Pressure used in Analysis	P 21260.812	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	98.31	247.00
Long. Stress at Bottom of Midspan	104.31	247.00
Long. Stress at Top of Saddles	103.87	247.00
Long. Stress at Bottom of Saddles	98.75	247.00



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Saddle Calcs: Test Case:

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Tangential Shear in Shell	1.84	138.58
Circ. Compressive Stress in Shell	1.30	247.00
Stiffener Circ. Stress at Shell	14.34	247.00
Stiffener Circ. Stress at Tip	84.12	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (1609/2 + 0) \cdot 970/1216$$

$$= 1924.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(136, 0) \cdot 970/7950$$

$$= 16.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 40739 + \max(16.6, 1925, 0, 0)$$

$$= 42663.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1799 (0 + \pi/4 (1.34 \cdot 1.2)^2)$$

$$= 4042.046 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	43356.84	Kgf
Transverse Shear Load	804.28	Kgf
Longitudinal Shear Load	136.02	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1210/2 + 67 + 30) \sin(120/2), 2 \cdot 638)$$

$$= 1215.900 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -42664 \cdot 2650 [1 - (1 - 2650/13000 + (638^2 - 0^2) / (2 \cdot 2650 \cdot 13000)) / (1 + (4 \cdot 0) / (3 \cdot 13000))]$$

$$= -22378.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 42664 \cdot 13000 / 4 ((1 + 2(638^2 - 0^2)/(13000^2)) / (1 + (4 \cdot 0) / (3 \cdot 13000)) - 4 \cdot 2650 / 13000)$$

$$= 26267.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 21261 \cdot 638 / (2 \cdot 67) - 26268 / (\pi \cdot 638^2 \cdot 67)$$

$$= 98.31 \text{ N./mm}^2$$



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Saddle Calcs: Test Case:

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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 21261 \cdot 638 / (2 \cdot 67) + 26268 / (\pi \cdot 638^2 \cdot 67)$$

$$= 104.31 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 21261 \cdot 638 / (2 \cdot 67) - 22378 / (\pi \cdot 638^2 \cdot 67)$$

$$= 103.87 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 21261 \cdot 638 / (2 \cdot 67) + 22378 / (\pi \cdot 638^2 \cdot 67)$$

$$= 98.75 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 42664 \cdot (13000 - 2 \cdot 2650) / (13000 + (4 \cdot 0 / 3))$$

$$= 25270.2 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 25270 / (\pi \cdot 638 \cdot 67)$$

$$= 1.84 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(638 \cdot 67)}$$

$$= 161.329 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.76 \cdot 42664 \cdot 0.1/244$$

$$= -1.30 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.34 \cdot 42664 / 244 + 0.053 \cdot 42664 \cdot 638 \cdot 41.3 / 2888$$

$$= 14.34 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.34 \cdot 42664 / 244 - 0.053 \cdot 42664 \cdot 638 \cdot 160 / 2888$$

$$= -84.12 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	45.0000	mm
Web Thickness	Webtk	45.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(605 \cdot 67)} + 350)^3) / 12$$

$$= 1664.420 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1664 + 445 (103 - 33.5)^2$$

$$= 23146.137 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	664.1	67.0	33.5	444.9	1490526.2	0.231E+05
Wearplate	350.0	30.0	82.0	105.0	861000.0	541.
Web	45.0	193.0	193.5	86.9	1680547.6	0.981E+04
BasePlate	150.0	75.0	327.5	112.5	3684374.8	0.572E+05
Totals	...	...	...	749.3	7716448.5	0.907E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7716448/749$$

$$= 102.984 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 42664$$

$$= 8683.0283 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.7979 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 970 - 605 \cdot \sin(120/2) / 1.05$$

$$= 469.669 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4078.2297 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.5393 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(42664 + 693)150 / (2 \cdot 1260 \cdot 175)}$$

$$= 20.830 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (605 + 67 + 30)$$

$$= 1215.900 \text{ mm}$$

Distance between Ribs [e]:



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= Web Length / (Nribs - 1)
 = 1216 / (5 - 1)
 = 303.975 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 304 • 150 / 2
 = 227.981 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 43357 / (1260 • 150)
 = 22.940 Kgf/cm²

Axial Load [P]:

= Ap • Bp
 = 228 • 22.9
 = 5229.916 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 115 + 68.4
 = 183.144 cm²

Compressive Stress [Sc]:

= P / Ar
 = 5230 / 183
 = 2.800 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	45.0	255.0	172.5	114.8	1979437.2	0.982E+04
Web	152.0	45.0	22.5	68.4	153887.3	0.616E+04
Totals	...	...	...	183.1	2133324.8	0.160E+05

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 300 - 45
 = 255.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 2133324 / 183
 = 116.483 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(116, 300 - 116)
 = 183.517 mm

Radius of Gyration [r]:

= √(Total Inertia / Total Area)
 = √(15975 / 183)
 = 93.396 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 970 - cos(120/2)(605 + 67 + 30) - 75
 = 544.000 mm

Intermediate Term [Cc]:



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$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 544 / 93.4$$

$$= 5.825$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 136 / (2 \cdot 1260) \cdot 304 \cdot 544 / 2$$

$$= 4.463 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($5.82 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.82)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (5.82) / (8 \cdot 123) - (5.82^3) / (8 \cdot 123^3))$$

$$= 154.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.8 / 154 + (4.46 \cdot 184 / 159753152) / 173$$

$$= 0.018$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	45.0	255.0	172.5	114.8	1979437.2	0.139E+05
Web	304.0	45.0	22.5	136.8	307774.6	0.664E+04
Totals	...	...	...	251.5	2287212.0	0.205E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2287212 / 252$$

$$= 90.929 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(90.9, \text{abs}(300 - 90.9))$$

$$= 209.071 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 970 - \cos(702/2) (75 + 0 + 0) - 0$$

$$= 193.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(20489 / 252)}$$

$$= 90.253 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 193 / 90.3$$

$$= 2.138$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 136 / (2 \cdot 1260)$$

$$= 0.054 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:



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$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.054 \cdot 304 \cdot 193 \\
 &= 3.167 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 3.17/980015 \\
 &= 0.032 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($2.14 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (2.14)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (2.14) / (8 \cdot 123) - (2.14^3) / (8 \cdot 123^3)) \\
 &= 155.4 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba} \\
 &= 5.94/155 + (3.17 \cdot 209/20489) / 173 \\
 &= 0.038
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	7.8800 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	6.4579 cm ²
Saddle Load Q0 (Weight)	Q0	41432.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	16.6 Kg
Maximum Transverse Force	Ft	804.3 Kg
Maximum Longitudinal Force	F1	136.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$\begin{aligned}
 &= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 136 / (6.46 \cdot 2) \\
 &= 1.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$\begin{aligned}
 &= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 804 / (6.46 \cdot 2) \\
 &= 6.1 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction) }$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 136 / (103 \cdot 2) \\
 &= 0.0646 \text{ cm}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned}
 &= B \cdot F_t + \text{Sum of X Moments} \\
 &= 970 \cdot 804 + 0 \\
 &= 780.17 \text{ Kg-m.}
 \end{aligned}$$

Eccentricity (e):



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Saddle Calcs: Test Case:

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= Rmom / QO
 = 780/41432
 = 18.83 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.065, 0]
 = 0.0646 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(300 + 1.56 \cdot \sqrt{638 \cdot 67}, 2 \cdot 2400)$
 = 622.6580 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	638.50	mm
Shell Thickness used in this Case	t	67.000	mm
Stiffened Vessel Length per 4.15.6	L	13000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2400.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	970.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		40255.06	Kgf
Pressure used in Analysis	P	21260.812	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		97.33	247.00
Long. Stress at Bottom of Midspan		105.29	247.00
Long. Stress at Top of Saddles		103.28	247.00
Long. Stress at Bottom of Saddles		99.35	247.00
Tangential Shear in Shell		1.85	138.58
Circ. Compressive Stress in Shell		1.23	247.00
Stiffener Circ. Stress at Shell		13.53	247.00
Stiffener Circ. Stress at Tip		79.37	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = $Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$



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Saddle Calcs: Test Case:

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$$= 3 (1609/2 + 0) \cdot 970/1216$$

$$= 1924.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(136, 0) \cdot 970/7950$$

$$= 16.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 38330 + \max(16.6, 1925, 0, 0)$$

$$= 40255.1 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1799 (0 + \pi/4 (1.34 \cdot 1.2)^2)$$

$$= 4042.046 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	40948.00	Kgf
Transverse Shear Load	804.28	Kgf
Longitudinal Shear Load	136.02	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1210/2 + 67 + 30) \sin(120/2), 2 \cdot 638)$$

$$= 1215.900 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -40255 \cdot 2400 [1 - (1 - 2400/13000 + (638^2 - 0^2) / (2 \cdot 2400 \cdot 13000)) / (1 + (4 \cdot 0) / (3 \cdot 13000))]$$

$$= -17205.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 40255 \cdot 13000/4 ((1 + 2(638^2 - 0^2)/(13000^2)) / (1 + (4 \cdot 0) / (3 \cdot 13000)) - 4 \cdot 2400/13000)$$

$$= 34848.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 21261 \cdot 638 / (2 \cdot 67) - 34849 / (\pi \cdot 638^2 \cdot 67)$$

$$= 97.33 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 21261 \cdot 638 / (2 \cdot 67) + 34849 / (\pi \cdot 638^2 \cdot 67)$$

$$= 105.29 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 21261 \cdot 638 / (2 \cdot 67) - 17205 / (\pi \cdot 638^2 \cdot 67)$$

$$= 103.28 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:



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$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= 21261 \cdot 638 / (2 \cdot 67) + -17205 / (\pi \cdot 638^2 \cdot 67)$$

$$= 99.35 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 40255 (13000 - 2 \cdot 2400) / (13000 + (4 \cdot 0 / 3))$$

$$= 25391.7 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 25392 / (\pi \cdot 638 \cdot 67)$$

$$= 1.85 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(638 \cdot 67)}$$

$$= 161.329 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K5 \cdot Q \cdot k / A$$

$$= -0.76 \cdot 40255 \cdot 0.1 / 244$$

$$= -1.23 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K8 \cdot Q / A + K6 \cdot Q \cdot R_m \cdot c1 / I$$

$$= -0.34 \cdot 40255 / 244 + 0.053 \cdot 40255 \cdot 638 \cdot 41.3 / 2888$$

$$= 13.53 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K8 \cdot Q / A - K6 \cdot Q \cdot R_m \cdot c2 / I$$

$$= -0.34 \cdot 40255 / 244 - 0.053 \cdot 40255 \cdot 638 \cdot 160 / 2888$$

$$= -79.37 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	75.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	45.0000	mm
Web Thickness	Webtk	45.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(605 \cdot 67)} + 350)^3) / 12$$

$$= 1664.420 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 1664 + 445 (103 - 33.5)^2$$

$$= 23146.137 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	664.1	67.0	33.5	444.9	1490526.2	0.231E+05



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Saddle Calcs: Test Case: Step: 20 9:36am Aug 12,2026

Wearplate	350.0	30.0	82.0	105.0	861000.0	541.
Web	45.0	193.0	193.5	86.9	1680547.6	0.981E+04
BasePlate	150.0	75.0	327.5	112.5	3684374.8	0.572E+05
Totals	...	...	...	749.3	7716448.5	0.907E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7716448/749$$

$$= 102.984 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 40255$$

$$= 8192.7773 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.6399 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 970 - 605 \cdot \sin(120/2) / 1.05$$

$$= 469.669 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3847.9697 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.2830 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(40255 + 693)150 / (2 \cdot 1260 \cdot 175)}$$

$$= 20.243 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (605 + 67 + 30)$$

$$= 1215.900 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1216 / (5 - 1)$$

$$= 303.975 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 304 \cdot 150/2$$

$$= 227.981 \text{ cm}^2$$

Bearing Pressure [Bp]:



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$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 40948 / (1260 \cdot 150)$$

$$= 21.666 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 228 \cdot 21.7$$

$$= 4939.351 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 115 + 68.4$$

$$= 183.144 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 4939 / 183$$

$$= 2.645 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	45.0	255.0	172.5	114.8	1979437.2	0.982E+04
Web	152.0	45.0	22.5	68.4	153887.3	0.616E+04
Totals	...	...	...	183.1	2133324.8	0.160E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300 - 45$$

$$= 255.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2133324 / 183$$

$$= 116.483 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(116, 300 - 116)$$

$$= 183.517 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(15975 / 183)}$$

$$= 93.396 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 970 - \cos(120/2) (605 + 67 + 30) - 75$$

$$= 544.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 544 / 93.4$$

$$= 5.825$$

Bending Moment [Rm]:



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Saddle Calcs: Test Case:

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$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 136 / (2 \cdot 1260) \cdot 304 \cdot 544 / 2$$

$$= 4.463 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (5.82 < 123) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.82)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (5.82) / (8 \cdot 123) - (5.82^3) / (8 \cdot 123^3))$$

$$= 154.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.64 / 154 + (4.46 \cdot 184 / 159753152) / 173$$

$$= 0.017$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	45.0	255.0	172.5	114.8	1979437.2	0.139E+05
Web	304.0	45.0	22.5	136.8	307774.6	0.664E+04
Totals	...	...	...	251.5	2287212.0	0.205E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2287212 / 252$$

$$= 90.929 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(90.9, \text{abs}(300 - 90.9))$$

$$= 209.071 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 970 - \cos(702/2) (75 + 0 + 0) - 0$$

$$= 193.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (20489 / 252) }$$

$$= 90.253 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 193 / 90.3$$

$$= 2.138$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 136 / (2 \cdot 1260)$$

$$= 0.054 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.054 \cdot 304 \cdot 193$$

$$= 3.167 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3.17 / 980015$$

$$= 0.032 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (2.14 < 123) per AISC E2-1 [Sca]:



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Saddle Calcs: Test Case: Step: 20 9:36am Aug 12, 2026

$$\begin{aligned}
 &= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3)) \\
 &= (1 - (2.14)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (2.14) / (8 \cdot 123) - (2.14^3) / (8 \cdot 123^3)) \\
 &= 155.4 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Mbase,c \cdot C1/I) / Sba \\
 &= 5.61/155 + (3.17 \cdot 209/20489) / 173 \\
 &= 0.036
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	7.8800 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	6.4579 cm ²
Saddle Load QO (Weight)	QO	39023.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	16.6 Kg
Maximum Transverse Force	Ft	804.3 Kg
Maximum Longitudinal Force	F1	136.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$\begin{aligned}
 &= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 136 / (6.46 \cdot 2) \\
 &= 1.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$\begin{aligned}
 &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 804 / (6.46 \cdot 2) \\
 &= 6.1 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 136 / (103 \cdot 2) \\
 &= 0.0646 \text{ cm}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned}
 &= B \cdot Ft + \text{Sum of X Moments} \\
 &= 970 \cdot 804 + 0 \\
 &= 780.17 \text{ Kg-m.}
 \end{aligned}$$

Eccentricity (e):

$$\begin{aligned}
 &= Rmom / QO \\
 &= 780/39023 \\
 &= 19.99 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}
 \end{aligned}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned}
 &= \max[Bltarearl, Bltarears, Bltareart] \\
 &= \max[0, 0.065, 0]
 \end{aligned}$$



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Saddle Calcs: Test Case:

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= 0.0646 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

Input, Nozzle Desc: N3- Channel inlet

From: 30

Pressure for Reinforcement Calculations	P	1011.860	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1210.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		924.35	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	151.00	N./mm ²
Allowable Stress At Ambient	Sna	161.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	757.2000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		150.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N3- Channel inlet

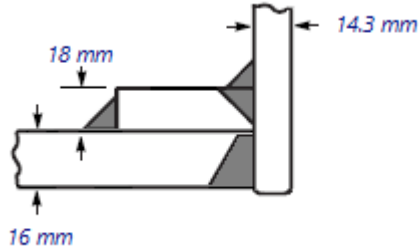
ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	16.876	in.
--	--------	-----



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 Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

Actual Thickness Used in Calculation 0.562 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 435 \cdot (\exp(1012/(151 \cdot 1)) - 1) + 3 + 0$
 $= 1.46 + 3 + 0 = 4.46 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 151 \cdot 1 \cdot \ln((2 \cdot 11.3 + 435) / 435) - 0$
 $= 7637 - 0 = 7637 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.340 mm
Min. required shell thickness, internal p + c :	6.763 mm

Nozzle neck minimum required thickness	: 6.763 mm
--	------------

Minimum nozzle neck thickness 12.497 mm , is greater than required thickness 6.763 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 36.6,	Allowable	: 151.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 353.4 N./mm ²	Passed
Occasional	: 11.3,	Allowable	: 200.8 N./mm ²	Passed
Shear	: 19.3,	Allowable	: 105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[151/164, 1]$
 $= 0.921$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[164/164, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$



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$$= 14.3 - 3$$

$$= 11.275 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1216/13$$

$$= 93.538 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1210/2 + 3$$

$$= 608.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(R_{eff} \cdot t) + W}$$

$$= \sqrt{(608 \cdot 13) + 150}$$

$$= 238.904 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((R_{eff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((608 + 13) \cdot (13 + 18))}$$

$$= 138.748 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 217$$

$$= 434.650 \text{ mm}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(239, 139, 435)$$

$$= 138.748 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 13, 18] + \sqrt{(217 \cdot 11.3)}$$

$$= 67.500 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(13 + 18)$$

$$= 248.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[67.5, 250, 248] + 13$$

$$= 80.500 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 13 + ((25 \cdot 1) / 139)$$

$$= 31.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{((D_i + t_{eff}) \cdot t_{eff})))$$

$$= \min(12, (2 \cdot 217 + 11.3) / (\sqrt{((1216 + 31) \cdot 31)}))$$



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= 2.268

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((L/5)0.85, 1) \\
 &= 13 \cdot 139 \cdot \max((2.27/5)0.85, 1) \\
 &= 18.037 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 11.3 \cdot 80.5 \\
 &= 9.076 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 10^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned}
 &= \min(W \cdot t_e, LR \cdot t_e) \\
 &= \min(150 \cdot 18, 139 \cdot 18) \\
 &= 24.975 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned}
 &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\
 &= 18 + 0.92(9.08 + 0) + 0.5 + 0 + 0 + 1(25) \\
 &= 51.869 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 11.3 / \ln[1 + 11.3/217] \\
 &= 222.915 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 31 / \ln[1 + 31/608] \\
 &= 623.371 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 164 \cdot 1 \\
 &= 246.0 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot LH \\
 &= 1012 \cdot 223 \cdot 80.5 \\
 &= 1851.6 \text{ Kg}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (LR + t_n) \\
 &= 1012 \cdot 623 (139 + 11.3) \\
 &= 9649.9 \text{ Kg}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 1012 \cdot 623 \cdot 217 \\
 &= 13978.9 \text{ Kg}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (9650 + 13979 + 1852) / 1012 \\
 &= 2469.4 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



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= Sallow / (2 • Ap/AT - Rxs/teff)
 = 246 / (2 • 2469/51.9 - 623/31)
 = 3275.1 KPa.

Maximum Allowable Working Pressure Candidate [Pmax2]:

= S[t/Rxs]
 = 164 [13/623]
 = 3419.9 KPa.

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
 = min(3275, 3420)
 = 3275.084 KPa.

Average Primary Membrane Stress [σAvg]:

= (fN + fS + fY) / AT
 = (1852 + 9650 + 13979)/51.9
 = 48.176 N./mm²

General Primary Membrane Stress [σCirc]:

= P • Rxs / teff
 = 1012 • 623/31
 = 20.3 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max(2 • σAvg - σCirc, σCirc)
 = max(2 • 48.2 - 20.3, 20.3)
 = 76.0 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm ²
Local Primary Membrane Stress	PL	76.00	N./mm ²
Maximum Allowable Working Pressure	Pmax	3275.08	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= (Rnc + tn) / Rnc
 = (217 + 11.3)/217
 = 1.052 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= π/2 • (Rn + tn)
 = π/2 • (217 + 11.3)
 = 359.084 mm

Weld Length of Pad to Shell Weld [LtP]:

= π/2 • (Rn + tn + W)
 = π/2 • (217 + 11.3 + 150)
 = 594.703 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 8.485, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be >= min(0.7 * 11.275, 6.000) Passed
 tf1, L42T = 8.485, must be >= min(0.6 * 18.000, 0.6 * 13.000) Passed

Weld Load Value [fwelds]:

= min(fy • ky, 1.5 • Sn(A2 + A3), π/4•P•Rn²•ky²)
 = min(13979•1.05, 1.5•151(9.08+0), π/4•1012•217²•1.05²)
 = 4235.060 Kgf

Discontinuity Force [fws]:

= fwelds • t • S / (t • S + te • Sp)



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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

$$= 4235 \cdot 13 \cdot 164 / (13 \cdot 164 + 18 \cdot 164)$$

$$= 1775.993 \text{ Kgf}$$

Discontinuity Force [fwp]:

$$= f_{welds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 4235 \cdot 18 \cdot 164 / (13 \cdot 164 + 18 \cdot 164)$$

$$= 2459.067 \text{ Kgf}$$

Shear Stress [τ_1]:

$$= f_{ws} / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 1776 / (359 \cdot (0.6 \cdot 13 + 0.49 \cdot 0))$$

$$= 6.218 \text{ N./mm}^2$$

Shear Stress [τ_2]:

$$= f_{wp} / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 2459 / (359 \cdot (0.6 \cdot 18 + 0.49 \cdot 7.07))$$

$$= 4.708 \text{ N./mm}^2$$

Shear Stress [τ_3]:

$$= f_{wp} / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 2459 / (595 \cdot (0.49 \cdot 8.49))$$

$$= 9.753 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(6.22, 4.71, 9.75)$$

$$= 9.8 \text{ must be less than or equal to } 164.0 \text{ N./mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is : -104 °C

The computed stress ratio is : 0.144

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (1.37 \cdot 1) / (12.5 - 3 - 0)$$

$$= 0.144$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is : -104 °C

The computed stress ratio is : 0.144

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (1.37 \cdot 1) / (12.5 - 3 - 0)$$

$$= 0.144$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C



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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.144

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (1.37 • 1)/(12.5 - 3 - 0)
= 0.144

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -23 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 6 °C
Temperature reduction per Graph 3.12 : 29 °C
The computed stress ratio is : 0.271

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (3.53 • 1)/(16 - 3 - 0)
= 0.271

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.144

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (1.37 • 1)/(12.5 - 3 - 0)
= 0.144

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -23 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -23 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 44.8509 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 310.8509 mm

Input Echo, WRC297 Item 1, Description: N3- Channel in:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1210.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	164.00 N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm



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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

Nozzle Diameter		428.650	mm
Nozzle Thickness		14.2748	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	161.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	151.00	N./mm ²
Diameter of Reinforcing Pad		757.200	mm
Thickness of Reinforcing Pad		18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1011.86	KPa.
Radial Load	P	-3441.77	Kgf
Circumferential Shear	Vc	2294.51	Kgf
Longitudinal Shear	Vl	2294.51	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{623.5(16.0 - 3.0)}$
 $= 754.301 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1247.000	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	31.000	mm
Nozzle Thickness used	(t) =	11.275	mm
T / t	=	2.749	
d / t	=	40.551	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.325$$

Nr/P	=	0.059
Mr/P	=	0.068
M0/P	=	0.053
N0/P	=	0.178
MrD/Mc	=	0.154
NrDL/Mc	=	0.093
M0d/Mc	=	0.143
N0DL/Mc	=	0.238
MrD/Ml	=	0.085
NrDL/Ml	=	0.058
M0D/Ml	=	0.074
N0DL/Ml	=	0.339

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	11	-11	11	-11



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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-21	-21	21	21
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	19	20	19	20

Outplane Stress Summary	-13	22	87	8
-------------------------	-----	----	----	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-32	32	32	-32
Inplane Pressure Stress	9	9	9	9

Inplane Stress Summary	-10	26	62	-31
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	17	29	87	40
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-15	-15	15	15
Outplane Bending (Mc)	-55	55	55	-55
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9

Outplane Stress Summary	-43	44	97	-35
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-59	59	59	-59



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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	20	19	20

Inplane Stress Summary	-29	61	101	-45
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	44	62	104	47
------------------------	----	----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	24	22	24

Hoop Stress Summary	7	8	50	52
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-163	163	163	-163
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	-105	99	253	-193
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10

Shear Stress Summary	13	13	8	8
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Nozzle Stresses (N./mm²):



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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	B1 Inside
Two * Max Shear Stress	116	101	253	246

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-15	-15	15	15
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	24	22	24
Hoop Stress Summary	13	14	44	45

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-313	313	313	-313
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11
Axial Stress Summary	-255	250	403	-343

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	D1 Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	8	8	13	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	D1 Inside
Two * Max Shear Stress	270	250	404	390

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{614.5(16.0 - 3.0)}$$

$$= 752.149 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1229.000 mm
Nozzle Outside Diameter	(d) =	757.200 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	161.275 mm



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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

T / t = 0.081
d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.990$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
Mr/P = 0.071
M0/P = 0.019
N0/P = 0.027
MrD/Mc = 0.239
NrDL/Mc = 0.094
M0d/Mc = 0.069
N0DL/Mc = 0.051
MrD/Ml = 0.043
NrDL/Ml = 0.040
M0D/Ml = 0.013
N0DL/Ml = 0.024

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	46	47	46	47
Outplane Stress Summary	52	42	98	17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	84	-84	84	-84
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-56	56	56	-56
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	51	-4	182	-99

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

Two * Max Shear Stress	56	47	182	117
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-91	91	91	-91
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23

Outplane Stress Summary	-50	85	154	-74
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	84	-84	84	-84
Inplane Membrane (Mc)	-20	-20	20	20
Inplane Bending (Mc)	-317	317	317	-317
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	46	47	46	47

Inplane Stress Summary	-198	269	478	-324
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	198	269	478	324
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-15	-15	27	27	-4	-4	8	8
Circ. Q (SUS)	-17	17	39	-39	-45	45	73	-73
Long. Pm (SUS)	9	9	9	9	9	9	9	9



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Nozzle Calcs.: N3- Channel inl Nozl: 9 9:36am Aug 12,2026

Long. P1 (SUS)	-1	-1	5	5	-9	-9	21	21
Long. Q (SUS)	-18	18	46	-46	-44	44	66	-66
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+P1 (SUS)	8.2	8.3	46.0	47.0	15.1	16.1	30.3	30.4
Pm+P1+Q (Total)	15.8	28.5	85.2	40.2	44.3	61.2	102.5	46.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	164.00	Passed
Pm+P1 (SUS)	47.03	246.00	Passed
Pm+P1+Q (TOTAL)	102.47	492.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	47	46	47	46	47	46	47
Circ. P1 (SUS)	0	0	10	10	-12	-12	28	28
Circ. Q (SUS)	6	-6	40	-40	-233	233	401	-401
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. P1 (SUS)	0	0	16	16	-6	-6	16	16
Long. Q (SUS)	28	-28	140	-140	-68	68	114	-114
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	46.0	47.0	46.0	47.0	46.0	47.0	46.0	47.0
Pm+P1 (SUS)	46.0	47.0	56.1	57.1	34.1	35.1	74.0	75.0
Pm+P1+Q (Total)	55.5	46.7	179.0	118.1	199.0	268.0	475.0	326.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	47.00	164.00	Passed
Pm+P1 (SUS)	75.03	246.00	Passed
Pm+P1+Q (TOTAL)	475.05	492.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Nozzle Calcs.: N3- Channel in1

Nozl: 9 9:36am Aug 12,2026

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	24	22	24	22	24	22	24
Circ. Pl (SUS)	-15	-15	27	27	-9	-9	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-14	-14	18	18	-14	-14	18	18
Long. Q (SUS)	-103	103	223	-223	-253	253	373	-373
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	22.0	24.0	22.0	24.0	22.0	24.0	22.0	24.0
Pm+Pl (SUS)	10.8	12.6	49.2	51.2	16.5	18.4	43.3	45.2
Pm+Pl+Q (Total)	115.5	101.6	252.3	245.5	269.5	250.3	402.4	389.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	151.00	Passed
Pm+Pl (SUS)	51.18	226.50	Passed
Pm+Pl+Q (TOTAL)	402.40	453.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2- Shell outlet

Nozl: 10 9:36am Aug 12,2026

Input, Nozzle Desc: N2- Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	16000.594	KPa.
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	175.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1210.00	mm
Shell Finished (Minimum) Thickness	t	67.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2924.35	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	149.80	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		15.9528	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Hub Height of Integral Nozzle	h	220.0000	mm
Height of Beveled Transition	L'	89.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		115.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		67.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2- Shell outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.953	in.
Actual Thickness Used in Calculation	1.024	in.



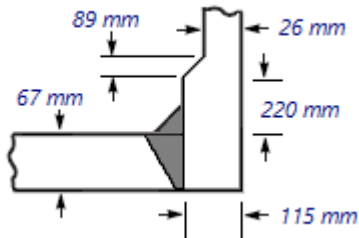
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Nozzle Calcs.: N2- Shell outle

Nozl: 10 9:36am Aug 12,2026



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 408 \cdot (\exp(16001/(150 \cdot 1)) - 1) + 1.5 + 0$$

$$= 23 + 1.5 + 0 = 24.5 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 150 \cdot 1 \cdot \ln((2 \cdot 24.5 + 408) / 408) - 0$$

$$= 16981 - 0 = 16981 \text{ KPa.}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 476.34 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 516.91 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm

Min. required shell thickness, internal p + c : 59.615 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 26.010 mm , is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 86.3,	Allowable	: 149.8 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 307.2 N./mm ²	Passed
Occasional	: 62.9,	Allowable	: 199.2 N./mm ²	Passed
Shear	: 15.1,	Allowable	: 104.9 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[150/175, 1]$$

$$= 0.856$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 115 - 1.5$$

$$= 113.500 \text{ mm}$$



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Nozzle Calcs.: N2- Shell outle Nozl: 10 9:36am Aug 12,2026

Thickness of Nozzle at Top [tn2]:
 = thickness - corrosion allowance
 = 26 - 1.5
 = 24.500 mm

Shell Diameter to Thickness ratio [D/t]:
 = Di/t
 = 1213/65.5
 = 18.519 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 1210/2 + 1.5
 = 606.500 mm

Effective Length of Vessel Wall [LR]:
 = $\min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$
 = $\min(\sqrt{606 \cdot 65.5}, 2 \cdot 204)$
 = 199.313 mm

Thickness Limit Candidate [LH1]:
 = $\min[1.5t, te] + \sqrt{Rn \cdot tn}$
 = $\min[1.5 \cdot 65.5, 0] + \sqrt{204 \cdot 114}$
 = 152.202 mm

Thickness Limit Candidate [LH2]:
 = Lpr1
 = 300
 = 300.000 mm

Thickness Limit Candidate [LH3]:
 = 8(t + te)
 = 8(65.5 + 0)
 = 524.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
 = $\min[LH1, LH2, LH3] + t$ for set-in nozzles
 = $\min[152, 300, 524] + 65.5$
 = 217.702 mm

Effective Vessel Thickness [teff]:
 = $t + ((A5 \cdot frp) / LR)$
 = 65.5 + ((0.1)/199)
 = 65.500 mm

Parameter [λ]:
 = $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
 = $\min(12, (2 \cdot 204 + 114) / (\sqrt{(1213 + 65.5) \cdot 65.5}))$
 = 1.803

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 = $t \cdot LR \cdot \max((N/5)^{0.85}, 1)$
 = 65.5 · 199 · $\max((1.8/5)^{0.85}, 1)$
 = 130.550 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 = $tn \cdot LH$
 = 114 · 218
 = 247.091 cm²

Area Contributed by the Outside Fillet Weld [A41]:
 = 0.5 · Leg41²
 = 0.5 · 10²



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Nozzle Calcs.: N2- Shell outle Nozl: 10 9:36am Aug 12,2026

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 131 + 0.86(247 + 0) + 0.5 + 0 + 0 \\ &= 342.562 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 114 / \ln[1 + 114/204] \\ &= 256.681 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 65.5 / \ln[1 + 65.5/606] \\ &= 638.690 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 175 \cdot 1 \\ &= 262.5 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 16001 \cdot 257 \cdot 218 \\ &= 91177.7 \text{ Kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 16001 \cdot 639 (199 + 114) \\ &= 325992.8 \text{ Kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 16001 \cdot 639 \cdot 204 \\ &= 212699.3 \text{ Kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (325993 + 212699 + 91178) / 16001 \\ &= 3860.3 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 262 / (2 \cdot 3860 / 343 - 639 / 65.5) \\ &= 20527.6 \text{ KPa.} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / (t / R_{xs}) \\ &= 175 [65.5 / 639] \\ &= 17945.5 \text{ KPa.} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(20528, 17946) \\ &= 17945.516 \text{ KPa.} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (91178 + 325993 + 212699) / 343 \\ &= 180.318 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 16001 \cdot 639 / 65.5 \end{aligned}$$



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Nozzle Calcs.: N2- Shell outle Nozl: 10 9:36am Aug 12,2026

$$= 156.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 180 - 156, 156)$$

$$= 204.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	262.50	N./mm ²
Local Primary Membrane Stress	PL	204.61	N./mm ²
Maximum Allowable Working Pressure	Pmax	17945.52	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (204 + 114) / 204$$

$$= 1.556 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (204 + 114)$$

$$= 498.885 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 113.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(212699 \cdot 1.56, 1.5 \cdot 150(247+0), \pi/4 \cdot 16001 \cdot 204^2 \cdot 1.56^2)$$

$$= 129265.773 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 129266 / (499(0.49 \cdot 7.07 + 0.6 \cdot 65.5 + 0.49 \cdot 0))$$

$$= 59.419 < \text{or} = 174.997 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	9 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	14 °C
Temperature reduction per Graph 3.13	:	6 °C
The computed stress ratio is	:	0.887

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (58.1 \cdot 1) / (67 - 1.5 - 0)$$

$$= 0.887$$

Gov. MDMT of the nozzle to shell joint welded assembly : 9 °C

The Drop for this Nozzle is : 90.0678 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 457.0677 mm

Input Echo, WRC297 Item 1, Description: N2- Shell outl:

Diameter Basis for Cylindrical Shell ID



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Nozzle Calcs.: N2- Shell outle Nozl: 10 9:36am Aug 12,2026

Shell Internal Corrosion Allowance		1.5000	mm
Shell External Corrosion Allowance		0.0000	mm
Shell Diameter		1210.000	mm
Shell Thickness		67.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	175.00	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		1.5000	mm
Nozzle Diameter		405.200	mm
Nozzle Thickness		115.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	149.80	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	16000.59	KPa.
Radial Load	P	-5736.27	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	6192.80	Kg-m.
Longitudinal Moment	Ml	6192.80	Kg-m.
Torsional Moment	Mt	7224.93	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.523) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 635.2 + 2 \cdot 1.65 \cdot \sqrt{(639.25(67.0 - 1.5))}$$

$$= 1310.458 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1278.500 mm
Nozzle Outside Diameter	(d) =	635.200 mm
Vessel Thickness used	(T) =	65.500 mm
Nozzle Thickness used	(t) =	113.500 mm
T / t	=	0.577
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.195$$

Nr/P	=	0.069
Mr/P	=	0.130
M0/P	=	0.039
N0/P	=	0.096
MrD/Mc	=	0.274
NrDL/Mc	=	0.107
M0d/Mc	=	0.086



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Nozzle Calcs.: N2- Shell outle

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NØDL/Mc = 0.071
MrD/Ml = 0.108
NrDL/Ml = 0.062
MØD/Ml = 0.035
NØDL/Ml = 0.119

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	140	156	140	156
Outplane Stress Summary	137	156	152	152

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	70	70	70	70
Inplane Stress Summary	65	74	97	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	137	156	152	152

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



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Nozzle Calcs.: N2- Shell outle Nozl: 10 9:36am Aug 12,2026

Normal Pressure Stress	70	70	70	70
------------------------	----	----	----	----

Outplane Stress Summary	61	78	87	58
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-36	36	36	-36
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	140	156	140	156

Inplane Stress Summary	112	181	190	112
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	112	181	190	113
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	46	27	46

Hoop Stress Summary	25	44	30	50
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	13	13	13	13



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Nozzle Calcs.: N2- Shell outle Nozl: 10 9:36am Aug 12,2026

Axial Stress Summary	9	11	22	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	25	44	30	50
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	46	27	46

Hoop Stress Summary	26	45	29	49
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	13	13	13	13

Axial Stress Summary	3	18	29	4
----------------------	---	----	----	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	26	45	30	49
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



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Nozzle Calcs.: N2- Shell outle

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Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	140	156	140	156	140	156	140	156
Circ. Pl (SUS)	-1	-1	3	3	-2	-2	2	2
Circ. Q (SUS)	-1	1	7	-7	-26	26	46	-46
Long. Pm (SUS)	70	70	70	70	70	70	70	70
Long. Pl (SUS)	-1	-1	1	1	0	0	2	2
Long. Q (SUS)	-4	4	24	-24	-8	8	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	140.0	156.0	140.0	156.0	140.0	156.0	140.0	156.0
Pm+Pl (SUS)	139.0	155.0	143.0	159.0	138.0	154.0	142.0	158.0
Pm+Pl+Q (Total)	138.0	156.0	150.0	152.0	112.0	180.0	188.0	112.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	156.00	175.00	Passed
Pm+Pl (SUS)	159.00	262.50	Passed
Pm+Pl+Q (TOTAL)	188.01	524.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	46	27	46	27	46	27	46
Circ. Pl (SUS)	-1	-1	3	3	0	0	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-1	1	5	-5	-7	7	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	27.0	46.0	27.0	46.0	27.0	46.0	27.0	46.0
Pm+Pl (SUS)	26.0	45.0	30.0	49.0	27.0	46.0	29.0	48.0
Pm+Pl+Q (Total)	26.0	45.0	30.0	49.0	27.0	46.0	29.0	48.0

Nozzle Stress Summation Comparison (N./mm²):



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Nozzle Calcs.: N2- Shell outle

Nozl: 10

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	149.80	Passed
Pm+Pl (SUS)	49.00	224.70	Passed
Pm+Pl+Q (TOTAL)	49.00	449.40	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N1- Shell inlet

Nozl: 11 9:36am Aug 12,2026

Input, Nozzle Desc: N1- Shell inlet

From: 40

Pressure for Reinforcement Calculations	P	16000.000	KPa.
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	175.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1210.00	mm
Shell Finished (Minimum) Thickness	t	67.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		13924.35	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	149.80	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		15.9528	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Hub Height of Integral Nozzle	h	220.0000	mm
Height of Beveled Transition	L'	89.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		115.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		67.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1- Shell inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.953	in.
Actual Thickness Used in Calculation	1.024	in.



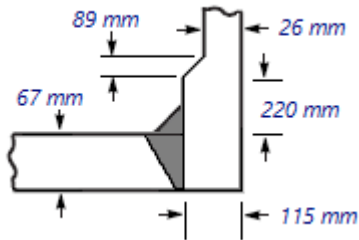
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Nozzle Calcs.: N1- Shell inlet

Noz1: 11 9:36am Aug 12,2026



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 408 \cdot (\exp(16000/(150 \cdot 1)) - 1) + 1.5 + 0$$

$$= 23 + 1.5 + 0 = 24.5 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 150 \cdot 1 \cdot \ln((2 \cdot 24.5 + 408) / 408) - 0$$

$$= 16981 - 0 = 16981 \text{ KPa.}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm

Min. required shell thickness, internal p + c : 59.612 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 26.010 mm , is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 86.3,	Allowable	: 149.8 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 307.2 N./mm ²	Passed
Occasional	: 62.9,	Allowable	: 199.2 N./mm ²	Passed
Shear	: 15.1,	Allowable	: 104.9 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[150/175, 1]$$

$$= 0.856$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 115 - 1.5$$

$$= 113.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 26 - 1.5$$

$$= 24.500 \text{ mm}$$



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Nozzle Calcs.: N1- Shell inlet

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Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1213/65.5$$

$$= 18.519 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1210/2 + 1.5$$

$$= 606.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(R_{eff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(606 \cdot 65.5)}, 2 \cdot 204)$$

$$= 199.313 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 65.5, 0] + \sqrt{(204 \cdot 114)}$$

$$= 152.202 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(65.5 + 0)$$

$$= 524.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[152, 300, 524] + 65.5$$

$$= 217.702 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 65.5 + ((0 \cdot 1) / 199)$$

$$= 65.500 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 204 + 114) / (\sqrt{((1213 + 65.5) \cdot 65.5)}))$$

$$= 1.803$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 65.5 \cdot 199 \cdot \max((1.8/5)^{0.85}, 1)$$

$$= 130.550 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 114 \cdot 218$$

$$= 247.091 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$



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Nozzle Calcs.: N1- Shell inlet

Nozl: 11

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$$= 131 + 0.86(247 + 0) + 0.5 + 0 + 0$$

$$= 342.562 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 114 / \ln[1 + 114/204]$$

$$= 256.681 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 65.5 / \ln[1 + 65.5/606]$$

$$= 638.690 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 175 \cdot 1$$

$$= 262.5 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 16000 \cdot 257 \cdot 218$$

$$= 91174.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 16000 \cdot 639 (199 + 114)$$

$$= 325980.7 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 16000 \cdot 639 \cdot 204$$

$$= 212691.3 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (325981 + 212691 + 91174) / 16000$$

$$= 3860.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 262 / (2 \cdot 3860 / 343 - 639 / 65.5)$$

$$= 20527.6 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 175 [65.5 / 639]$$

$$= 17945.5 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(20528, 17946)$$

$$= 17945.516 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (91174 + 325981 + 212691) / 343$$

$$= 180.312 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 16000 \cdot 639 / 65.5$$

$$= 156.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$



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Nozzle Calcs.: N1- Shell inlet Nozl: 11 9:36am Aug 12,2026

$$= \max(2 \cdot 180 - 156, 156)$$

$$= 204.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	262.50	N./mm ²
Local Primary Membrane Stress	PL	204.60	N./mm ²
Maximum Allowable Working Pressure	Pmax	17945.52	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (204 + 114) / 204$$

$$= 1.556 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (204 + 114)$$

$$= 498.885 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 113.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(212691 \cdot 1.56, 1.5 \cdot 150(247+0), \pi/4 \cdot 16000 \cdot 204^2 \cdot 1.56^2)$$

$$= 129260.969 \text{ Kg}$$

Weld Stress Value [T]:

$$= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 129261 / (499(0.49 \cdot 7.07 + 0.6 \cdot 65.5 + 0.49 \cdot 0))$$

$$= 59.417 < \text{or} = \text{to } 174.997 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	9 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	14 °C
Temperature reduction per Graph 3.13	:	6 °C
The computed stress ratio is	:	0.887

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (58.1 \cdot 1) / (67 - 1.5 - 0)$$

$$= 0.887$$

Gov. MDMT of the nozzle to shell joint welded assembly : 9 °C

The Drop for this Nozzle is : 90.0678 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 457.0677 mm

Input Echo, WRC297 Item 1, Description: N1- Shell inle:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.5000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	1210.000 mm
Shell Thickness	67.0000 mm



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Nozzle Calcs.: N1- Shell inlet Nozl: 11 9:36am Aug 12,2026

Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	175.00	N./mm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		1.5000	mm
Nozzle Diameter		405.200	mm
Nozzle Thickness		115.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	149.80	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	16000.00	KPa.
Radial Load	P	-5736.27	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	6192.80	Kg-m.
Longitudinal Moment	Ml	6192.80	Kg-m.
Torsional Moment	Mt	7224.93	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.523) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
= 635.2 + 2 • 1.65 • √(639.25(67.0 - 1.5))
= 1310.458 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1278.500 mm
Nozzle Outside Diameter	(d) =	635.200 mm
Vessel Thickness used	(T) =	65.500 mm
Nozzle Thickness used	(t) =	113.500 mm
T / t	=	0.577
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.195$$

Nr/P	=	0.069
Mr/P	=	0.130
M0/P	=	0.039
N0/P	=	0.096
MrD/Mc	=	0.274
NrDL/Mc	=	0.107
M0d/Mc	=	0.086
N0DL/Mc	=	0.071
MrD/Ml	=	0.108
NrDL/Ml	=	0.062
M0D/Ml	=	0.035



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Nozzle Calcs.: N1- Shell inlet

Nozl: 11 9:36am Aug 12,2026

NØDL/M1 = 0.119

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	140	156	140	156
Outplane Stress Summary	137	156	152	152

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	70	70	70	70
Inplane Stress Summary	65	74	97	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	137	156	152	152

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	70	70	70	70
Outplane Stress Summary	61	78	87	58



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Nozzle Calcs.: N1- Shell inlet

Nozl: 11 9:36am Aug 12,2026

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-36	36	36	-36
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	140	156	140	156
Inplane Stress Summary	112	181	190	112

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	112	181	190	113

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	46	27	46
Hoop Stress Summary	25	44	30	50

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	9	11	22	10

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N1- Shell inlet Nozl: 11 9:36am Aug 12,2026

(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	25	44	30	50

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	46	27	46
Hoop Stress Summary	26	45	29	49

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	3	18	29	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	26	45	30	49

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of	Stress Values at
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Nozzle Calcs.: N1- Shell inlet

Nozl: 11 9:36am Aug 12,2026

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	140	156	140	156	140	156	140	156
Circ. Pl (SUS)	-1	-1	3	3	-2	-2	2	2
Circ. Q (SUS)	-1	1	7	-7	-26	26	46	-46
Long. Pm (SUS)	70	70	70	70	70	70	70	70
Long. Pl (SUS)	-1	-1	1	1	0	0	2	2
Long. Q (SUS)	-4	4	24	-24	-8	8	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	140.0	156.0	140.0	156.0	140.0	156.0	140.0	156.0
Pm+Pl (SUS)	139.0	155.0	143.0	159.0	138.0	154.0	142.0	158.0
Pm+Pl+Q (Total)	138.0	156.0	150.0	152.0	112.0	180.0	188.0	112.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	156.00	175.00	Passed
Pm+Pl (SUS)	159.00	262.50	Passed
Pm+Pl+Q (TOTAL)	188.01	524.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	46	27	46	27	46	27	46
Circ. Pl (SUS)	-1	-1	3	3	0	0	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-1	1	5	-5	-7	7	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	27.0	46.0	27.0	46.0	27.0	46.0	27.0	46.0
Pm+Pl (SUS)	26.0	45.0	30.0	49.0	27.0	46.0	29.0	48.0
Pm+Pl+Q (Total)	26.0	45.0	30.0	49.0	27.0	46.0	29.0	48.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	149.80	Passed



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Nozzle Calcs.: N1- Shell inlet

Nozl: 11

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Pm+Pl (SUS)	49.00	224.70	Passed
Pm+Pl+Q (TOTAL)	49.00	449.40	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N4- Channel out

Nozl: 12

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Input, Nozzle Desc: N4- Channel outlet

From: 50

Pressure for Reinforcement Calculations	P	1011.860	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	164.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1210.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		15924.35	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	151.00	N./mm ²
Allowable Stress At Ambient	Sna	161.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	164.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	175.00	N./mm ²
Diameter of Pad along vessel surface	Dp	857.2000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		200.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N4- Channel outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	16.876	in.
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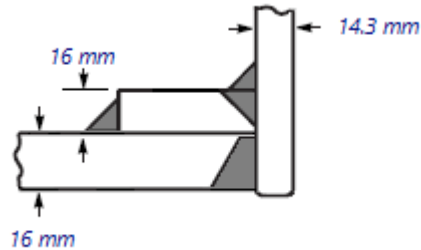
Nozzle Calcs.: N4- Channel out

Nozl: 12

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Actual Thickness Used in Calculation

0.562 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 435 \cdot (\exp(1012/(151 \cdot 1)) - 1) + 3 + 0$$

$$= 1.46 + 3 + 0 = 4.46 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 151 \cdot 1 \cdot \ln((2 \cdot 11.3 + 435) / 435) - 0$$

$$= 7637 - 0 = 7637 \text{ KPa.}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 362.80 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm

Min. required shell thickness, internal p + c : 6.763 mm

Nozzle neck minimum required thickness : 6.763 mm

Minimum nozzle neck thickness 12.497 mm , is greater than required thickness 6.763 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 36.6,	Allowable	: 151.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 353.4 N./mm ²	Passed
Occasional	: 11.3,	Allowable	: 200.8 N./mm ²	Passed
Shear	: 19.3,	Allowable	: 105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[151/164, 1]$$

$$= 0.921$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[164/164, 1]$$



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Nozzle Calcs.: N4- Channel out Nozl: 12 9:36am Aug 12,2026

= 1.000

Thickness of Nozzle [tn]:

= thickness - corrosion allowance

= 14.3 - 3

= 11.275 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t

= 1216/13

= 93.538 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance

= 1210/2 + 3

= 608.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

= $\sqrt{(Reff \cdot t)} + W$

= $\sqrt{(608 \cdot 13)} + 200$

= 288.904 mm

Limit [LR2]:

= $\sqrt{((Reff + t) \cdot (t + te))}$

= $\sqrt{((608 + 13) \cdot (13 + 16))}$

= 134.198 mm

Limit [LR3]:

= 2 • Rn

= 2 • 217

= 434.650 mm

Final value of [LR]:

= min(LR1, LR2, LR3)

= min(289, 134, 435)

= 134.198 mm

Thickness Limit Candidate [LH1]:

= min[1.5t, te] + $\sqrt{(Rn \cdot tn)}$

= min[1.5•13, 16] + $\sqrt{(217 \cdot 11.3)}$

= 65.500 mm

Thickness Limit Candidate [LH2]:

= Lpr1

= 250

= 250.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)

= 8(13 + 16)

= 232.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles

= min[65.5, 250, 232] + 13

= 78.500 mm

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)

= 13 + ((21.5•1)/134)

= 29.000 mm



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Nozzle Calcs.: N4- Channel out Nozl: 12 9:36am Aug 12,2026

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{((D_i + t_{eff}) \cdot t_{eff})}))$$

$$= \min(12, (2 \cdot 217 + 11.3) / (\sqrt{((1216 + 29) \cdot 29)}))$$

$$= 2.347$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 13 \cdot 134 \cdot \max((2.35/5)^{0.85}, 1)$$

$$= 17.446 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 11.3 \cdot 78.5$$

$$= 8.851 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(200 \cdot 16, 134 \cdot 16)$$

$$= 21.472 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 17.4 + 0.92(8.85 + 0) + 0.5 + 0 + 0 + 1(21.5)$$

$$= 47.566 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 11.3 / \ln[1 + 11.3/217]$$

$$= 222.915 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 29 / \ln[1 + 29/608]$$

$$= 622.388 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 164 \cdot 1$$

$$= 246.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 1012 \cdot 223 \cdot 78.5$$

$$= 1805.6 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 1012 \cdot 622 (134 + 11.3)$$

$$= 9342.4 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1012 \cdot 622 \cdot 217$$

$$= 13956.9 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$



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$$= (9342 + 13957 + 1806) / 1012$$

$$= 2433.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap} / \text{AT} - \text{Rxs} / \text{teff})$$

$$= 246 / (2 \cdot 2433 / 47.6 - 622 / 29)$$

$$= 3043.0 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 164 [13/622]$$

$$= 3425.3 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{\text{max1}}, P_{\text{max2}})$$

$$= \min(3043, 3425)$$

$$= 3042.976 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (1806 + 9342 + 13957) / 47.6$$

$$= 51.759 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 1012 \cdot 622 / 29$$

$$= 21.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 51.8 - 21.7, 21.7)$$

$$= 81.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	246.00	N./mm ²
Local Primary Membrane Stress	PL	81.80	N./mm ²
Maximum Allowable Working Pressure	Pmax	3042.98	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (217 + 11.3) / 217$$

$$= 1.052 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (217 + 11.3)$$

$$= 359.084 \text{ mm}$$

Weld Length of Pad to Shell Weld [L_TP]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (217 + 11.3 + 200)$$

$$= 673.243 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 11.275, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 13.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(13957 \cdot 1.05, 1.5 \cdot 151(8.85+0), \pi/4 \cdot 1012 \cdot 217^2 \cdot 1.05^2)$$



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= 4235.060 Kgf

Discontinuity Force [fws]:

= fwelds • t • S / (t • S + te • Sp)

= 4235 • 13 • 164 / (13 • 164 + 16 • 164)

= 1898.475 Kgf

Discontinuity Force [fwp]:

= fwelds • te • Sp / (t • S + te • Sp)

= 4235 • 16 • 164 / (13 • 164 + 16 • 164)

= 2336.585 Kgf

Shear Stress [τ1]:

= fws / (Lt • (0.6 • tw1 + 0.49 • L43T))

= 1898 / (359 • (0.6 • 13 + 0.49 • 0))

= 6.647 N./mm²

Shear Stress [τ2]:

= fwp / (Lt • (0.6 • tw2 + 0.49 • L41T))

= 2337 / (359 • (0.6 • 16 + 0.49 • 7.07))

= 4.884 N./mm²

Shear Stress [τ3]:

= fwp / (LtP • (0.49 • L42T))

= 2337 / (673 • (0.49 • 8.49))

= 8.186 N./mm²

Maximum Shear Stress in the Welds:

= max(τ1, τ2, τ3)

= max(6.65, 4.88, 8.19)

= 8.2 must be less than or equal to 164.0 N./mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C

The computed stress ratio is : 0.144

The computed thickness ratio [sr]:

= (tr • E*) / (tn timer - ci - cext)

= (1.37 • 1) / (12.5 - 3 - 0)

= 0.144

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C

The computed stress ratio is : 0.144

The computed thickness ratio [sr]:

= (tr • E*) / (tn timer - ci - cext)

= (1.37 • 1) / (12.5 - 3 - 0)

= 0.144

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: N4- Channel out Nozl: 12 9:36am Aug 12,2026

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.144

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (1.37 \cdot 1) / (12.5 - 3 - 0)$$

$$= 0.144$$

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -23 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 6 °C
 Temperature reduction per Graph 3.12 : 29 °C
 The computed stress ratio is : 0.271

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.53 \cdot 1) / (16 - 3 - 0)$$

$$= 0.271$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.144

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (1.37 \cdot 1) / (12.5 - 3 - 0)$$

$$= 0.144$$

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -23 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -23 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 44.8509 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 310.8509 mm

Input Echo, WRC297 Item 1, Description: N4- Channel ou:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1210.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00 N./mm ²



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Nozzle Calcs.: N4- Channel out Nozl: 12 9:36am Aug 12,2026

Vessel Hot S.I. Allowable	Smh	164.00	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		428.650	mm
Nozzle Thickness		14.2748	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	161.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	151.00	N./mm ²
Diameter of Reinforcing Pad		857.200	mm
Thickness of Reinforcing Pad		16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1011.86	KPa.
Radial Load	P	-3441.77	Kgf
Circumferential Shear	Vc	2294.51	Kgf
Longitudinal Shear	Vl	2294.51	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{622.5(16.0 - 3.0)}$$

$$= 754.062 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1245.000 mm
Nozzle Outside Diameter	(d) =	457.200 mm
Vessel Thickness used	(T) =	29.000 mm
Nozzle Thickness used	(t) =	11.275 mm
T / t	=	2.572
d / t	=	40.551

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.406$$

Nr/P	=	0.060
Mr/P	=	0.069
M0/P	=	0.050
N0/P	=	0.171
MrD/Mc	=	0.159
NrDL/Mc	=	0.094
M0d/Mc	=	0.135
N0DL/Mc	=	0.229
MrD/Ml	=	0.085
NrDL/Ml	=	0.057
M0D/Ml	=	0.068
N0DL/Ml	=	0.323

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



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Nozzle Calcs.: N4- Channel out

Nozl: 12 9:36am Aug 12,2026

longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-23	-23	23	23
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	20	21	20	21

Outplane Stress Summary	-13	22	93	10
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-37	37	37	-37
Inplane Pressure Stress	10	10	10	10

Inplane Stress Summary	-12	29	71	-37
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	18	32	93	48

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-16	-16	16	16
Outplane Bending (Mc)	-59	59	59	-59
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	10	10	10	10

Outplane Stress Summary	-47	48	106	-37
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N4- Channel out Nozl: 12 9:36am Aug 12,2026

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-70	70	70	-70
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	20	21	20	21
Inplane Stress Summary	-37	70	116	-55

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	48	71	119	57

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-23	-23	23	23
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	24	22	24
Hoop Stress Summary	5	7	53	54

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	62	-62	62	-62
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-166	166	166	-166
Axial Pressure Stress	11	11	11	11
Axial Stress Summary	-107	101	259	-199

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10



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Shear Stress Summary	13	13	8	8
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	116	103	259	254

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-16	-16	16	16
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	24	22	24

Hoop Stress Summary	12	14	46	47
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	62	-62	62	-62
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-327	327	327	-327
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	-267	261	419	-359
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10

Shear Stress Summary	8	8	13	13
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	281	262	420	408

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{614.5(16.0 - 3.0)}$$

$$= 752.149 \text{ mm}$$

WRC 297 Curve Access Parameters:



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Vessel Mean Diameter (D) = 1229.000 mm
 Nozzle Outside Diameter (d) = 857.200 mm
 Vessel Thickness used (T) = 13.000 mm
 Nozzle Thickness used (t) = 211.275 mm
 T / t = 0.062
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.782$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
 Mr/P = 0.071
 M0/P = 0.019
 N0/P = 0.024
 MrD/Mc = 0.238
 NrDL/Mc = 0.092
 M0d/Mc = 0.067
 N0DL/Mc = 0.045
 MrD/Ml = 0.037
 NrDL/Ml = 0.035
 M0D/Ml = 0.012
 N0DL/Ml = 0.020

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	46	47	46	47
Outplane Stress Summary	56	40	92	19

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	84	-84	84	-84
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	67	-15	167	-89

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1



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Nozzle Calcs.: N4- Channel out

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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	68	56	167	108
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses Normal to
circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-78	78	78	-78
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23

Outplane Stress Summary	-36	75	137	-64
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses parallel to
circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	84	-84	84	-84
Inplane Membrane (Mc)	-18	-18	18	18
Inplane Bending (Mc)	-279	279	279	-279
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	46	47	46	47

Inplane Stress Summary	-156	233	438	-289
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Shear stress normal to
circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	156	233	438	289
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	21	20	21	20	21	20	21



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Nozzle Calcs.: N4- Channel out Nozl: 12 9:36am Aug 12,2026

Circ. P1 (SUS)	-17	-17	29	29	-4	-4	8	8
Circ. Q (SUS)	-17	17	41	-41	-54	54	86	-86
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. P1 (SUS)	-2	-2	6	6	-10	-10	22	22
Long. Q (SUS)	-21	21	53	-53	-47	47	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	20.0	21.0	20.0	21.0	20.0	21.0	20.0	21.0
Pm+P1 (SUS)	8.2	8.2	49.0	50.0	16.1	17.1	32.2	32.3
Pm+P1+Q (Total)	18.5	31.4	90.4	46.4	47.9	71.4	115.9	58.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	21.00	164.00	Passed
Pm+P1 (SUS)	50.03	246.00	Passed
Pm+P1+Q (TOTAL)	115.93	492.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	47	46	47	46	47	46	47
Circ. P1 (SUS)	1	1	7	7	-9	-9	27	27
Circ. Q (SUS)	9	-9	35	-35	-195	195	363	-363
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. P1 (SUS)	3	3	15	15	-4	-4	12	12
Long. Q (SUS)	41	-41	127	-127	-56	56	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	46.0	47.0	46.0	47.0	46.0	47.0	46.0	47.0
Pm+P1 (SUS)	47.0	48.0	53.1	54.1	37.1	38.1	73.0	74.0
Pm+P1+Q (Total)	67.8	54.3	165.0	108.0	158.0	233.0	436.0	289.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	47.00	164.00	Passed
Pm+P1 (SUS)	74.03	246.00	Passed
Pm+P1+Q (TOTAL)	436.03	492.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Nozzle Calcs.: N4- Channel out

Nozl: 12 9:36am Aug 12,2026

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	24	22	24	22	24	22	24
Circ. Pl (SUS)	-17	-17	29	29	-10	-10	22	22
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-14	-14	18	18	-14	-14	18	18
Long. Q (SUS)	-104	104	228	-228	-265	265	389	-389
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	22.0	24.0	22.0	24.0	22.0	24.0	22.0	24.0
Pm+Pl (SUS)	8.9	10.8	51.2	53.2	15.5	17.5	44.3	46.2
Pm+Pl+Q (Total)	114.5	102.5	257.3	252.5	280.5	262.3	418.4	406.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	151.00	Passed
Pm+Pl (SUS)	53.17	226.50	Passed
Pm+Pl+Q (TOTAL)	418.38	453.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12, 2026

Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	16000.00 KPa.
Shell Maximum Operating Pressure	Psox,max	16000.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	67.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1210.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	110.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	158.20 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1210.000 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	164.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1417
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13000.00 mm
Design Temperature of the Tubes		110.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		125.20 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	187.80 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	575.174 mm



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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12,2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	7296.3 cm ²
Perimeter of Tube Layout	Cp	3400.0 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	110.00 °C
Tubesheet Material		SA-350 LF1
Tubesheet Material UNS Number		K03009
Tubesheet Allowable Stress at Temperature	S	125.20 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	250.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1344.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



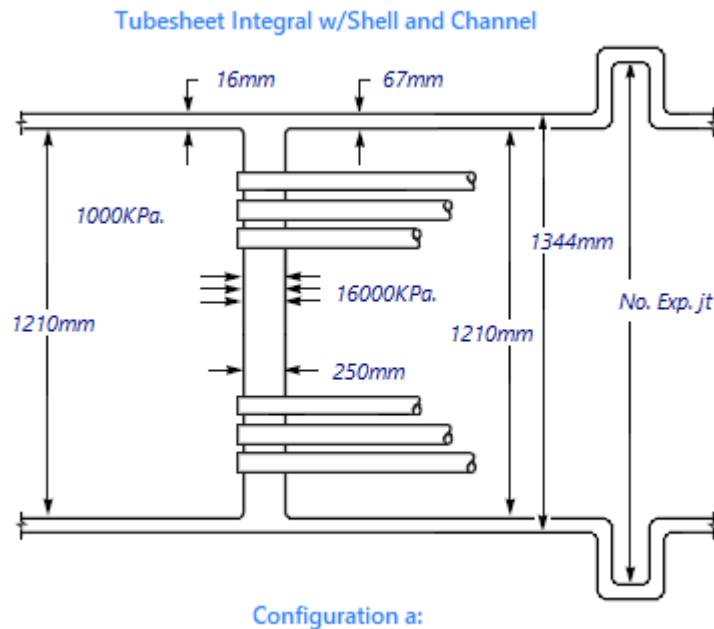
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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12, 2026

**Messages regarding the Tubesheet & Exchanger Input Data:**

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025**Elasticity/Expansion Material Properties:**

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	110.0 °C	0.19753E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12, 2026

Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(1000/(125 \cdot 1)) - 1) + 0$
 $= 0.056 + 0$
 $= 0.056 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 13000 / \sqrt{(9.5 \cdot 2.5)}$
 $= 2667.544$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.072 \cdot 197530928 (2.5/19)$
 $= 3009.648 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= F_{he} / F_y$
 $= 3009.6/187.8$
 $= 16.026$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 188/379$
 $= 0.495$

Ratio [AE]:

$= F_{he} / E$
 $= 3010/197530928$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.153$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 188 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15306604}$
 $= 487.243 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.303$



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ASME TS Calc: DESIGN CASE

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Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (379 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$$

$$= 737.130 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$$

$$= 0.351$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-1.946	0.362E-07	0.323E-07	0.984E-07	-0.813E-07	0.110E+08	0.195E-02
2	94	-4.768	0.781E-09	0.781E-09	0.200E-07	-0.199E-07	0.151E+09	0.146E-01
3	142	-3.357	0.736E-08	0.732E-08	0.507E-07	-0.503E-07	0.495E+08	0.124E-01
4	165	-2.652	0.173E-07	0.169E-07	0.723E-07	-0.695E-07	0.238E+08	0.826E-02
5	177	-2.299	0.253E-07	0.240E-07	0.848E-07	-0.777E-07	0.163E+08	0.431E-02
6	183	-2.122	0.304E-07	0.281E-07	0.914E-07	-0.804E-07	0.134E+08	0.155E-02
7	186	-2.034	0.332E-07	0.302E-07	0.949E-07	-0.811E-07	0.122E+08	0.976E-04
8	185	-2.078	0.317E-07	0.291E-07	0.932E-07	-0.808E-07	0.128E+08	0.750E-03
9	185	-2.056	0.324E-07	0.297E-07	0.940E-07	-0.810E-07	0.125E+08	0.332E-03
10	186	-2.045	0.328E-07	0.299E-07	0.944E-07	-0.811E-07	0.123E+08	0.119E-03
11	186	-2.040	0.330E-07	0.300E-07	0.947E-07	-0.811E-07	0.122E+08	0.110E-04
12	186	-2.037	0.331E-07	0.301E-07	0.948E-07	-0.811E-07	0.122E+08	0.432E-04
13	186	-2.038	0.330E-07	0.301E-07	0.947E-07	-0.811E-07	0.122E+08	0.161E-04
14	186	-2.039	0.330E-07	0.301E-07	0.947E-07	-0.811E-07	0.122E+08	0.252E-05
15	186	-2.039	0.330E-07	0.301E-07	0.947E-07	-0.811E-07	0.122E+08	0.425E-05
16	186	-2.039	0.330E-07	0.301E-07	0.947E-07	-0.811E-07	0.122E+08	0.864E-06

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 186 / 188$$

$$= 1.671$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 186 / 1.67$$

$$= 111.586 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 111580 (2.5 / 19)$$

$$= 29363.088 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.6314 mm
Allowable Internal Pressure at Corroded thickness	38231.12 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	29363.09 KPa.
Required External Design Pressure	16000.00 KPa.
Required Thickness due to Shell Side pressure	1.6314 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:



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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12, 2026

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/250$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter $[d^*]$:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 2.5 (197530928/197530928) \cdot (125/125) \cdot (0); 19 - 2 \cdot 2.5)$$

$$= 19.0000 \text{ mm}$$

Equivalent outer tube limit circle diameter $[Do]$:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 575 + 19$$

$$= 1169.348 \text{ mm}$$

Determine the basic ligament efficiency for shear $[\mu]$:

$$= (p - dt) / p$$

$$= (25 - 19) / 25$$

$$= 0.240$$

Equivalent outer tube limit radius $[ao]$:

$$= Do / 2$$

$$= 1169/2$$

$$= 584.6740 \text{ mm}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 25 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1169 \cdot 25) / (\pi \cdot 1169^2) }$$

$$= 25.0000 \text{ mm}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (25 - 19) / 25$$

$$= 0.2400$$

Ratio $[\rho_s]$:

$$= a_s / a_o$$

$$= 605/585$$

$$= 1.0348$$

Ratio $[\rho_c]$:

$$= a_c / a_o$$

$$= 605/585$$

$$= 1.0348$$

Compute parameter $[xt]$:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 1417 ((19 - 2 \cdot 2.5) / (2 \cdot 585))^2$$

$$= 0.7969$$

Determine parameter $[xs]$:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 1417 (19 / (2 \cdot 585))^2$$

$$= 0.6259$$

Determine the value $[h'g]$:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:



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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12, 2026

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / L$$

$$= \pi \cdot 67 (1210 + 67) 197530928 / 13000$$

$$= 416489.2500 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 2.5 (19 - 2.5) 197530928 / 13000$$

$$= 200.7992 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 416489 / (1417 \cdot 201)$$

$$= 1.46377$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 416489/200)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1210 + 67)67)$$

$$= 0.0062 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.0062 \cdot 197530928 \cdot 67^3 / (6(1 - 0.3^2))$$

$$= 67622252544.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1210 + 67) \cdot 67.62E+09 / 250^3 \cdot (1 + 250 \cdot 0.0062 + (250 \cdot 0.0062)^2 / 2)$$

$$= 124702408.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1210^2 / (4 \cdot 197530928 \cdot 67) \cdot (1 - 1210 \cdot 0.3 / (2(1210 + 67)))$$

$$= 0.23725E-01 \text{ mm/N./mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc)tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1210 + 16)16)$$

$$= 0.0130 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.013 \cdot 200187728 \cdot 16^3 / (6(1 - 0.3^2))$$

$$= 1949194624.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1210 + 16 / \cos(0)) \cdot 1949194624 / 250^3 \cdot (1 + 250 \cdot 0.013 \cdot (1 + 1) / 2 + (250 \cdot 0.013)^2 / 2 \cdot 1)$$

$$= 8726153.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= Dc^2 / (4 \cdot Ec \cdot Tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc)))$$

$$= 1210^2 / (4 \cdot 200187728 \cdot 16) \cdot (1 - 1210 \cdot 0.3 / (2(1210 + 16)))$$

$$= 0.97352E-01 \text{ mm/N./mm}^2$$



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Channel coefficient [zetac]:

$$\text{zetac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 10.000000$; $\mu^* = 0.240000$; $E^* = 40302092$ KPa.

$E^*/E = 0.204029$; $\nu^* = 0.407183$

Note: As h/p (10.000) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$\begin{aligned} &= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4} \\ &= \left(\left(24(1 - 0.41^2)1417 \cdot 197530928 \cdot 2.5 \right. \right. \\ &\quad \left. \left. (19 - 2.5)585^2 \right) / (40302092 \cdot 13000 \cdot 250^3) \right)^{1/4} \\ &= 1.7626 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.256793; & Z_v &= 0.162234; & Z_m &= 0.664956 \\ Z_a &= 0.696172E+00; & Z_w &= 0.162234 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 1344/1169 \\ &= 1.1494 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.41) / 40302092 \cdot (124702408 + 8726153 + \\ &\quad 197530928 \cdot \ln(1.15)) \\ &= 2.3671 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu^*)F \\ &= (1 + 0.41)2.37 \\ &= 3.3310 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.03 - 1 - 3.33 \cdot 0.16) / (1 + 3.33 \cdot 0.66) \\ &= -0.157276 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4 \\ &= (0.26 + -0.16 \cdot 0.16) / 2 \cdot 1.76^4 \\ &= 1.1162 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4 \\ &= (0.16 + -0.16 \cdot 0.66) / 2 \cdot 1.76^4 \\ &= 0.2783 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\psi_s - 1)Z_m)Xa^4 / (1 + \phi \cdot Z_m) \\ &= (0.16 + (1.03 - 1)0.66)1.76^4 / (1 + 3.33 \cdot 0.66) \\ &= 0.5565 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$



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Calculate parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ = 1.03 \cdot 67.62E+09 \cdot 0.0062 \cdot 0.023724504 (1 + 250 \cdot 0.0062) \\ = 26347.7500 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= a_o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s \\ = 585^2 (1.03^2 - 1) \cdot (1.03 - 1) / 4 - 26348 \\ = -26137.5879 \text{ mm}^2$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)] \\ = 1.03 \cdot 1949194624 [0.097352259 (1 + 0.013 \cdot 1 \cdot 250) - \\ 0.000 (1/250 + 1 \cdot 0.013/2)] \\ = 10818.7617 \text{ mm}^2$$

Parameter [ω*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c \\ = 585^2 [(1.03^2 + 1) \cdot (1.03 - 1) / 4 - (1.03 - 1) / 2] - 10819 \\ = -10608.5986 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - \\ [(\rho_s^2 - 1) / (J \cdot K_{st})] - [(1 - J) / (2J \cdot K_{st})] [(D_j^2 - (D_s)^2) / D_o^2] \} \\ = 16000 \{ 0.63 + 2(1 - 0.63) 0.3 + [2/1.46 \cdot \\ (1210/1169)^2] 0.3 - [(1.03^2 - 1) / (1 \cdot 1.46)] - \\ [(1 - 1) / (2 \cdot 1 \cdot 1.46)] [(0^2 - (1210)^2) / 1169^2] \} \\ = 19854.8867 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t \\ = [0.8 + 2(1 - 0.8) 0.3 + 1/(1 \cdot 1.46)] 1000 \\ = 1601.9235 \text{ KPa.}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2) \\ = 1417 \cdot 201 \cdot 0 / (3.14 \cdot 585^2) \\ = 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2) \\ = -0 \cdot 0.56 \cdot 0 / (2 \cdot \pi \cdot 585^2) \\ = -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t) \\ = -(0.56/585^2) \cdot (-26138 \cdot 16000 - -10609 \cdot 1000) \\ = 663.5446 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P^* s - \omega_c \cdot P^* c) \\ = (0.56/585^2) \cdot (26348 \cdot 0 - 10819 \cdot 0) \\ = 0.0000 \text{ KPa.}$$

Determine the effective pressure [P_e]:



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$$\begin{aligned}
 &= J \cdot Kst / ((1 + J \cdot Kst(Qz1 + (ps - 1) Qz2)) \cdot \\
 &\quad (P's - P't + P_y + P_w + P_w + Prim)) \\
 &= 0.1000E+01 \cdot 1.46 / ((1 + 1 \cdot 1.46 (1.12 + (1.03 - \\
 &\quad 1) 0.28)) \cdot (19855 - 1602 + 0 + 0 + -0 + 664)) \\
 &= 10456.4902 \text{ KPa.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned}
 &= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + \\
 &\quad W \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\
 &= [((-26138 \cdot 16000 - -10609 \cdot 1000) - \\
 &\quad (26348 \cdot 0 - 10819 \cdot 0)) \cdot 0.0001 + \\
 &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 3.33 \cdot 0.66) \\
 &= -12928.5615 \text{ Kgf}
 \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2) \\
 &= -0.16 + 2 \cdot -12929 / (10456 \cdot 585^2) \\
 &= -0.22821
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.114106

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.11 / 0.24) \cdot (2 \cdot 585 / (250 - 0))^2 \cdot 10456 \\
 &= 163.1576 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 125 = 188 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.11 / 0.24) \cdot (2 \cdot 585 / (231 - 0))^2 \cdot 10358 \\
 &= 187.7916 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 231 + 0 + 0 \\
 &= 230.7237 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [Tlimit]:

$$\begin{aligned}
 &= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533Sy) \\
 &= (2 \cdot 0.24 \cdot 250 / 585) \cdot 100 \\
 &= 20.5440 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress - final thickness [τ_f]:

$$\begin{aligned}
 &= (A_p \cdot Pe) / (\mu \cdot h \cdot C_p) \\
 &= (7296 \cdot 6645) / (0.24 \cdot 60 \cdot 3400) \\
 &= 99.0618 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$\begin{aligned}
 &= H + Cats + Catc \\
 &= 60 + 0 + 0 \\
 &= 59.9796 \text{ mm}
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(231; 60) \\
 &= 230.7237 \text{ mm}
 \end{aligned}$$



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ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = 0.89, F_{tmax} = 1.06$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\ &= ((16000 \cdot 0.63 - 1000 \cdot 0.8) - (10456) \cdot 0.89) / \\ &\quad (0.8 - 0.63) \\ &= -0.6561 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\ &= ((16000 \cdot 0.63 - 1000 \cdot 0.8) - (10456) \cdot 1.06) / \\ &\quad (0.8 - 0.63) \\ &= -10.9577 \text{ N./mm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 11 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 125.1985 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.26 + -0.23 \cdot 0.16)1.76^4); 1.25) \\ &= 2.0000 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4 \\ &= (\sqrt{ (19^2 + (19 - 2 \cdot 2.5)^2) }) / 4 \\ &= 5.9002 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot E / S_{yt}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 197530928 / 188) } \\ &= 144.0953 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 0 \cdot 0 / 5.9 \\ &= 0.0000 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\ &= (188 / 2) \cdot (1 - 0 / (2 \cdot 144)) \\ &= 93.899 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)} \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 11 \cdot 1.3 \\ &= 144.80 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19 - 2.5) 2.5 \\ &= 1.296 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:



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$$\begin{aligned}
 &= \pi \cdot t (d_o - t) S \\
 &= \pi \cdot 2.5 (19 - 2.5) 125 \\
 &= 1654.415 \text{ Kgf}
 \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 &= 0.85 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 125 \\
 &= 1386.8209 \text{ Kgf}
 \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 125/125 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned}
 &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\
 &= 0.55 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 125 \\
 &= 897.3547 \text{ Kgf}
 \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned}
 &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\
 &= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.5 \cdot (19 - 2.5) 1 \cdot 1)} - 0.75 \cdot 19 \right) \\
 &= 2.945 \text{ mm}
 \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.5000 mm

Minimum Required Groove Weld Leg [agr]: 1.5000 mm

Tube-Tubesheet Jt allowable, 1654 is $\geq$ tube strength 1654 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 158 \cdot 1 = 158 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$\begin{aligned}
 &= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s) \\
 &= 585^2 / ((1210 + 67) 67) \cdot [10456 + (1.03^2 - 1) \\
 &\quad (16000 - 1000)] + 605^2 \cdot 1000 / ((1210 + 67) 67) \\
 &= 50.2984 \text{ N./mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
For a given Shell thickness of 67 mm, the minimum Shell length adjacent to the tubesheet should be 512.510 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$\begin{aligned}
 &= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s) \\
 &= 585^2 / ((1210 + 67) 67) [10456 + (1.03^2 - 1) \\
 &\quad (16000 - 1000)] + 605^2 \cdot 1000 / ((1210 + 67) 67) \\
 &= 50.2984 \text{ N./mm}^2
 \end{aligned}$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$\begin{aligned}
 &= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + \\
 &\quad 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \} \\
 &= 6 \cdot 67.62 \text{E}+09 / 67^2 \{ 0.0062 [0.023724504 \cdot 16000 + 605^2 \\
 &\quad \cdot 0 / (13.23 \text{E}+09)] + \\
 &\quad 6(1 - 0.41^2) / (40302092) (585/250)^3 (1 + 250 \cdot
 \end{aligned}$$



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$$0.0062/2) [10456 (0.16 + 0.66 \cdot -0.16) + 2/585^2 \cdot 0.66 \cdot -12929] \}$$

$$= 241.2151 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$$

$$\therefore |50.3| + |241| \leq 237 \text{ N./mm}^2$$

$$\therefore 292, \text{ must be } \leq 237 \text{ N./mm}^2 \text{ (See Elastic-Plastic iteration below)}$$

Note:

For a given Channel thickness of 16 mm, the minimum Channel length adjacent to the tubesheet should be 250.452 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_{c,m}]$:

$$= a c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$$

$$= 605^2 \cdot 1000 / ((1210 + 16) 16 \cdot \cos(0))$$

$$= 18.661 \text{ N./mm}^2$$

Channel Bending Stress due to Joint Interaction $[\sigma_{c,b}]$:

$$= 6 \cdot k c / t c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6 ((1 - \nu^*) / E^*) \cdot (a o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 1949194624 / 16^2 \cdot \{ (0.097352259 \cdot 1 \cdot 0.013 - 0.000 \cdot 1) \cdot 1000 - 6 ((1 - 0.41^2) / 40302092) \cdot (585 / 250)^3 \cdot (1 + 250 \cdot 0.013 / 2 \cdot 1) \cdot [10456 (0.16 + 0.66 \cdot -0.16) + 2 / 585^2 \cdot 0.66 \cdot -12929] \}$$

$$= 36.8675 \text{ N./mm}^2$$

Channel Stress Summation vs. Allowable

$$\therefore |\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c$$

$$\therefore |18.7| + |36.9| \leq 246 \text{ N./mm}^2$$

$$\therefore 55.5 \text{ must be } \leq 246 \text{ N./mm}^2$$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	163.2 <=	187.8 N./mm ²	Pass
Tubesheet Shear Stress	37.4 <=	100.1 N./mm ²	Pass
Maximum Tube Stress	11.0 <=	125.2 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-11.0 <=	-93.9 N./mm ²	Pass
Maximum Force on any one Tube	144.8 <=	1654.4 Kg _f	Pass
Axial Membrane Stress in Shell	50.3 <=	158.2 N./mm ²	Pass
Shell Stress (jt. inter.)	291.5 <=	237.3 N./mm ²	See below
Channel Stress (jt. inter.)	55.5 <=	246.0 N./mm ²	Pass

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable $[S_{stars}]$:

$$= \min(Sys, SPS, s/2)$$

$$= \min(237, 475/2) = 237 \text{ N./mm}^2$$

Compute Allowable $[S^*c]$:

$$= \min(Syc, SPS, c/2)$$

$$= \min(246, 492/2) = 246 \text{ N./mm}^2$$

Compute Factor $[facts]$:

$$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_{s,b}) / S_{stars}], 1)$$



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$$= \min([1.4 - 0.4 \cdot \text{abs}(241)/237], 1) = 0.99$$

Compute Factor [factc]:

$$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_c, b) / S_{\text{starc}}], 1)$$

$$= \min([1.4 - 0.4 \cdot \text{abs}(36.9)/246], 1) = 1$$

Compute Modified Modulus [Es]:

$$= E_s \cdot \text{factc}$$

$$= 197530928 \cdot 0.99 = 196264032 \text{ KPa.}$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.11/0.24) \cdot (2 \cdot 585 / (231 - 0))^2 \cdot 10357$$

$$= 187.7915 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + \text{Cats} + \text{Catc}$$

$$= 231 + 0 + 0$$

$$= 230.6850 \text{ mm}$$

For the Elastic-Plastic iteration, Shell Allowable = SPS,s

$$\text{ShellAllow} = 474.8170 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 12:

For the Elastic-Platic iteration, the channel allowable stress = SPS,c

$$\text{Challow} = 492.0227 \text{ N./mm}^2$$

[Computations completed for ASME tubesheet configuration: a](#)

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	163.1 <=	187.8 N./mm ²	Pass
Tubesheet Shear Stress	37.4 <=	100.1 N./mm ²	Pass
Maximum Tube Stress	11.0 <=	125.2 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-11.0 <=	-93.9 N./mm ²	Pass
Maximum Force on any one Tube	144.8 <=	1654.4 Kg _f	Pass
Axial Membrane Stress in Shell	50.3 <=	158.2 N./mm ²	Pass
Shell Stress (jt. inter.)	291.5 <=	474.8 N./mm ²	Pass
Channel Stress (jt. inter.)	55.5 <=	492.0 N./mm ²	Pass

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	24.214 ...	14	188	3 100	Ps+Pt-Th	D1 Ok
D2uc	239.544 ...	174	188	41 100	Ps+Pt-Th	D2 Ok
D3uc	230.724 ...	163	188	37 100	Ps+Pt-Th	D3 Ok
O1uc	8.690 ...	14	376	3 100	Ps+Pt+Th	O1 Ok
O2uc	159.876 ...	174	376	41 100	Ps+Pt+Th	O2 Ok
O3uc	152.755 ...	163	376	37 100	Ps+Pt+Th	O3 Ok
O4uc	3.810 ...	...	376	... 100	Ps+Pt+Th	O4 Ok
D1c	28.688 ...	14	188	3 100	Ps+Pt-Th-c	D1 Ok
D2c	242.277 ...	177	188	41 100	Ps+Pt-Th-c	D2 Ok
D3c	233.714 ...	166	188	38 100	Ps+Pt-Th-c	D3 Ok
O1c	13.246 ...	14	376	3 100	Ps+Pt+Th-c	O1 Ok
O2c	163.175 ...	177	376	41 100	Ps+Pt+Th-c	O2 Ok
O3c	156.212 ...	167	376	38 100	Ps+Pt+Th-c	O3 Ok
O4c	8.310 ...	...	376	... 100	Ps+Pt+Th-c	O4 Ok



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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12, 2026

Max: 242.2773 ... mm 0.943 0.411 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures

derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	...	158	...	...	:	...	...	...	...	Ok
D2uc	50	158	...	...	:	...	...	...	...	Ok
D3uc	50	158	...	...	:	...	...	...	...	Ok
O1uc	...	474	...	...	:	...	...	...	...	Ok
O2uc	50	474	...	...	:	...	...	...	...	Ok
O3uc	50	474	...	...	:	...	...	...	...	Ok
O4uc	...	474	...	...	:	...	...	...	...	Ok
D1c	...	158	...	...	:	...	...	...	...	Ok
D2c	51	158	...	...	:	...	...	...	...	Ok
D3c	51	158	...	...	:	...	...	...	...	Ok
O1c	...	474	...	...	:	...	...	...	...	Ok
O2c	51	474	...	...	:	...	...	...	...	Ok
O3c	51	474	...	...	:	...	...	...	...	Ok
O4c	...	474	...	...	:	...	...	...	...	Ok
Max RATIO	0.325	...	...	...	:	...	...	...	...	...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress	Stress Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd Fail
D1uc	1	125	...	-93	14	1654	10	237	84	245 Ok
D2uc	12	125	-12	-93	159	1654	301	474	28	492 Ok
D3uc	11	125	-11	-93	145	1654	292	474	56	492 Ok
O1uc	1	250	...	-93	14	3308	10	474	84	492 Ok
O2uc	12	250	-12	-93	159	3308	301	474	28	492 Ok
O3uc	11	250	-11	-93	145	3308	292	474	56	492 Ok
O4uc	...	250	...	...	...	3308	...	474	...	492 Ok
D1c	1	125	...	-93	15	1654	12	237	103	245 Ok
D2c	12	125	-12	-93	158	1654	313	474	31	492 Ok
D3c	11	125	-11	-93	143	1654	303	474	71	492 Ok
O1c	1	250	...	-93	15	3308	12	474	103	492 Ok
O2c	12	250	-12	-93	158	3308	313	474	31	492 Ok



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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12, 2026

O3c	11	250	-11	-93	143	3308	303	474	71	492	Ok
O4c	...	250	...	...	...	3308	...	474	...	492	Ok

Max RATIO 0.096 0.128 0.096 0.660 0.417

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	242.277	250.000	Ok
Tube thickness :	1.631	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (67.000) adj. to tubesheet: 512.510 mm

Min Channel length of thk, (16.000) adj. to tubesheet: 250.452 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -1.104 mm
 Due to Pressure + Thermal Shell Compresses by : -1.104 mm

Tubesheet MAWP used to Compute Hydrottest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	13234.64	1.000	17003.64	1.000
Tubesheet Shear Stress	30868.32	1.000	38970.20	1.000
Tube Tensile Stress	112351.43	1.000	166212.52	1.000
Tube Compressive Stress	26469.29	0.077	124660.07	1.000
Tube-Tubesheet Joint load	112354.66	1.000	166213.42	1.000
Shell Stress (Axial, Junction)	5861.06	0.142	17003.64	1.000
Tubesheet-Channel Junction	4792.96	1.000	17003.64	1.000
Tube Pressure Stress	38231.11	1.000	29363.08	1.000
Minimum MAWP	4792.96		17003.64	

Tubesheet MAPnc used to Compute Hydrottest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	15294.78	1.000	19023.83	0.997
Tubesheet Shear Stress	34130.32	1.000	43057.11	1.000
Tube Tensile Stress	126124.51	1.000	183203.08	1.000
Tube Compressive Stress	551236.81	1.000	136074.80	1.000
Tube-Tubesheet Joint load	126124.44	1.000	183203.16	1.000
Shell Stress (Axial, Junction)	15294.78	0.300	19023.83	0.997
Tubesheet-Channel Junction	6248.46	0.996	19023.83	0.997
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	6248.46		19023.83	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded



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ASME TS Calc: DESIGN CASE

Case: 1 9:36a Aug 12, 2026

1	0.0130	0.0130	...	...	...	...
2	-1.1034	-1.1037	...	...	...	...
3	-1.0904	-1.0906	...	...	...	...
4	...	...	...	...	...	...
5	0.0130	0.0130	...	...	...	...
6	-1.1034	-1.1037	...	...	...	...
7	-1.0904	-1.0906	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(250/4, \min(250, 67)) \\
 &= 67.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio:

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(16000/19024, 1) \\
 &= 0.841
 \end{aligned}$$

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 6 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 14 °C
 Temperature reduction per Graph 3.13 : 8 °C
 The computed stress ratio is : 0.841

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(250/4, \min(250, 16)) \\
 &= 62.500 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio:

$$\begin{aligned}
 &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\
 &= \min(1000/6248, 1) \\
 &= 0.160
 \end{aligned}$$

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 The Curve for this Material is : B
 As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.160

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ASME TS Calc: 100% SOR

Case: 2 9:36a Aug 12, 2026

Input Echo, Tubesheet Number 2, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	13260.00 KPa.
Shell Maximum Operating Pressure	Psox,max	13260.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	67.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1210.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	110.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	158.20 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	450.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	450.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1210.000 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	164.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1417
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13000.00 mm
Design Temperature of the Tubes		110.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		125.20 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	187.80 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	575.174 mm



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ASME TS Calc: 100% SOR

Case: 2 9:36a Aug 12,2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	7296.3 cm ²
Perimeter of Tube Layout	Cp	3400.0 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	110.00 °C
Tubesheet Material		SA-350 LF1
Tubesheet Material UNS Number		K03009
Tubesheet Allowable Stress at Temperature	S	125.20 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	250.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1344.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	62.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	44.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



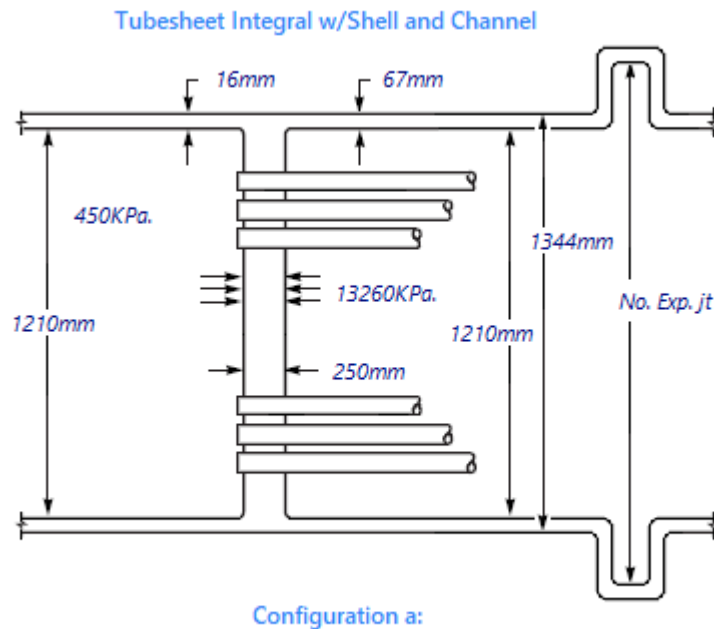
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ASME TS Calc: 100% SOR

Case: 2 9:36a Aug 12,2026

**Messages regarding the Tubesheet & Exchanger Input Data:**

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025**Elasticity/Expansion Material Properties:**

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	62.0 °C	0.0000118560 /°C
Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	62.0 °C	0.20036E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	44.0 °C	0.0000117393 /°C
Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Elastic Mod. at Metal Temp. along Len	44.0 °C	0.20139E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	110.0 °C	0.19753E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%



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ASME TS Calc: 100% SOR

Case: 2 9:36a Aug 12, 2026

Elastic Mod. at Design Temperature	110.0 °C	0.19615E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(450/(125 \cdot 1)) - 1) + 0$
 $= 0.025 + 0$
 $= 0.025 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 13000 / \sqrt{(9.5 \cdot 2.5)}$
 $= 2667.544$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.072 \cdot 197530928 (2.5/19)$
 $= 3009.648 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3009.6/187.8$
 $= 16.026$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 188/379$
 $= 0.495$

Ratio [AE]:

$= Fhe / E$
 $= 3010/197530928$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.153$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 188 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15306604}$
 $= 487.243 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.303$



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ASME TS Calc: 100% SOR

Case: 2 9:36a Aug 12, 2026

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (379 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$
 $= 737.130 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$
 $= 0.351$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-1.946	0.362E-07	0.323E-07	0.984E-07	-0.813E-07	0.110E+08	0.195E-02
2	94	-4.768	0.781E-09	0.781E-09	0.200E-07	-0.199E-07	0.151E+09	0.146E-01
3	142	-3.357	0.736E-08	0.732E-08	0.507E-07	-0.503E-07	0.495E+08	0.124E-01
4	165	-2.652	0.173E-07	0.169E-07	0.723E-07	-0.695E-07	0.238E+08	0.826E-02
5	177	-2.299	0.253E-07	0.240E-07	0.848E-07	-0.777E-07	0.163E+08	0.431E-02
6	183	-2.122	0.304E-07	0.281E-07	0.914E-07	-0.804E-07	0.134E+08	0.155E-02
7	186	-2.034	0.332E-07	0.302E-07	0.949E-07	-0.811E-07	0.122E+08	0.976E-04
8	185	-2.078	0.317E-07	0.291E-07	0.932E-07	-0.808E-07	0.128E+08	0.750E-03
9	185	-2.056	0.324E-07	0.297E-07	0.940E-07	-0.810E-07	0.125E+08	0.332E-03
10	186	-2.045	0.328E-07	0.299E-07	0.944E-07	-0.811E-07	0.123E+08	0.119E-03
11	186	-2.040	0.330E-07	0.300E-07	0.947E-07	-0.811E-07	0.122E+08	0.110E-04
12	186	-2.037	0.331E-07	0.301E-07	0.948E-07	-0.811E-07	0.122E+08	0.432E-04
13	186	-2.038	0.330E-07	0.301E-07	0.947E-07	-0.811E-07	0.122E+08	0.161E-04
14	186	-2.039	0.330E-07	0.301E-07	0.947E-07	-0.811E-07	0.122E+08	0.252E-05
15	186	-2.039	0.330E-07	0.301E-07	0.947E-07	-0.811E-07	0.122E+08	0.425E-05
16	186	-2.039	0.330E-07	0.301E-07	0.947E-07	-0.811E-07	0.122E+08	0.864E-06

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 186/188$
 $= 1.671$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 186/1.67$
 $= 111.586 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 111580 (2.5/19)$
 $= 29363.088 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4331 mm
Allowable Internal Pressure at Corroded thickness	38231.12 KPa.
Required Internal Design Pressure	450.00 KPa.
Allowable External Pressure at Corroded thickness	29363.09 KPa.
Required External Design Pressure	13260.00 KPa.
Required Thickness due to Shell Side pressure	1.4331 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	7.354	...	6	188	1	100	Ps+Pt-Th	D1	Ok
D2uc	215.391	...	144	188	34	100	Ps+Pt-Th	D2	Ok



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ASME TS Calc: 100% SOR

Case: 2 9:36a Aug 12, 2026

D3uc	210.973	...	139	188	32	100	Ps+Pt-Th	D3	Ok
O1uc	68.104	...	72	376	16	100	Ps+Pt+Th	O1	Ok
O2uc	97.815	...	101	376	19	100	Ps+Pt+Th	O2	Ok
O3uc	91.688	...	96	376	18	100	Ps+Pt+Th	O3	Ok
O4uc	61.501	...	66	376	14	100	Ps+Pt+Th	O4	Ok
D1c	11.909	...	6	188	1	100	Ps+Pt-Th-c	D1	Ok
D2c	218.293	...	147	188	34	100	Ps+Pt-Th-c	D2	Ok
D3c	213.997	...	142	188	33	100	Ps+Pt-Th-c	D3	Ok
O1c	71.516	...	74	376	16	100	Ps+Pt+Th-c	O1	Ok
O2c	102.328	...	103	376	20	100	Ps+Pt+Th-c	O2	Ok
O3c	96.370	...	98	376	18	100	Ps+Pt+Th-c	O3	Ok
O4c	65.058	...	68	376	14	100	Ps+Pt+Th-c	O4	Ok

Max: 218.2925 ... mm 0.783 0.340 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					---- Shell Band Stress ----					Pass
Case#	Ten	Allwd	Cmp	Allwd	Ten	Allwd	Cmp	Allwd	Fail	
D1uc	...	158	...	...	...	...	...	...	...	Ok
D2uc	41	158	...	...	...	...	...	...	...	Ok
D3uc	42	158	...	...	...	...	...	...	...	Ok
O1uc	16	474	-16	-157	...	...	...	...	...	Ok
O2uc	25	474	...	...	...	...	...	...	...	Ok
O3uc	25	474	...	...	...	...	...	...	...	Ok
O4uc	16	474	-16	-157	...	...	...	...	...	Ok
D1c	...	158	...	...	...	...	...	...	...	Ok
D2c	42	158	...	...	...	...	...	...	...	Ok
D3c	42	158	...	...	...	...	...	...	...	Ok
O1c	16	474	-16	-157	...	...	...	...	...	Ok
O2c	26	474	...	...	...	...	...	...	...	Ok
O3c	26	474	...	...	...	...	...	...	...	Ok
O4c	16	474	-16	-157	...	...	...	...	...	Ok
Max RATIO	0.269		0.103		...			...		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail



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ASME TS Calc: 100% SOR

Case: 2 9:36a Aug 12, 2026

D1uc	...	125	...	-93	7	1654	5	237	38	245	Ok
D2uc	10	125	-10	-93	132	1654	249	474	23	245	Ok
D3uc	10	125	-10	-93	126	1654	245	474	14	245	Ok
O1uc	27	250	...	...	355	3308	80	474	82	492	Ok
O2uc	19	250	...	...	257	3308	174	474	21	492	Ok
O3uc	19	250	...	...	255	3308	170	474	59	492	Ok
O4uc	26	250	...	...	348	3308	76	474	44	492	Ok
D1c	1	125	...	-93	7	1654	5	237	46	245	Ok
D2c	10	125	-10	-93	131	1654	260	474	26	245	Ok
D3c	9	125	-9	-93	125	1654	255	474	20	245	Ok
O1c	27	250	...	...	353	3308	83	474	91	492	Ok
O2c	20	250	...	...	259	3308	181	474	19	492	Ok
O3c	20	250	...	...	258	3308	176	474	65	492	Ok
O4c	26	250	...	...	346	3308	79	474	45	492	Ok

Max RATIO 0.107 0.106 0.107 0.547 0.188

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	218.293	250.000	Ok
Tube thickness :	1.433	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (67.000) adj. to tubesheet: 512.510 mm

Min Channel length of thk, (16.000) adj. to tubesheet: 250.452 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -2.810 mm

Due to Pressure Shell Compresses by : -0.915 mm

Due to Pressure + Thermal Shell Compresses by : -3.724 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	13260.30	1.000	17004.01	1.000
Tubesheet Shear Stress	26405.76	1.000	38978.43	1.000
Tube Tensile Stress	112264.59	1.000	166091.56	1.000
Tube Compressive Stress	338806.44	1.000	124568.11	1.000
Tube-Tubesheet Joint load	112262.72	1.000	166090.81	1.000
Shell Stress (Axial, Junction)	5342.65	0.286	17004.01	1.000
Tubesheet-Channel Junction	4354.67	1.000	17004.01	1.000
Tube Pressure Stress	38231.11	1.000	29363.08	1.000
Minimum MAWP	4354.67		17004.01	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	15322.20	1.000	19022.74	0.997
Tubesheet Shear Stress	34139.02	1.000	43065.52	1.000
Tube Tensile Stress	126029.26	1.000	183071.36	1.000
Tube Compressive Stress	536900.75	1.000	135976.83	1.000
Tube-Tubesheet Joint load	126029.21	1.000	183071.44	1.000
Shell Stress (Axial, Junction)	15322.20	0.302	19022.74	0.997



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ASME TS Calc: 100% SOR

Case: 2 9:36a Aug 12,2026

Tubesheet-Channel Junction	6247.28	0.996	19022.74	0.997
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	6247.28		19022.74	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0059	0.0059	...	...	...	...
2	-0.9145	-0.9147	...	...	...	...
3	-0.9086	-0.9088	...	...	...	...
4	...	...	...	...	...	...
5	0.0059	0.0059	-2.8090	-2.8096	...	...
6	-0.9145	-0.9147	-2.8090	-2.8096	...	...
7	-0.9086	-0.9088	-2.8090	-2.8096	...	...
8	...	...	-2.8090	-2.8096	...	...

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ASME TS Calc: 110% EOR

Case: 3 9:36a Aug 12, 2026

Input Echo, Tubesheet Number 3, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	13460.00 KPa.
Shell Maximum Operating Pressure	Psox,max	13460.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	67.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1210.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	110.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	158.20 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	450.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	450.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1210.000 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	164.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1417
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13000.00 mm
Design Temperature of the Tubes		65.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		130.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	195.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	575.174 mm



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ASME TS Calc: 110% EOR

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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	7296.3 cm ²
Perimeter of Tube Layout	Cp	3400.0 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	65.00 °C
Tubesheet Material		SA-350 LF1
Tubesheet Material UNS Number		K03009
Tubesheet Allowable Stress at Temperature	S	130.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	250.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1344.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	58.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	47.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



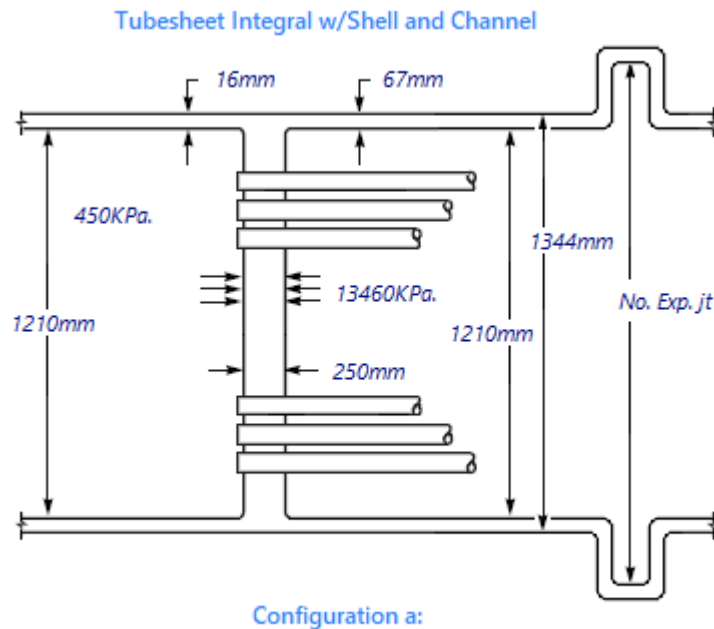
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ASME TS Calc: 110% EOR

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**Messages regarding the Tubesheet & Exchanger Input Data:**

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025**Elasticity/Expansion Material Properties:**

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	58.0 °C	0.0000118301 /°C
Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	58.0 °C	0.20059E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	47.0 °C	0.0000117588 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	47.0 °C	0.20122E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: 110% EOR

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Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(450/(130 \cdot 1)) - 1) + 0$
 $= 0.024 + 0$
 $= 0.024 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{R_o \cdot t}$
 $= 13000 / \sqrt{9.5 \cdot 2.5}$
 $= 2667.544$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y(t / Do)$
 $= 1.6 \cdot 0.072 \cdot 200187728 (2.5/19)$
 $= 3050.128 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3050.1/195.0$
 $= 15.642$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 195/379$
 $= 0.514$

Ratio [AE]:

$= Fhe / E$
 $= 3050/200187728$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.51) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.145$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 195(1 + 0.0020) / \ln(1 + 0.0020)^{0.14489266}$
 $= 480.867 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f(m2f2 - R)$
 $= 0.6(1 - 0.51)$
 $= 0.291$



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Curve Fitting Constant [A2]:

$$= (\text{outs} \cdot \exp(m2)) / (m2 \cdot m2)$$

$$= (379 \cdot \exp(0.29)) / (0.29 \cdot 0.29)$$

$$= 727.008 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.511.5 - 0.5 \cdot 0.512.5 - 0.513.5$$

$$= 0.361$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-2.162	0.296E-07	0.275E-07	0.900E-07	-0.799E-07	0.139E+08	0.155E-02
2	284	0.691	0.324E-06	-0.451E-06	0.241E-06	0.530E-06	0.154E+07	0.169E+00
3	237	-0.736	0.110E-06	-0.120E-09	0.155E-06	0.984E-07	0.271E+07	0.721E-01
4	213	-1.449	0.592E-07	0.421E-07	0.120E-06	-0.629E-07	0.612E+07	0.196E-01
5	201	-1.806	0.423E-07	0.362E-07	0.104E-06	-0.799E-07	0.927E+07	0.650E-02
6	195	-1.984	0.354E-07	0.319E-07	0.969E-07	-0.812E-07	0.114E+08	0.199E-02
7	192	-2.073	0.324E-07	0.297E-07	0.934E-07	-0.808E-07	0.125E+08	0.111E-03
8	191	-2.118	0.310E-07	0.286E-07	0.917E-07	-0.804E-07	0.132E+08	0.747E-03
9	191	-2.095	0.317E-07	0.292E-07	0.925E-07	-0.806E-07	0.129E+08	0.324E-03
10	192	-2.084	0.320E-07	0.294E-07	0.930E-07	-0.807E-07	0.127E+08	0.108E-03
11	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.112E-05
12	192	-2.081	0.321E-07	0.295E-07	0.931E-07	-0.808E-07	0.127E+08	0.536E-04
13	192	-2.080	0.322E-07	0.295E-07	0.931E-07	-0.808E-07	0.126E+08	0.263E-04
14	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.126E-04
15	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.573E-05
16	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.230E-05

Factor of Safety [FS]:

Since Fic > 0.55 Fy and Fic < Fy

$$= 2.407 - 0.741 \cdot \text{Fic} / \text{Fy}$$

$$= 2.407 - 0.741 \cdot 192/195$$

$$= 1.676$$

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic} / \text{FS}$$

$$= 192 / 1.68$$

$$= 114.792 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot \text{Fha} (t / \text{Do})$$

$$= 2 \cdot 114785 (2.5/19)$$

$$= 30206.646 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4090 mm
Allowable Internal Pressure at Corroded thickness	39697.23 KPa.
Required Internal Design Pressure	450.00 KPa.
Allowable External Pressure at Corroded thickness	30206.65 KPa.
Required External Design Pressure	13460.00 KPa.
Required Thickness due to Shell Side pressure	1.4090 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		Fail	
D1uc	7.021	...	6	195	1	104	Ps+Pt-Th	D1	Ok
D2uc	212.454	...	146	195	34	104	Ps+Pt-Th	D2	Ok



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Case: 3 9:36a Aug 12, 2026

D3uc	208.148	...	141	195	33	104	Ps+Pt-Th	D3	Ok
O1uc	41.353	...	47	390	10	104	Ps+Pt+Th	O1	Ok
O2uc	115.738	...	120	390	25	104	Ps+Pt+Th	O2	Ok
O3uc	111.034	...	115	390	24	104	Ps+Pt+Th	O3	Ok
O4uc	34.816	...	41	390	9	104	Ps+Pt+Th	O4	Ok
D1c	11.576	...	6	195	1	104	Ps+Pt-Th-c	D1	Ok
D2c	215.366	...	149	195	35	104	Ps+Pt-Th-c	D2	Ok
D3c	211.179	...	144	195	33	104	Ps+Pt-Th-c	D3	Ok
O1c	45.413	...	48	390	10	104	Ps+Pt+Th-c	O1	Ok
O2c	119.720	...	122	390	26	104	Ps+Pt+Th-c	O2	Ok
O3c	115.128	...	117	390	24	104	Ps+Pt+Th-c	O3	Ok
O4c	38.955	...	42	390	9	104	Ps+Pt+Th-c	O4	Ok

Max: 215.3662 ... mm 0.764 0.332 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	...	158	...	...	...	...	...	...	...	Ok
D2uc	42	158	...	...	...	...	...	...	...	Ok
D3uc	42	158	...	...	...	...	...	...	...	Ok
O1uc	10	474	-10	-157	...	...	...	...	...	Ok
O2uc	32	474	...	...	...	...	...	...	...	Ok
O3uc	32	474	...	...	...	...	...	...	...	Ok
O4uc	10	474	-10	-157	...	...	...	...	...	Ok
D1c	...	158	...	...	...	...	...	...	...	Ok
D2c	43	158	...	...	...	...	...	...	...	Ok
D3c	43	158	...	...	...	...	...	...	...	Ok
O1c	10	474	-10	-157	...	...	...	...	...	Ok
O2c	33	474	...	...	...	...	...	...	...	Ok
O3c	33	474	...	...	...	...	...	...	...	Ok
O4c	10	474	-10	-157	...	...	...	...	...	Ok
Max RATIO	0.272			0.064		...			...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	



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ASME TS Calc: 110% EOR

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D1uc	...	129	...	-97	7	1717	5	237	38	245	Ok
D2uc	10	129	-10	-97	133	1717	253	474	24	245	Ok
D3uc	10	129	-10	-97	127	1717	249	474	14	245	Ok
O1uc	17	259	...	...	221	3435	50	474	65	492	Ok
O2uc	12	259	...	...	156	3435	207	474	4	492	Ok
O3uc	12	259	...	...	155	3435	202	474	41	492	Ok
O4uc	16	259	...	...	214	3435	46	474	27	492	Ok
D1c	1	129	...	-97	7	1717	5	237	46	245	Ok
D2c	10	129	-10	-97	132	1717	263	474	26	245	Ok
D3c	10	129	-10	-97	126	1717	258	474	20	245	Ok
O1c	17	259	...	...	220	3435	53	474	74	492	Ok
O2c	12	259	...	...	162	3435	215	474	1	492	Ok
O3c	12	259	...	...	160	3435	210	474	48	492	Ok
O4c	16	259	...	...	213	3435	48	474	27	492	Ok

Max RATIO 0.078 0.103 0.078 0.554 0.188

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	215.366	250.000	Ok
Tube thickness :	1.409	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (67.000) adj. to tubesheet: 512.510 mm

Min Channel length of thk, (16.000) adj. to tubesheet: 250.452 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.716 mm

Due to Pressure Shell Compresses by : -0.916 mm

Due to Pressure + Thermal Shell Compresses by : -2.632 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	13709.72	1.000	17703.32	1.000
Tubesheet Shear Stress	29289.02	1.000	40507.75	1.000
Tube Tensile Stress	116212.62	1.000	173602.88	1.000
Tube Compressive Stress	357882.50	1.000	130201.54	1.000
Tube-Tubesheet Joint load	116209.28	1.000	173602.08	1.000
Shell Stress (Axial, Junction)	5552.58	0.225	17703.32	1.000
Tubesheet-Channel Junction	4527.74	1.000	17703.32	1.000
Tube Pressure Stress	39697.22	1.000	30206.64	1.000
Minimum MAWP	4527.74		17703.32	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	15294.78	1.000	19023.83	0.997
Tubesheet Shear Stress	34130.32	1.000	43057.11	1.000
Tube Tensile Stress	126124.52	1.000	183203.08	1.000
Tube Compressive Stress	551235.69	1.000	136074.80	1.000
Tube-Tubesheet Joint load	126124.44	1.000	183203.16	1.000
Shell Stress (Axial, Junction)	15294.78	0.300	19023.83	0.997



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Tubesheet-Channel Junction	6248.46	0.996	19023.83	0.997
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	6248.46		19023.83	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0058	0.0058	...	...	...	...
2	-0.9159	-0.9161	...	...	...	...
3	-0.9101	-0.9104	...	...	...	...
4	...	...	...	...	...	...
5	0.0058	0.0058	-1.7157	-1.7161	...	...
6	-0.9159	-0.9161	-1.7157	-1.7161	...	...
7	-0.9101	-0.9104	-1.7157	-1.7161	...	...
8	...	...	-1.7157	-1.7161	...	...

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ASME TS Calc: 10% SOR

Case: 4 9:36a Aug 12, 2026

Input Echo, Tubesheet Number 4, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	13830.00 KPa.
Shell Maximum Operating Pressure	Psox,max	13830.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	67.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1210.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	110.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	158.20 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	450.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	450.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1210.000 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	164.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1417
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13000.00 mm
Design Temperature of the Tubes		65.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		130.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	195.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	575.174 mm



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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	7296.3 cm ²
Perimeter of Tube Layout	Cp	3400.0 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	65.00 °C
Tubesheet Material		SA-350 LF1
Tubesheet Material UNS Number		K03009
Tubesheet Allowable Stress at Temperature	S	130.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	250.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1344.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	54.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	35.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



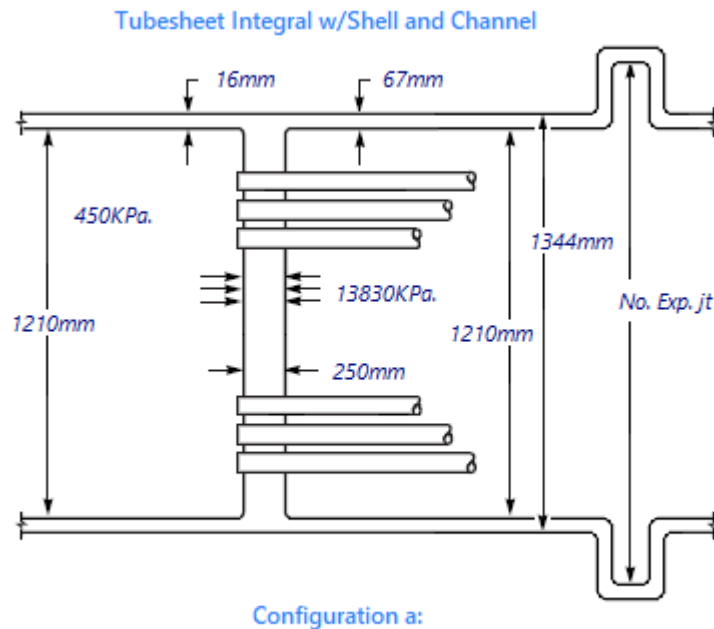
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Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	54.0 °C	0.0000118041 /°C
Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	54.0 °C	0.20082E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	35.0 °C	0.0000116690 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	35.0 °C	0.20191E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%



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Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(450/(130 \cdot 1)) - 1) + 0$
 $= 0.024 + 0$
 $= 0.024 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 13000 / \sqrt{(9.5 \cdot 2.5)}$
 $= 2667.544$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / Do)$
 $= 1.6 \cdot 0.072 \cdot 200187728 (2.5/19)$
 $= 3050.128 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3050.1/195.0$
 $= 15.642$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 195/379$
 $= 0.514$

Ratio [AE]:

$= Fhe / E$
 $= 3050/200187728$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.51) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.145$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 195 (1 + 0.0020) / \ln(1 + 0.0020)^{0.14489266}$
 $= 480.867 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.51)$
 $= 0.291$



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Curve Fitting Constant [A2]:

$$= (\text{outs} \cdot \exp(m2)) / (m2 \cdot m2)$$

$$= (379 \cdot \exp(0.29)) / (0.29 \cdot 0.29)$$

$$= 727.008 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.511.5 - 0.5 \cdot 0.512.5 - 0.513.5$$

$$= 0.361$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-2.162	0.296E-07	0.275E-07	0.900E-07	-0.799E-07	0.139E+08	0.155E-02
2	284	0.691	0.324E-06	-0.451E-06	0.241E-06	0.530E-06	0.154E+07	0.169E+00
3	237	-0.736	0.110E-06	-0.120E-09	0.155E-06	0.984E-07	0.271E+07	0.721E-01
4	213	-1.449	0.592E-07	0.421E-07	0.120E-06	-0.629E-07	0.612E+07	0.196E-01
5	201	-1.806	0.423E-07	0.362E-07	0.104E-06	-0.799E-07	0.927E+07	0.650E-02
6	195	-1.984	0.354E-07	0.319E-07	0.969E-07	-0.812E-07	0.114E+08	0.199E-02
7	192	-2.073	0.324E-07	0.297E-07	0.934E-07	-0.808E-07	0.125E+08	0.111E-03
8	191	-2.118	0.310E-07	0.286E-07	0.917E-07	-0.804E-07	0.132E+08	0.747E-03
9	191	-2.095	0.317E-07	0.292E-07	0.925E-07	-0.806E-07	0.129E+08	0.324E-03
10	192	-2.084	0.320E-07	0.294E-07	0.930E-07	-0.807E-07	0.127E+08	0.108E-03
11	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.112E-05
12	192	-2.081	0.321E-07	0.295E-07	0.931E-07	-0.808E-07	0.127E+08	0.536E-04
13	192	-2.080	0.322E-07	0.295E-07	0.931E-07	-0.808E-07	0.126E+08	0.263E-04
14	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.126E-04
15	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.573E-05
16	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.230E-05

Factor of Safety [FS]:

Since Fic > 0.55 Fy and Fic < Fy

$$= 2.407 - 0.741 \cdot \text{Fic} / \text{Fy}$$

$$= 2.407 - 0.741 \cdot 192/195$$

$$= 1.676$$

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic} / \text{FS}$$

$$= 192 / 1.68$$

$$= 114.792 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot \text{Fha} (t / \text{Do})$$

$$= 2 \cdot 114785 (2.5/19)$$

$$= 30206.646 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4359 mm
Allowable Internal Pressure at Corroded thickness	39697.23 KPa.
Required Internal Design Pressure	450.00 KPa.
Allowable External Pressure at Corroded thickness	30206.65 KPa.
Required External Design Pressure	13830.00 KPa.
Required Thickness due to Shell Side pressure	1.4359 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/ Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	7.021	...	6	195	1	104	Ps+Pt-Th	D1	Ok
D2uc	215.757	...	150	195	35	104	Ps+Pt-Th	D2	Ok



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Case: 4 9:36a Aug 12, 2026

D3uc	211.528	...	145	195	34	104	Ps+Pt-Th	D3	Ok
O1uc	69.087	...	76	390	17	104	Ps+Pt+Th	O1	Ok
O2uc	97.749	...	105	390	20	104	Ps+Pt+Th	O2	Ok
O3uc	91.859	...	100	390	18	104	Ps+Pt+Th	O3	Ok
O4uc	62.763	...	70	390	15	104	Ps+Pt+Th	O4	Ok
D1c	11.576	...	6	195	1	104	Ps+Pt-Th-c	D1	Ok
D2c	218.645	...	153	195	35	104	Ps+Pt-Th-c	D2	Ok
D3c	214.534	...	148	195	34	104	Ps+Pt-Th-c	D3	Ok
O1c	72.465	...	78	390	17	104	Ps+Pt+Th-c	O1	Ok
O2c	102.274	...	107	390	20	104	Ps+Pt+Th-c	O2	Ok
O3c	96.548	...	102	390	19	104	Ps+Pt+Th-c	O3	Ok
O4c	66.281	...	72	390	15	104	Ps+Pt+Th-c	O4	Ok

Max: 218.6453 ... mm 0.785 0.341 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	...	158	...	...	...	...	...	...	...	Ok
D2uc	43	158	...	...	...	...	...	...	...	Ok
D3uc	43	158	...	...	...	...	...	...	...	Ok
O1uc	17	474	-17	-157	...	...	...	...	...	Ok
O2uc	26	474	...	...	...	...	...	...	...	Ok
O3uc	26	474	...	...	...	...	...	...	...	Ok
O4uc	17	474	-17	-157	...	...	...	...	...	Ok
D1c	...	158	...	...	...	...	...	...	...	Ok
D2c	44	158	...	...	...	...	...	...	...	Ok
D3c	44	158	...	...	...	...	...	...	...	Ok
O1c	17	474	-17	-157	...	...	...	...	...	Ok
O2c	27	474	...	...	...	...	...	...	...	Ok
O3c	27	474	...	...	...	...	...	...	...	Ok
O4c	17	474	-17	-157	...	...	...	...	...	Ok
Max RATIO	0.280		0.109		...					

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	



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ASME TS Calc: 10% SOR

Case: 4 9:36a Aug 12, 2026

D1uc	...	129	...	-97	7	1717	5	237	38	245	Ok
D2uc	10	129	-10	-97	137	1717	260	474	24	245	Ok
D3uc	10	129	-10	-97	130	1717	256	474	14	245	Ok
O1uc	28	259	...	...	374	3435	83	474	84	492	Ok
O2uc	21	259	...	...	271	3435	181	474	22	492	Ok
O3uc	20	259	...	...	270	3435	176	474	60	492	Ok
O4uc	28	259	...	...	367	3435	79	474	46	492	Ok
D1c	1	129	...	-97	7	1717	5	237	46	245	Ok
D2c	10	129	-10	-97	136	1717	270	474	27	245	Ok
D3c	10	129	-10	-97	129	1717	266	474	20	245	Ok
O1c	28	259	...	...	372	3435	87	474	93	492	Ok
O2c	21	259	...	...	274	3435	188	474	20	492	Ok
O3c	21	259	...	...	272	3435	183	474	66	492	Ok
O4c	28	259	...	...	365	3435	82	474	47	492	Ok

Max RATIO 0.109 0.106 0.109 0.569 0.189

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	218.645	250.000	Ok
Tube thickness :	1.436	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (67.000) adj. to tubesheet: 512.510 mm

Min Channel length of thk, (16.000) adj. to tubesheet: 250.452 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -2.941 mm

Due to Pressure Shell Compresses by : -0.941 mm

Due to Pressure + Thermal Shell Compresses by : -3.882 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	13709.72	1.000	17703.32	1.000
Tubesheet Shear Stress	27325.36	1.000	40507.75	1.000
Tube Tensile Stress	116212.62	1.000	173602.88	1.000
Tube Compressive Stress	357882.50	1.000	130201.54	1.000
Tube-Tubesheet Joint load	116209.28	1.000	173602.08	1.000
Shell Stress (Axial, Junction)	5321.63	0.292	17703.32	1.000
Tubesheet-Channel Junction	4337.06	1.000	17703.32	1.000
Tube Pressure Stress	39697.22	1.000	30206.64	1.000
Minimum MAWP	4337.06		17703.32	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	15294.78	1.000	19023.83	0.997
Tubesheet Shear Stress	34130.32	1.000	43057.11	1.000
Tube Tensile Stress	126124.52	1.000	183203.08	1.000
Tube Compressive Stress	551235.69	1.000	136074.80	1.000
Tube-Tubesheet Joint load	126124.44	1.000	183203.16	1.000
Shell Stress (Axial, Junction)	15294.78	0.300	19023.83	0.997



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Tubesheet-Channel Junction	6248.46	0.996	19023.83	0.997
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	6248.46		19023.83	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0058	0.0058	...	...	...	...
2	-0.9411	-0.9413	...	...	...	...
3	-0.9353	-0.9355	...	...	...	...
4	...	...	...	...	...	...
5	0.0058	0.0058	-2.9400	-2.9407	...	...
6	-0.9411	-0.9413	-2.9400	-2.9407	...	...
7	-0.9353	-0.9355	-2.9400	-2.9407	...	...
8	...	...	-2.9400	-2.9407	...	...

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ASME TS Calc: START UP

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Input Echo, Tubesheet Number 5, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	10000.00 KPa.
Shell Maximum Operating Pressure	Psox,max	10000.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	67.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1210.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	110.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	158.20 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	500.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	500.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1210.000 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	164.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1417
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13000.00 mm
Design Temperature of the Tubes		65.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		130.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	195.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	575.174 mm



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ASME TS Calc: START UP

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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	7296.3 cm ²
Perimeter of Tube Layout	Cp	3400.0 mm

Tubesheet Data:**Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A**

Tubesheet Design Metal Temperature	T	65.00 °C
Tubesheet Material		SA-350 LF1
Tubesheet Material UNS Number		K03009
Tubesheet Allowable Stress at Temperature	S	130.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	250.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1344.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	36.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	36.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



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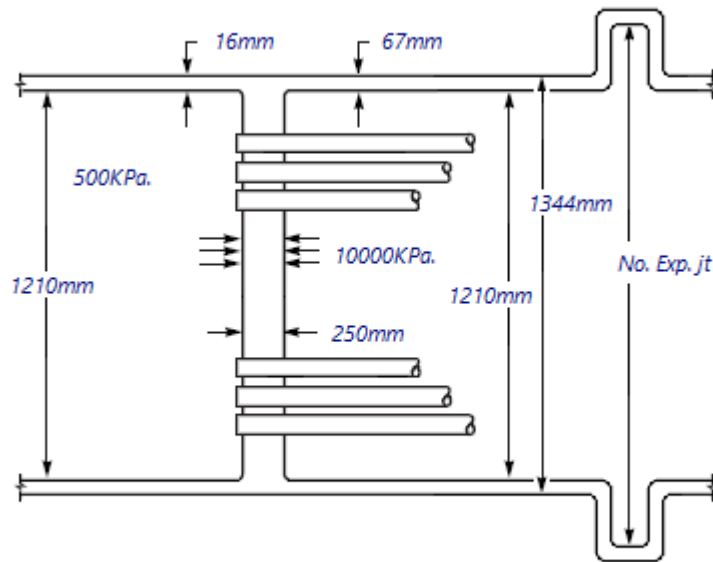
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ASME TS Calc: START UP

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Tubesheet Integral w/Shell and Channel



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	36.0 °C	0.0000116798 /°C
Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	36.0 °C	0.20185E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	36.0 °C	0.0000116798 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	36.0 °C	0.20185E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%



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Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(500/(130 \cdot 1)) - 1) + 0$
 $= 0.027 + 0$
 $= 0.027 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 13000 / \sqrt{(9.5 \cdot 2.5)}$
 $= 2667.544$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / Do)$
 $= 1.6 \cdot 0.072 \cdot 200187728 (2.5/19)$
 $= 3050.128 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3050.1/195.0$
 $= 15.642$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 195/379$
 $= 0.514$

Ratio [AE]:

$= Fhe / E$
 $= 3050/200187728$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.51) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.145$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 195 (1 + 0.0020) / \ln(1 + 0.0020)^{0.14489266}$
 $= 480.867 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.51)$
 $= 0.291$



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Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (379 \cdot \exp(0.29)) / (0.29 \cdot 0.29)$$

$$= 727.008 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.511.5 - 0.5 \cdot 0.512.5 - 0.513.5$$

$$= 0.361$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-2.162	0.296E-07	0.275E-07	0.900E-07	-0.799E-07	0.139E+08	0.155E-02
2	284	0.691	0.324E-06	-0.451E-06	0.241E-06	0.530E-06	0.154E+07	0.169E+00
3	237	-0.736	0.110E-06	-0.120E-09	0.155E-06	0.984E-07	0.271E+07	0.721E-01
4	213	-1.449	0.592E-07	0.421E-07	0.120E-06	-0.629E-07	0.612E+07	0.196E-01
5	201	-1.806	0.423E-07	0.362E-07	0.104E-06	-0.799E-07	0.927E+07	0.650E-02
6	195	-1.984	0.354E-07	0.319E-07	0.969E-07	-0.812E-07	0.114E+08	0.199E-02
7	192	-2.073	0.324E-07	0.297E-07	0.934E-07	-0.808E-07	0.125E+08	0.111E-03
8	191	-2.118	0.310E-07	0.286E-07	0.917E-07	-0.804E-07	0.132E+08	0.747E-03
9	191	-2.095	0.317E-07	0.292E-07	0.925E-07	-0.806E-07	0.129E+08	0.324E-03
10	192	-2.084	0.320E-07	0.294E-07	0.930E-07	-0.807E-07	0.127E+08	0.108E-03
11	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.112E-05
12	192	-2.081	0.321E-07	0.295E-07	0.931E-07	-0.808E-07	0.127E+08	0.536E-04
13	192	-2.080	0.322E-07	0.295E-07	0.931E-07	-0.808E-07	0.126E+08	0.263E-04
14	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.126E-04
15	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.573E-05
16	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.230E-05

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 192/195$$

$$= 1.676$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 192/1.68$$

$$= 114.792 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 114785 (2.5/19)$$

$$= 30206.646 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1481 mm
Allowable Internal Pressure at Corroded thickness	39697.23 KPa.
Required Internal Design Pressure	500.00 KPa.
Allowable External Pressure at Corroded thickness	30206.65 KPa.
Required External Design Pressure	10000.00 KPa.
Required Thickness due to Shell Side pressure	1.1481 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	8.265	...	7	195	2	104	Ps+Pt-Th	D1	Ok
D2uc	178.807	...	109	195	25	104	Ps+Pt-Th	D2	Ok



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Case: 5 9:36a Aug 12, 2026

D3uc	172.890	...	103	195	24	104	Ps+Pt-Th	D3	Ok
O1uc	3.810	...	7	390	2	104	Ps+Pt+Th	O1	Ok
O2uc	112.180	...	109	390	25	104	Ps+Pt+Th	O2	Ok
O3uc	106.892	...	103	390	24	104	Ps+Pt+Th	O3	Ok
O4uc	3.810	...	...	390	...	104	Ps+Pt+Th	O4	Ok
D1c	12.822	...	7	195	2	104	Ps+Pt-Th-c	D1	Ok
D2c	181.964	...	111	195	26	104	Ps+Pt-Th-c	D2	Ok
D3c	176.189	...	105	195	24	104	Ps+Pt-Th-c	D3	Ok
O1c	8.310	...	7	390	2	104	Ps+Pt+Th-c	O1	Ok
O2c	115.834	...	111	390	26	104	Ps+Pt+Th-c	O2	Ok
O3c	110.656	...	105	390	24	104	Ps+Pt+Th-c	O3	Ok
O4c	8.310	...	...	390	...	104	Ps+Pt+Th-c	O4	Ok

Max: 181.9643 ... mm 0.568 0.247 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	...	158	...	...	...	...	...	...	...	Ok
D2uc	31	158	...	...	...	...	...	...	...	Ok
D3uc	31	158	...	...	...	...	...	...	...	Ok
O1uc	...	474	...	...	...	...	...	...	...	Ok
O2uc	31	474	...	...	...	...	...	...	...	Ok
O3uc	31	474	...	...	...	...	...	...	...	Ok
O4uc	...	474	...	...	...	...	...	...	...	Ok
D1c	...	158	...	...	...	...	...	...	...	Ok
D2c	32	158	...	...	...	...	...	...	...	Ok
D3c	32	158	...	...	...	...	...	...	...	Ok
O1c	...	474	...	...	...	...	...	...	...	Ok
O2c	32	474	...	...	...	...	...	...	...	Ok
O3c	32	474	...	...	...	...	...	...	...	Ok
O4c	...	474	...	...	...	...	...	...	...	Ok
Max RATIO	0.203	...	...	...	...	...	...	...	...	...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	



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ASME TS Calc: START UP

Case: 5 9:36a Aug 12, 2026

D1uc	1	129	...	-97	7	1717	5	237	42	245	Ok
D2uc	7	129	-7	-97	99	1717	188	237	17	245	Ok
D3uc	7	129	-7	-97	92	1717	183	237	24	245	Ok
O1uc	1	259	...	-97	7	3435	5	474	42	492	Ok
O2uc	7	259	-7	-97	99	3435	188	474	17	492	Ok
O3uc	7	259	-7	-97	92	3435	183	474	24	492	Ok
O4uc	...	259	...	...	...	3435	...	474	...	492	Ok
D1c	1	129	...	-97	7	1717	6	237	51	245	Ok
D2c	7	129	-7	-97	98	1717	195	237	19	245	Ok
D3c	7	129	-7	-97	91	1717	190	237	32	245	Ok
O1c	1	259	...	-97	7	3435	6	474	51	492	Ok
O2c	7	259	-7	-97	98	3435	195	474	19	492	Ok
O3c	7	259	-7	-97	91	3435	190	474	32	492	Ok
O4c	...	259	...	...	...	3435	...	474	...	492	Ok

Max RATIO 0.058 0.077 0.058 0.824 0.209

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	181.964	250.000	Ok
Tube thickness :	1.148	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (67.000) adj. to tubesheet: 512.510 mm

Min Channel length of thk, (16.000) adj. to tubesheet: 250.452 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.681 mm
 Due to Pressure + Thermal Shell Compresses by : -0.681 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	13709.72	1.000	17703.32	1.000
Tubesheet Shear Stress	32040.75	1.000	40507.75	1.000
Tube Tensile Stress	116212.62	1.000	173602.88	1.000
Tube Compressive Stress	27419.45	0.077	130201.45	1.000
Tube-Tubesheet Joint load	116209.28	1.000	173602.08	1.000
Shell Stress (Axial, Junction)	5864.21	0.142	17703.32	1.000
Tubesheet-Channel Junction	4794.97	1.000	17703.32	1.000
Tube Pressure Stress	39697.22	1.000	30206.64	1.000
Minimum MAWP	4794.97		17703.32	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	15294.78	1.000	19023.83	0.997
Tubesheet Shear Stress	34130.32	1.000	43057.11	1.000
Tube Tensile Stress	126124.51	1.000	183203.08	1.000
Tube Compressive Stress	551236.81	1.000	136074.80	1.000
Tube-Tubesheet Joint load	126124.47	1.000	183203.16	1.000
Shell Stress (Axial, Junction)	15294.78	0.300	19023.83	0.997



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Case: 5 9:36a Aug 12,2026

Tubesheet-Channel Junction	6248.46	0.996	19023.83	0.997
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	6248.46		19023.83	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0064	0.0064	...	...	...	...
2	-0.6805	-0.6806	...	...	...	...
3	-0.6741	-0.6742	...	...	...	...
4	...	...	...	...	...	...
5	0.0064	0.0064	...	...	...	...
6	-0.6805	-0.6806	...	...	...	...
7	-0.6741	-0.6742	...	...	...	...
8	...	...	...	...	...	...

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ASME TS Calc:

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Input Echo, Tubesheet Number 6, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	13260.00 KPa.
Shell Maximum Operating Pressure	Psox,max	13260.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	67.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1210.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	110.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	158.20 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	500.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	500.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1210.000 mm
Channel Design Temperature	TEMPC	65.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	164.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1417
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13000.00 mm
Design Temperature of the Tubes		65.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		130.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	195.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	575.174 mm



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ASME TS Calc:

Case: 6 9:36a Aug 12,2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	7296.3 cm ²
Perimeter of Tube Layout	Cp	3400.0 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	65.00 °C
Tubesheet Material		SA-350 LF1
Tubesheet Material UNS Number		K03009
Tubesheet Allowable Stress at Temperature	S	130.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	250.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1344.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	62.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	44.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



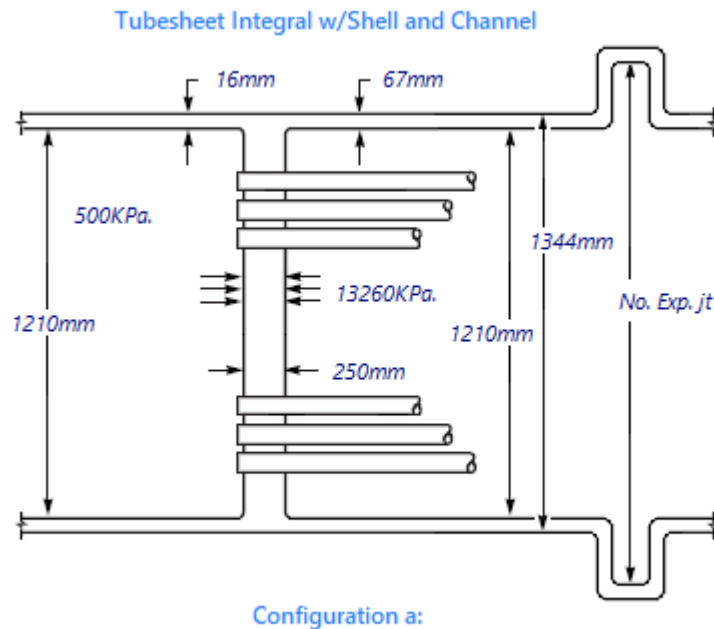
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ASME TS Calc:

Case: 6 9:36a Aug 12,2026

**Messages regarding the Tubesheet & Exchanger Input Data:**

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025**Elasticity/Expansion Material Properties:**

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	62.0 °C	0.0000118560 /°C
Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	62.0 °C	0.20036E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	44.0 °C	0.0000117393 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	44.0 °C	0.20139E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: Case: 6 9:36a Aug 12, 2026

Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(500/(130 \cdot 1)) - 1) + 0$
 $= 0.027 + 0$
 $= 0.027 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{R_o \cdot t}$
 $= 13000 / \sqrt{9.5 \cdot 2.5}$
 $= 2667.544$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / Do)$
 $= 1.6 \cdot 0.072 \cdot 200187728 (2.5/19)$
 $= 3050.128 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3050.1/195.0$
 $= 15.642$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 195/379$
 $= 0.514$

Ratio [AE]:

$= Fhe / E$
 $= 3050/200187728$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.51) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.145$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 195 (1 + 0.0020) / \ln(1 + 0.0020)^{0.14489266}$
 $= 480.867 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.51)$
 $= 0.291$



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ASME TS Calc:

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Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (379 \cdot \exp(0.29)) / (0.29 \cdot 0.29)$$

$$= 727.008 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.511.5 - 0.5 \cdot 0.512.5 - 0.513.5$$

$$= 0.361$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-2.162	0.296E-07	0.275E-07	0.900E-07	-0.799E-07	0.139E+08	0.155E-02
2	284	0.691	0.324E-06	-0.451E-06	0.241E-06	0.530E-06	0.154E+07	0.169E+00
3	237	-0.736	0.110E-06	-0.120E-09	0.155E-06	0.984E-07	0.271E+07	0.721E-01
4	213	-1.449	0.592E-07	0.421E-07	0.120E-06	-0.629E-07	0.612E+07	0.196E-01
5	201	-1.806	0.423E-07	0.362E-07	0.104E-06	-0.799E-07	0.927E+07	0.650E-02
6	195	-1.984	0.354E-07	0.319E-07	0.969E-07	-0.812E-07	0.114E+08	0.199E-02
7	192	-2.073	0.324E-07	0.297E-07	0.934E-07	-0.808E-07	0.125E+08	0.111E-03
8	191	-2.118	0.310E-07	0.286E-07	0.917E-07	-0.804E-07	0.132E+08	0.747E-03
9	191	-2.095	0.317E-07	0.292E-07	0.925E-07	-0.806E-07	0.129E+08	0.324E-03
10	192	-2.084	0.320E-07	0.294E-07	0.930E-07	-0.807E-07	0.127E+08	0.108E-03
11	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.112E-05
12	192	-2.081	0.321E-07	0.295E-07	0.931E-07	-0.808E-07	0.127E+08	0.536E-04
13	192	-2.080	0.322E-07	0.295E-07	0.931E-07	-0.808E-07	0.126E+08	0.263E-04
14	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.126E-04
15	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.573E-05
16	192	-2.079	0.322E-07	0.296E-07	0.932E-07	-0.808E-07	0.126E+08	0.230E-05

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 192/195$$

$$= 1.676$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 192/1.68$$

$$= 114.792 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / D_o)$$

$$= 2 \cdot 114785 (2.5/19)$$

$$= 30206.646 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.3944 mm
Allowable Internal Pressure at Corroded thickness	39697.23 KPa.
Required Internal Design Pressure	500.00 KPa.
Allowable External Pressure at Corroded thickness	30206.65 KPa.
Required External Design Pressure	13260.00 KPa.
Required Thickness due to Shell Side pressure	1.3944 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/ Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	8.265	...	7	195	2	104	Ps+Pt-Th	D1	Ok
D2uc	210.649	...	144	195	34	104	Ps+Pt-Th	D2	Ok



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ASME TS Calc:

Case: 6 9:36a Aug 12, 2026

D3uc	205.809	...	139	195	32	104	Ps+Pt-Th	D3	Ok
O1uc	66.888	...	73	390	16	104	Ps+Pt+Th	O1	Ok
O2uc	94.055	...	101	390	19	104	Ps+Pt+Th	O2	Ok
O3uc	87.240	...	95	390	17	104	Ps+Pt+Th	O3	Ok
O4uc	59.788	...	67	390	15	104	Ps+Pt+Th	O4	Ok
D1c	12.822	...	7	195	2	104	Ps+Pt-Th-c	D1	Ok
D2c	213.575	...	147	195	34	104	Ps+Pt-Th-c	D2	Ok
D3c	208.867	...	141	195	32	104	Ps+Pt-Th-c	D3	Ok
O1c	70.323	...	76	390	16	104	Ps+Pt+Th-c	O1	Ok
O2c	98.608	...	103	390	19	104	Ps+Pt+Th-c	O2	Ok
O3c	91.984	...	98	390	18	104	Ps+Pt+Th-c	O3	Ok
O4c	63.380	...	68	390	15	104	Ps+Pt+Th-c	O4	Ok

Max: 213.5746 ... mm 0.753 0.327 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					---- Shell Band Stress ----					Pass
Case#	Ten	Allwd	Cmp	Allwd	Ten	Allwd	Cmp	Allwd	Fail	
D1uc	...	158	...	...	...	...	...	...	...	Ok
D2uc	41	158	...	...	...	...	...	...	...	Ok
D3uc	42	158	...	...	...	...	...	...	...	Ok
O1uc	16	474	-16	-157	...	...	...	...	...	Ok
O2uc	25	474	...	...	...	...	...	...	...	Ok
O3uc	25	474	...	...	...	...	...	...	...	Ok
O4uc	16	474	-16	-157	...	...	...	...	...	Ok
D1c	...	158	...	...	...	...	...	...	...	Ok
D2c	42	158	...	...	...	...	...	...	...	Ok
D3c	42	158	...	...	...	...	...	...	...	Ok
O1c	16	474	-16	-157	...	...	...	...	...	Ok
O2c	26	474	...	...	...	...	...	...	...	Ok
O3c	26	474	...	...	...	...	...	...	...	Ok
O4c	16	474	-16	-157	...	...	...	...	...	Ok
Max RATIO	0.268			0.104	...			...		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail



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ASME TS Calc:

Case: 6 9:36a Aug 12, 2026

D1uc	1	129	...	-97	7	1717	5	237	42	245	Ok
D2uc	10	129	-10	-97	131	1717	249	474	23	245	Ok
D3uc	9	129	-9	-97	124	1717	244	474	19	245	Ok
O1uc	27	259	...	...	358	3435	80	474	86	492	Ok
O2uc	20	259	...	...	259	3435	173	474	21	492	Ok
O3uc	20	259	...	...	258	3435	169	474	63	492	Ok
O4uc	27	259	...	...	351	3435	76	474	44	492	Ok
D1c	1	129	...	-97	7	1717	6	237	51	245	Ok
D2c	10	129	-10	-97	130	1717	259	474	26	245	Ok
D3c	9	129	-9	-97	123	1717	254	474	26	245	Ok
O1c	27	259	...	...	356	3435	84	474	96	492	Ok
O2c	20	259	...	...	262	3435	180	474	19	492	Ok
O3c	20	259	...	...	260	3435	175	474	71	492	Ok
O4c	26	259	...	...	349	3435	79	474	45	492	Ok

Max RATIO 0.104 0.102 0.104 0.546 0.209

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	213.575	250.000	Ok
Tube thickness :	1.394	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (67.000) adj. to tubesheet: 512.510 mm

Min Channel length of thk, (16.000) adj. to tubesheet: 250.452 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -2.810 mm

Due to Pressure Shell Compresses by : -0.903 mm

Due to Pressure + Thermal Shell Compresses by : -3.712 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	13709.72	1.000	17703.32	1.000
Tubesheet Shear Stress	27535.49	1.000	40507.75	1.000
Tube Tensile Stress	116212.62	1.000	173602.88	1.000
Tube Compressive Stress	357882.50	1.000	130201.54	1.000
Tube-Tubesheet Joint load	116209.28	1.000	173602.08	1.000
Shell Stress (Axial, Junction)	5346.34	0.285	17703.32	1.000
Tubesheet-Channel Junction	4357.47	1.000	17703.32	1.000
Tube Pressure Stress	39697.22	1.000	30206.64	1.000
Minimum MAWP	4357.47		17703.32	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	15294.78	1.000	19023.83	0.997
Tubesheet Shear Stress	34130.32	1.000	43057.11	1.000
Tube Tensile Stress	126124.51	1.000	183203.08	1.000
Tube Compressive Stress	551236.81	1.000	136074.80	1.000
Tube-Tubesheet Joint load	126124.47	1.000	183203.16	1.000
Shell Stress (Axial, Junction)	15294.78	0.300	19023.83	0.997



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ASME TS Calc:

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Tubesheet-Channel Junction	6248.46	0.996	19023.83	0.997
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	6248.46		19023.83	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0064	0.0064	...	...	...	...
2	-0.9023	-0.9025	...	...	...	...
3	-0.8959	-0.8961	...	...	...	...
4	...	...	...	...	...	...
5	0.0064	0.0064	-2.8090	-2.8096	...	...
6	-0.9023	-0.9025	-2.8090	-2.8096	...	...
7	-0.8959	-0.8961	-2.8090	-2.8096	...	...
8	...	...	-2.8090	-2.8096	...	...

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: